

5G RAN

ANR Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

Technical Changes

None

Editorial Changes

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

1.2 5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-12-14).

Technical Changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the configuration of the measurement period and the threshold expressed as the number of reportCGI measurements for intra-RAT neighboring cell class label management. For details, see: 4.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells 4.1.4.1.1 Data Preparation 4.1.4.1.2 Using MML Commands	Added parameters: gNodeBAlgo.NrHoSuccRateStatInterval NRCellAnr.NrNCellRptCgiNumThld Modified parameter: Added the NR_HO_SUCC_RATE_INTVL_CFG_SW option to the gNodeBAlgo.SelfConfigSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added optimized measurement of the success rate of handovers to neighboring cells in SA networking. For details, see the following sections in 4.1 Intra-RAT ANR and 5.1 Inter-RAT ANR: • 4.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells • 4.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate • 4.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells • 4.1.2.1 Benefits • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands • 5.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate • 5.1.1.6 Automatic Optimization of Neighbor Relationship	Modified parameter: Added the SA_NR_NCELL_SUCC_RT_STAT_SW and SA_LTE_NCELL_SUCC_RT_STAT_SW options to the NRCellAnr.AnrAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Attributes Based on Identification of Over-Distant Neighboring Cells • 5.1.2.1 Benefits • 5.1.4.2 Data Configuration • 5.1.4.2.2 Using MML Commands			
Added UE blacklist management in intra-RAT ANR. For details, see: • 4.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells • 4.1.2.2 Impacts • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands	Modified parameter: Added the NR_NR_ANR_SW_OFF option to the gNBUECompatSw.BlacklistCtrlExtSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added optimized measurement for automatic removal of neighboring cells. For details, see the following sections in 4.1 Intra-RAT ANR and 5.1 Inter-RAT ANR: • 4.1.1.3 Automatic Removal of Neighboring Cells • 4.1.2.1 Benefits • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands • 5.1.1.3 Automatic Removal of Neighboring Cells • 5.1.2.1 Benefits • 5.1.4.2.1 Data Preparation • 5.1.4.2.2 Using MML Commands	Modified parameter: Added the NR_NCELL_DEL_STAT_OPT_SW and LTE_NCELL_DEL_STAT_OPT_SW options to the gNodeBAlgo.SelfConfigSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added a switch for fast NR networking option query. For details, see: • 4.1.1.4 Automatic NCL Update • 4.1.1.7.2 Automatic External-Cell PLMN List Update • 4.1.2.2 Impacts • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands	Modified parameter: Added the NR_NTW_OPTN_FAST_QUERY_SW option to the gNodeBAlgo.SelfConfigSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for the configuration of the measurement periods for "automatic optimization of neighbor relationship attributes based on the handover success rate" and "automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells". For details, see the following sections in 4.1 Intra-RAT ANR and 5.1 Inter-RAT ANR: • 4.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate • 4.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands • 5.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate • 5.1.1.6 Automatic Optimization of	Added parameters: • gNodeBAlgo.NrHoSuccRateStatInterval • gNodeBAlgo.N2LhoSuccRateStatInterval Modified parameter: Added the NR_HO_SUCC_RATE_INTVL_CFG_SW and N2L_HO_SUCC_RATE_INTVL_CFG_SW options to the gNodeBAlgo.SelfConfigSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells • 5.1.4.2 Data Configuration • 5.1.4.2.2 Using MML Commands			
Added intra-RAT PLMN selection optimization. For details, see: • 4.1.1.7.1 Automatic External-Cell PLMN List Addition • 4.1.2.1 Benefits • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands	Modified parameter: Added the ANR_PLMN_SEL_OPT_SW option to the gNodeBAlgo.SelfConfigSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for forbidding automatic neighboring frequency configuration for a specific neighboring frequency. For details, see the following sections in 4.1 Intra-RAT ANR and 5.1 Inter-RAT ANR: • 4.1.1.8 Automatic Neighboring Frequency Configuration • 4.1.2.2 Impacts • 4.1.4.1.1 Data Preparation • 4.1.4.1.2 Using MML Commands • 5.1.1.8 Automatic Neighboring Frequency Configuration • 5.1.2.2 Impacts • 5.1.4.2 Data Configuration • 5.1.4.2.2 Using MML Commands	Modified parameters: • Added the AUTO_NR_FREQ_REL_FORBID_SW option to the NRCellFreqRelation.FrequencyCtrlSwitch parameter. • Added the AUTO_EUTRAN_FREQ_REL_FORBID_SW option to the NRCellEutranNFreq.FrequencyCtrlSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship of intra-RAT ANR with RF module fast deep dormancy. For details, see 4.1.3.2.3 Function Impacts.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added inter-RAT neighboring cell class label management. For details, see: • 5.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells • 5.1.2.1 Benefits • 5.1.4.2.1 Data Preparation • 5.1.4.2.2 Using MML Commands	Added parameters: • NRCellEutraNRelation.EutraNCellClassLabel • gNodeBAlgo.N2LhoSuccessRateStatInterval • NRCellAnr.EutraNCellHoSuccRateThld • NRCellAnr.EutraNCellRptCgiNumThld Modified parameters: • Added the EUTRAN_NCELL_CLASS_LABEL_SW and N2L_HO_SUCC_RATE_INTVL_CFG_SW options to the gNodeBAlgo.SelfConfigSwitch parameter. • Adapted the SA_LTE_NCELL_SUCC_RATE_STAT_SW option of the NRCellAnr.AnrAlgoSwitch parameter and the NRCellAnr.LteNCellHoNumThld parameter to inter-RAT neighboring cell class label management.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Editorial Changes

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-021204	Automatic Neighbour Relation (ANR)	4.1 Intra-RAT ANR 4.2 Intra-RAT PLMN ID Management 5.1 Inter-RAT ANR 5.2 Inter-RAT PLMN ID Management

2.3 Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Intra-RAT ANR	• Optimization mode: None • Control mode: None • Penalty on reportCGI measurements: None • Automatic detection and addition of missing neighboring cells: Supported in both NSA and SA networking, with the following differences: In NSA networking, an eNodeB forwards RRC messages between a gNodeB and a UE. In SA networking, a gNodeB and a UE can directly exchange RRC messages. In NSA networking, automatic neighboring cell class label adjustment is controlled by a function switch; in SA networking, this function is not controlled by any function switch. In addition, measurement methods are different between NSA and SA networking. • Automatic removal of neighboring cells: None • Automatic NCL update: Supported in both NSA and SA networking, with the following differences: In NSA networking,	4.1 Intra-RAT ANR

Function Name	Difference	Chapter/Section
	NCL update can be triggered by measurement reports from UEs or queries from the MAE-Access. In SA networking, NCL update can be triggered by measurement reports from UEs, messages transmitted over the Xn interface between gNodeBs, or queries from the MAE-Access. • Automatic optimization of neighbor relationship attributes based on the handover success rate: None • Automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells: None • Automatic external-cell PLMN list configuration: None • Automatic neighboring frequency configuration: None • NCL/NRT management based on the gNodeB blacklist: None	
Intra-RAT PLMN ID management	None	4.2 Intra-RAT PLMN ID Management
Inter-RAT ANR	Supported only in SA networking	5.1 Inter-RAT ANR
Inter-RAT PLMN ID management	Supported only in SA networking	5.2 Inter-RAT PLMN ID Management

2.4 Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Intra-RAT ANR	Supported only in low frequency bands	4.1 Intra-RAT ANR
Intra-RAT PLMN ID management	Supported only in low frequency bands	4.2 Intra-RAT PLMN ID Management
Inter-RAT ANR	Supported only in low frequency bands	5.1 Inter-RAT ANR
Inter-RAT PLMN ID management	Supported only in low frequency bands	5.2 Inter-RAT PLMN ID Management

3 Overview

3.1 Definition

In communications networks, neighboring cells are considered as possible target cells for UEs during handovers. During network O&M, such as network swapping or capacity expansion, neighbor relationships are constantly changing and need prompt adjustment. Manual neighbor relationship configuration has a risk of errors and is costly and inefficient. Improper neighbor relationships cause service drops, which adversely affect user experience. To address this issue, automatic neighbor relation (ANR) is introduced.

This feature automatically manages intra-NG-RAN neighboring cells as well as inter-RAT NG-RAN neighboring cells and E-UTRAN neighboring cells. This not only reduces network planning and optimization costs but also improves network optimization efficiency and user experience. This feature involves intra-RAT ANR and inter-RAT ANR. For details, see [4 Intra-RAT ANR Management](#) and [5 Inter-RAT ANR Management](#), respectively.

3.2 Related Concepts

3.2.1 NCL

A neighboring cell list (NCL) of a gNodeB contains information about external cells, specifically, neighboring cells of all the cells served by the gNodeB. Such neighboring cells are not served by this gNodeB.

The NCL is a gNodeB-specific configuration. Each gNodeB has an intra-RAT NCL and an inter-RAT NCL.

Intra-RAT NCL

An intra-RAT NCL contains all external NG-RAN cells of a gNodeB. [Table 3-1](#) lists the primary information in an intra-RAT NCL.

Table 3-1 Example of primary information in an intra-RAT NCL

Parameter	External NG-RAN Cell 1	External NG-RAN Cell 2	...
Mcc	460	460	...
Mnc	01	01	...
gNBId	1	2	...
CellId	1	2	...
Tac	8	8	...
PhysicalCellId	15	16	...
SsbFreqPos	620000	620000	...
SsbDescMethod	SSB_DESC_TYPE_NARF CN	SSB_DESC_TYPE_NARF CN	...
NrNetworkingOption	NSA	NSA	...
FrequencyBand	n77	n77	...
AdditionalFrequency- Band	null	null	...
PlmnReservedFlag	FALSE	FALSE	...

The parameters in [Table 3-1](#) are defined as follows:

- Mcc: mobile country code (MCC)
- Mnc: mobile network code (MNC)
- gNBId: gNodeB ID
- CellId: cell ID
- Tac: tracking area code (TAC)
- PhysicalCellId: physical cell identifier (PCI)
- SsbFreqPos: frequency-domain position of the synchronization signal and PBCH block (SSB)
- SsbDescMethod: description method of the SSB's frequency-domain position
- NrNetworkingOption: NR networking option
- FrequencyBand: frequency band
- AdditionalFrequencyBand: additional frequency band
- PlmnReservedFlag: reserved PLMN flag

Inter-RAT NCL

An inter-RAT NCL contains all external E-UTRAN cells of a gNodeB. [Table 3-2](#) lists the primary information in an inter-RAT NCL.

Table 3-2 Example of primary information in an inter-RAT NCL

Parameter	External E-UTRAN Cell 1	External E-UTRAN Cell 2	...
Mcc	460	460	...
Mnc	01	01	...
EnodebId	1	2	...
CellId	1	2	...
Tac	8	8	...
PhysicalCellId	15	16	...
DLearfcn	1300	1300	...
MASTER_PLMN_RESERV ED_FLAG	0	0	...

The parameters in [Table 3-2](#) are defined as follows:

- Mcc: MCC
- Mnc: MNC
- EnodebId: eNodeB ID
- CellId: cell ID
- Tac: TAC
- PhysicalCellId: PCI
- DLearfcn: downlink E-UTRA absolute radio frequency channel number (EARFCN)
- MASTER_PLMN_RESERVED_FLAG: reserved flag of the primary PLMN

3.2.2 NRT

The neighboring relation table (NRT) of a cell contains information about all neighboring cells of the cell.

The NRT is a cell-specific configuration. Each cell has an intra-RAT NRT and an inter-RAT NRT.

Intra-RAT NRT

An intra-RAT NRT contains all neighboring NG-RAN cells of a cell. [Table 3-3](#) lists the primary information in the intra-RAT NRT.

Table 3-3 Example of primary information in an intra-RAT NRT

Parameter	Neighboring NG-RAN Cell 1	Neighboring NG-RAN Cell 2	...
NrCellID	1	1	...
Mcc	460	460	...
Mnc	01	01	...
gNBId	1	2	...
CellId	1	2	...

The parameters in [Table 3-3](#) are defined as follows:

- NrCellID: NG-RAN cell ID
- Mcc: MCC
- Mnc: MNC
- gNBId: gNodeB ID
- CellId: cell ID

Inter-RAT NRT

An inter-RAT NRT contains all neighboring E-UTRAN cells of a cell. [Table 3-4](#) lists the primary information in the inter-RAT NRT.

Table 3-4 Example of primary information in an inter-RAT NRT

Parameter	Neighboring E-UTRAN Cell 1	Neighboring E-UTRAN Cell 2	...
NrCellID	1	1	...
Mcc	460	460	...
Mnc	01	01	...
EnodebId	1	2	...
CellId	1	2	...

The parameters in [Table 3-4](#) are defined as follows:

- NrCellID: NG-RAN cell ID
- Mcc: MCC
- Mnc: MNC
- EnodebId: eNodeB ID
- CellId: cell ID

4 Intra-RAT ANR Management

4.1 Intra-RAT ANR

4.1.1 Principles

Intra-RAT ANR enables automatic management of intra-NR NCLs and intra-NR NRTs (NCL and NRT for short). [Table 4-1](#) lists support for subfunctions of intra-RAT ANR depending on whether neighboring NG-RAN cells are shared among multiple operators.

Table 4-1 Subfunctions of intra-RAT ANR

Subfunction	Scenarios with Multi-Operator Sharing	Scenarios Without Multi-Operator Sharing
4.1.1.1.1 Optimization Mode	Supported	Supported
4.1.1.1.2 Control Mode	Supported	Supported
4.1.1.1.3 Penalty on reportCGI Measurements	Supported	Supported
4.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells	Supported	Supported
4.1.1.3 Automatic Removal of Neighboring Cells	Supported	Supported
4.1.1.4 Automatic NCL Update	Supported	Supported

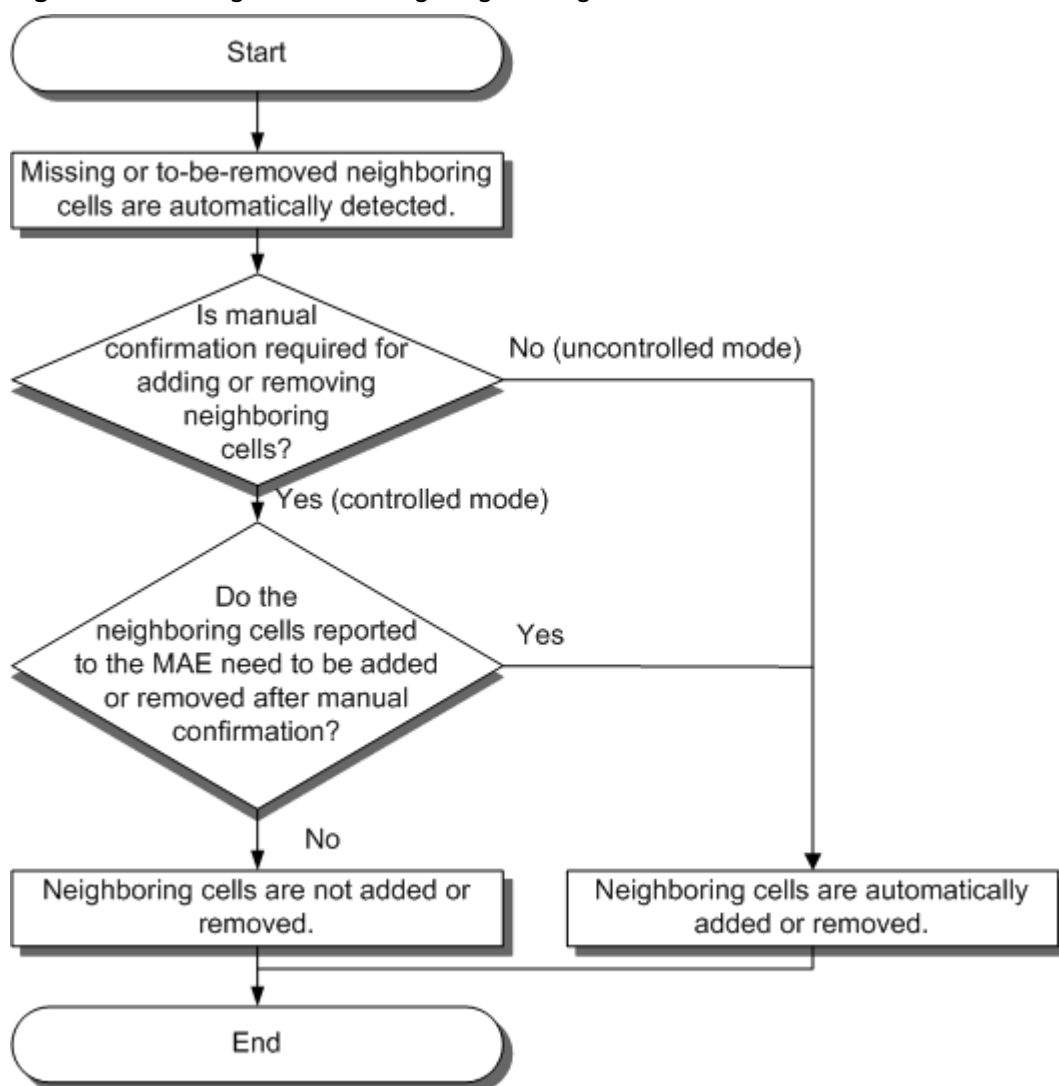
Subfunction	Scenarios with Multi-Operator Sharing	Scenarios Without Multi-Operator Sharing
4.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate	Supported	Supported
4.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells	Supported	Supported
4.1.1.7 Automatic External-Cell PLMN List Configuration	Supported	N/A
4.1.1.8 Automatic Neighboring Frequency Configuration	Supported	Supported
4.1.1.9 NCL/NRT Management Based on the gNodeB Blacklist	Supported	Supported

4.1.1.1 General Principles

4.1.1.1.1 Optimization Mode

The optimization mode is classified into uncontrolled mode and controlled mode, depending on whether manual confirmation is required in neighboring cell addition and removal. [Figure 4-1](#) shows how neighboring cells are added and removed in each mode. The **NR_NR_ANR_CTRL_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter specifies the optimization mode. When this option is selected, controlled mode is used. When this option is deselected, uncontrolled mode is used.

Figure 4-1 Adding and removing neighboring cells



Uncontrolled Mode

In uncontrolled mode, the gNodeB automatically adds or removes neighboring NG-RAN cells once missing or to-be-removed neighboring NG-RAN cells are detected.

Controlled Mode

In controlled mode, the gNodeB reports optimization advice on neighboring cells to the MAE-Access once missing or to-be-removed neighboring NG-RAN cells are detected. After manual confirmation on the MAE-Access, the gNodeB adds or removes neighboring NG-RAN cells. For details about operations on the MAE-Access, see descriptions of NR ANR in *SON Management* in *MAE Product Documentation (EulerOS, TaiShan)*.

The gNodeB periodically reports optimization advice on neighboring cells to the MAE-Access every night. Before the optimization advice is manually confirmed on the MAE-Access, the gNodeB temporarily stores the detected missing neighboring cells in its memory to reduce reportCGI measurements on the missing neighboring

cells. When detecting the PCIs of the missing neighboring cells again, the gNodeB does not deliver reportCGI measurement configurations of these cells to UEs. Therefore, if the NR cell global identifiers (NCGIs) of the missing neighboring cells change but their PCIs remain unchanged, the NCGIs of the missing neighboring cells in the optimization advice reported by the gNodeB are still the original ones. The NCGI consists of the public land mobile network (PLMN) ID, gNodeB ID, and cell ID. The PLMN ID consists of the MCC and MNC.

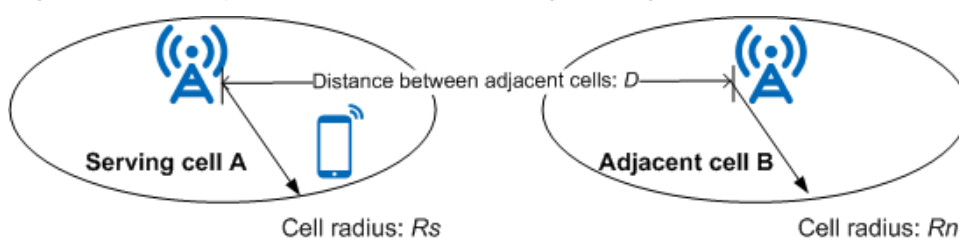
The **NR_NR_ANR_CTRL_MODE_HO_SW** option of the **NRCellAnr.AnrModeSwitch** parameter determines whether to allow handovers or redirections to unconfirmed neighboring NG-RAN cells (in SA networking) and allow EN-DC PSCell changes (in NSA networking).

- SA networking
 - When this option is selected, the gNodeB can initiate handovers or redirections to neighboring NG-RAN cells unconfirmed on the MAE-Access based on the NCGIs of the neighboring cells reported from UEs.
 - When this option is deselected, handovers and redirections to neighboring NG-RAN cells unconfirmed on the MAE-Access are prohibited.
- NSA networking
 - When this option is selected, the gNodeB can select neighboring NG-RAN cells unconfirmed on the MAE-Access as the target cells for EN-DC PSCell changes based on the NCGIs of the neighboring cells reported from UEs.
 - When this option is deselected, neighboring NG-RAN cells unconfirmed on the MAE-Access cannot serve as target cells for EN-DC PSCell changes.

In controlled mode, when the MAE-Access displays the optimization advice for adding a neighbor relationship, it also displays a message indicating whether the neighboring cell is an over-distant neighboring cell. The **NRCellAnr.OverDistantNCellThld** parameter is used to determine whether the neighboring cell is an over-distant neighboring cell in the preceding scenario.

- When the **NRCellAnr.OverDistantNCellThld** parameter is not set to **0**, an over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the parameter value.
- When the **NRCellAnr.OverDistantNCellThld** parameter is set to **0**, an over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the sum of the access radii of the two cells. The example shown in [Figure 4-2](#) indicates that D is greater than the sum of R_s and R_n . In conclusion, adjacent cell B is an over-distant neighboring cell of serving cell A.

Figure 4-2 Example of an over-distant neighboring cell



 NOTE

The MAE-Access calculates the distance between the local cell and a neighboring cell based on the longitude and latitude information contained in base station engineering parameters. If a multi-RRU cell has multiple latitudes and longitudes, the minimum distance is used for determination.

4.1.1.1.2 Control Mode

The control mode can be automatic or manual, depending on whether the gNodeB can automatically modify or remove neighbor relationships and external cells with ANR.

The control mode of a neighbor relationship is specified by the **NRCellRelation.CtrlMode** parameter. The control mode of an external cell is specified by the **NRExternalNCell.CtrlMode** parameter. The modification of external cells also includes the addition, modification, and removal of the **NRExternalNCellPlmn** MO instances.

Table 4-2 lists the application scenarios of different control modes.

Table 4-2 Application scenarios of different control modes

Value of Control Mode	Application Scenario
AUTO_MODE	- Manual modification or removal of neighbor relationships and external cells - ANR-based automatic modification or removal of neighbor relationships and external cells - ANR-based removal of neighbor relationships and external cells after manual confirmation
MANUAL_MODE	- Manual modification or removal of neighbor relationships and external cells - ANR-based removal of neighbor relationships and external cells after manual confirmation

4.1.1.1.3 Penalty on reportCGI Measurements

During the period of reportCGI measurements (short for measurements of report type *reportCGI*), the gNodeB cannot process the measurement reports sent by the UE.

To prevent continuous reportCGI measurements from affecting UE mobility, the gNodeB automatically imposes penalty on reportCGI measurements. If the current reportCGI measurement times out or the measurement report sent by the UE does not contain valid *CGI-Info*, the gNodeB does not deliver reportCGI measurement configurations to the UE when the next reportCGI measurement is triggered.

4.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells

Automatic detection and addition of missing neighboring cells can be triggered by event-triggered measurements or periodic measurements. Automatic detection

and addition of missing neighboring cells triggered by periodic measurements is applicable to scenarios with PCI-specific measurements and scenarios with non-PCI-specific measurements.

Based on the preceding triggering types and measurement scenarios, automatic detection and addition of missing neighboring cells is divided into three subfunctions: event-triggered ANR, fast ANR, and PCI-specific ANR. The gNodeB performs automatic detection and addition of missing neighboring cells at the cell and frequency levels, which are controlled by cell- and frequency-specific switches, respectively, as listed in [Table 4-3](#).

- Cell-specific switches: To enable automatic detection and addition of missing neighboring cells of a specific cell, operators need to turn on the corresponding cell-level switches.
- Frequency-specific switches: To enable automatic detection and addition of missing neighboring cells on a specific neighboring frequency of a cell after the corresponding cell-specific switches are turned on for the cell, operators need to turn on the frequency-specific switches. Frequency-specific switches are not designed for the operating frequency of the serving cell.

Table 4-3 Switches that control automatic detection and addition of missing neighboring cells

Subfunction	Triggering Type	PCI-specific Measurements	Cell-specific Switch	Frequency-specific Switch
Event-triggered ANR	Event-triggered measurements	No	NR_NR_ANR_SW option of the NRCCellAlgoSwitch . AnrSwitch parameter	NRCCellFreqRelation . AnrInd set to ALLOWED
Fast ANR	Periodic measurements	No	NR_NR_ANR_SW and NR_NR_FAST_ANR_SW options of the NRCCellAlgoSwitch . AnrSwitch parameter	
PCI-specific ANR	Periodic measurements	Yes	NR_NR_ANR_SW and PCI_SPECIFIC_NR_ANR_SW options of the NRCCellAlgoSwitch . AnrSwitch parameter	

Automatic detection and addition of missing neighboring cells triggered by event measurements is also called event-triggered ANR. With this function enabled, the

gNodeB detects missing neighboring cells based on the information in intra- and inter-frequency measurement reports.

Intra-RAT measurements include mobility measurements and CA measurements. Only measurements related to the functions or scenarios listed in [Table 4-4](#) can trigger event-triggered ANR:

Table 4-4 Functions or scenarios that can trigger event-triggered ANR

Function or Scenario	Reference
Coverage-based intra-frequency handover	• <i>NSA Mobility Management</i> • <i>SA Mobility Management in Connected Mode</i>
Coverage-based inter-frequency handover	
Frequency-priority-based inter-frequency handover	
Operator-specific-priority-based inter-frequency handover	
Service-based inter-frequency handover	
Protection of UEs under weak uplink coverage	<i>Multi-Frequency Smart Selection</i>
RFSP-based inter-frequency handover back to the HPLMN enabled with the BACK_TO_HPLMN_HO_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter selected	<i>Flexible User Steering</i>
Low-speed UE outmigration	<i>High Speed Mobility</i>
An SCell is added for CA based on measurements and unknown neighboring cells are detected based on inter-frequency measurements.	<i>Carrier Aggregation</i>

Depending on whether PCI-specific measurements are performed, automatic detection and addition of missing neighboring cells triggered by periodic measurements is classified into the following types:

- One is fast ANR, in which non-PCI-specific measurements are performed. This function can fast detect and add missing neighboring cells.
After this function is enabled, the gNodeB selects some UEs in RRC_CONNECTED mode in a serving cell and sends measurement configurations including intra-RAT frequencies to the UEs. The gNodeB detects missing neighboring cells based on the periodic measurement reports sent from these UEs. The UEs perform the measurements for four times and send measurement reports at an interval of 5120 ms.
- The other is PCI-specific ANR, in which PCI-specific measurements are performed. This function can fast detect and add PCI-specific missing neighboring cells. The specific PCI is specified by the **NRCellNcgiMeas.TargetPhysicalCellId** parameter. A maximum of 16 specific PCIs can be configured for a base station. The intra-RAT frequency associated with the specific PCI is specified by the

NRCellNcgiMeas.TargetSsbDescMethod and
NRCellNcgiMeas.TargetSsbFreqPos parameters.

After this function is enabled, the gNodeB selects some UEs in RRC_CONNECTED mode in a serving cell and sends measurement configurations including intra-RAT frequencies associated with the specific PCIs to the UEs. The gNodeB detects PCI-specific missing neighboring cells based on the periodic measurement reports sent from these UEs. The UEs perform the measurements for four times and send measurement reports at an interval of 5120 ms.

After fast ANR or PCI-specific ANR is enabled, the gNodeB randomly selects UEs that meet all of the following conditions and sends measurement configurations to them at an interval of 30s. A maximum of five UEs in the serving cell can be selected for measurements at a time.

- The UE is in RRC_CONNECTED mode and a PDU session has been set up for the UE.
- The UE supports the NR neighboring cell measurement with the report type *reportCGI*. For details, see [4.1.3.4 Others](#).
- The UE is only running one or more of the following types of non-GBR services, except for FWA Internet, FWA video, and FWA private line services (services for which the **gNBQciBearer.ServiceType** parameter is set to **FWA_INTERNET**, **FWA_VIDEO**, and **FWA_PRIVATE_LINE**, respectively).
 - In NSA networking, services with a QCI of 5, 6, 8, or 9
 - In SA networking, services with a 5QI of 5, 6, 8, or 9 or services with a mapped QCI of 5, 6, 8, or 9
- The UE has not been selected for periodic measurements within one hour while the UE context is set up and retained.

 NOTE

CA UEs that meet the preceding conditions but treat the local cell as their SCell cannot be selected for periodic measurements.

After fast ANR is enabled, the gNodeB instructs UEs to measure all intra-RAT frequencies. After PCI-specific ANR is enabled, the gNodeB instructs UEs to measure all intra-RAT frequencies associated with specific PCIs.

After fast ANR or PCI-specific ANR is enabled, if the number of frequencies to be measured is limited and not all the frequencies can therefore be delivered for measurements, the gNodeB selects frequencies based on their priorities. For details about configuring the candidate frequency list based on frequency priorities, see *NSA Mobility Management* and *SA Mobility Management in Connected Mode*. The maximum number of periodic measurements of each frequency is 100. The rules for calculating the number of periodic measurements are as follows:

- Once fast ANR is enabled, the gNodeB sets the number of periodic measurements of every intra-RAT frequency performed by UEs in the cell to zero.
- Once PCI-specific ANR is enabled, the gNodeB sets the number of periodic measurements of every intra-RAT frequency associated with a specific PCI performed by UEs in the cell to zero.

- After a periodic measurement of a frequency is complete, the gNodeB increases the number of periodic measurements for this frequency by one.
- If the neighbor relationship with a cell on a frequency is successfully added by ANR, the gNodeB sets the number of periodic measurements of the frequency to zero. If this neighbor relationship is added by running MML commands, the gNodeB does not set this number to zero.
- The number of periodic measurements of a new intra-RAT neighboring frequency starts from zero.
- The number of periodic measurements of a frequency corresponding to a PCI newly specified by the **NRCellNcgiMeas.TargetPhysicalCellId** parameter is counted from zero.

After fast ANR or PCI-specific ANR is enabled, the gNodeB delivers frequency measurement configurations based on the following rules:

- If the number of consecutive periodic measurements of the frequency is greater than or equal to 100 but no neighbor relationship with a cell on this frequency is added, this frequency will no longer be delivered for periodic measurements.
- If the number of intra-RAT neighbor relationships configured for a cell or the gNodeB reaches its maximum value, all intra-RAT frequencies will no longer be delivered for measurements. When the number of intra-RAT neighbor relationships configured for a cell or the gNodeB decreases from the maximum value, the counting of periodic measurements of these frequencies continues from the previous numbers.
- In controlled mode, if the number of to-be-confirmed intra-RAT neighbor relationships configured for a cell or the gNodeB reaches its maximum value, all intra-RAT frequencies will no longer be delivered for measurements. When the number of to-be-confirmed intra-RAT neighbor relationships configured for a cell or the gNodeB decreases from the maximum value, the counting of periodic measurements of these frequencies continues from the previous numbers.
- In the intra-RAT candidate frequency list, frequencies for which the **NRCellFreqRelation.MeasActvFlag** parameter is set to **INACTIVE** or the **NRCellFreqRelation.InterFreqPrdcMrType** parameter is set to **PERIODIC_MR_ONLY** or **FORBID_MEAS** are not delivered for measurements.

NOTE

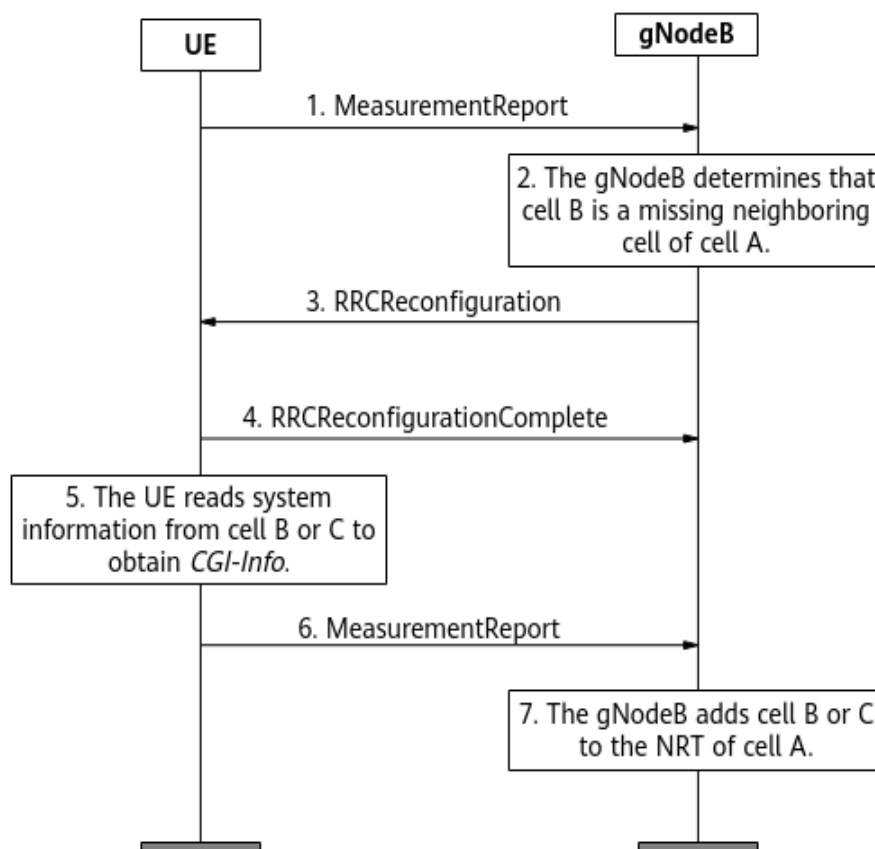
- If the base station is restarted, all counters (such as the numbers of measurements and handovers) related to ANR will be re-measured.
- If both fast ANR and PCI-specific ANR are enabled and a PCI specified through the **NRCellNcgiMeas** MO is removed, but the number of periodic measurements on the frequency corresponding to the PCI is less than 100, the base station still delivers measurement configurations of this frequency.
- Neighbor relationship addition triggered by fast ANR or PCI-specific ANR is not service-triggered. It does not trigger specification-based neighboring relationship replacement or specification-based external cell replacement. In controlled mode, no optimization advice is reported when the number of neighbor relationships reaches its maximum.
- NRT query triggered by fast ANR or PCI-specific ANR is not service-triggered. It is not counted in the number of times a neighbor relationship is used and does not affect specification-based neighbor relationship and external cell replacement and identification of redundant neighbor relationships.
- If a UE starts periodic measurements but the measurements conflict with other functions (such as cell handover, CA, and DC) that require measurement resources, the periodic measurements triggered by fast ANR or PCI-specific ANR automatically pauses to meet the measurement requirements of the functions.
- When a neighbor relationship is added in uncontrolled mode, the **NRCellRelation.AnrFlag** parameter is set to **TRUE** for the neighbor relationship by default. When the neighbor relationship is added in controlled mode after manual confirmation on the MAE-Access, the **NRCellRelation.AnrFlag** parameter is set to **FALSE** by default.
- Assume that an NSA UE measures NCGIs on the NR side. The **NSA_NCGI_MEAS_DRX_SW** option of the **gNodeBParam.NsaDcOptSwitch** parameter on the NR side and the **NSA_NR_NCGI_DRX_OPT_SW** option of the **EnodebAlgoExtSwitch.MultiNetworkingOptionOptSw** parameter on the LTE side must be both selected to ensure a long common discontinuous reception (DRX) sleep time on both the NR and LTE sides. This increases the probability that NSA UEs measure NCGIs. For details, see *EPC-based NSA Basics*.
- For details about the maximum allowed number of neighbor relationships in an intra-RAT NRT, see the descriptions of the **NRCellRelation** MO in *3900 & 5900 Series Base Station MO and Parameter Reference* in *3900 & 5900 Series Base Station Product Documentation*.

The gNodeB automatically detects and adds missing neighboring cells based on event-triggered or periodic measurement reports sent by UEs. The following provides the details in SA and NSA networking.

SA Networking

Assume that a UE is located in the coverage area of cell A, which provides NR services for the UE. Cell B and cell C are NG-RAN cells adjacent to cell A. The gNodeB sends measurement configurations to the UE. Based on the measurement report sent from the UE, the gNodeB detects and adds cell B, as shown in [Figure 4-3](#).

Figure 4-3 Automatically detecting and adding missing intra-RAT neighboring cells during movement in SA networking



1. The UE sends a measurement report.
The UE measures cells on the frequency specified by the gNodeB, and then sends the gNodeB a measurement report containing the PCIs of cells (cell B for example) that meet signal quality requirements.
 - If the UE sends an event-triggered measurement report, cell B has the best signal quality among all cells in the report. In this case, the procedure goes to 2.
 - If the UE sends a periodic measurement report, the procedure varies as follows:
 - If only fast ANR is enabled or both fast ANR and PCI-specific ANR are enabled, cell B has the best signal quality among all cells in the report. In this case, the procedure goes to 2.
 - If only PCI-specific ANR is enabled, the intersection of the cells contained in the measurement report and the cells each with the PCI specified by the **NRCellNcgiMeas.TargetPhysicalCellId** parameter is used. If there is an intersection, cell B has the best signal quality among all cells in the intersection. In addition, if its RSRP value is greater than or equal to the value of **NRCellAnr.IntraRatAnrRSRPThld**, the procedure goes to 3. Otherwise, the procedure ends.
2. The gNodeB obtains the PCI of cell B from the measurement report, queries the NCGIs of all neighboring cells in the NRT of cell A, and then searches its

NCL and served cells for the PCIs and frequencies of cells indicated by these NCGIs. The gNodeB then checks whether a cell in the NCL or a served cell has the same PCI and frequency as cell B.

If it does not, cell B is not in the NRT of cell A. That is, cell B is a missing neighboring cell of cell A. In this case, the procedure goes to 3. If it does, cell B is in the NRT of cell A. That is, cell B is a neighboring cell of cell A.

- If the measurement report is a periodic measurement report and only fast ANR is enabled, or if the measurement report is an event-triggered report, the procedure ends.
 - If the measurement report is a periodic measurement report and both fast ANR and PCI-specific ANR are enabled, the intersection of the cells contained in the measurement report and the cells each with the PCI specified by the **NRCeIIIncgI Meas. TargetPhysicalCellId** parameter is used. The procedure goes to 3 in any of the following scenarios. In other scenarios, the procedure ends.
 - If there is an intersection, cell B is in the intersection and its RSRP value must be greater than or equal to the value of **NRCeIIAnr. IntraRatAnrRSRPThId**.
 - If there is an intersection but cell B is not in the intersection, the gNodeB selects another cell (for example, cell C) with the best signal quality from the intersection. The RSRP value of cell C must be greater than or equal to the value of **NRCeIIAnr. IntraRatAnrRSRPThId**.
3. The gNodeB sends the UE an RRCReconfiguration message carrying the measurement report type of *reportCGI* and DRX configuration. With the configuration of this type of report, the UE performs measurements and reports *CGI-Info* of the cell with the specified PCI (*CGI-Info* of cell B or C). DRX terminates downlink data reception in the serving cell so that the UE can receive system information of the target cell.
 4. The UE responds to the gNodeB with an RRCReconfigurationComplete message.
 5. The UE reads SIB1 (system information) from cell B or C to obtain *CGI-Info* of cell B or C, respectively.
 6. The UE sends the gNodeB a measurement report carrying the *CGI-Info*.
 7. The gNodeB obtains the NCGI, TAC, PCI, PLMN list, and other information about cell B or C from the *CGI-Info* IE in the measurement report. In addition, the gNodeB adds cell B or C to the NCL and the neighbor relationship with cell B or C to the NRT or updates its NCL based on the information. The gNodeB supports automatic addition of missing neighboring cells based on the number of CGI-Info reporting times. It adds cell B or cell C to NCL and the neighbor relationship with cell B or C to the NRT only when the number of times *CGI-Info* of cell B or cell C is reported is greater than or equal to the **NRCeIIAnr. NCellAddCgiRptCountThId** parameter value. If this parameter is set to 1, this function does not take effect.
 - If cell B or C is not in the NRT of cell A and is served by the same gNodeB as cell A, the gNodeB adds cell B or C to the NRT of cell A. If cell A is not a neighboring cell of cell B or C, the gNodeB also adds cell A to the NRT of cell B or C.

- If cell B or C is not in the NRT of cell A and is not served by the same gNodeB as cell A, the gNodeB checks whether cell B or C is in its NCL.
 - If it is, the gNodeB adds cell B or C to the NRT of cell A.
 - If it is not, the gNodeB adds cell B or C to its NCL and then adds cell B or C to the NRT of cell A.

If the *CGI-Info* IE contains the TAC information about neighboring cells, **NRExternalNCell.NrNetworkingOption** is automatically set to **UNLIMITED** (default value) for cell B or C when the cell is being added to the NCL. The gNodeB queries the MAE-Access for the networking option of cell B or C and updates this option in the NCL.

If the *CGI-Info* IE does not contain the TAC information about neighboring cells, **NRExternalNCell.NrNetworkingOption** and **NRExternalNCell.Tac** are automatically set to **NSA** and **4294967295** (invalid value) respectively for cell B or C when the cell is being added to the NCL.
- If cell B or C is in the NRT of cell A and is served by the same gNodeB as cell A, the procedure ends.
- If cell B or C is in the NRT of cell A and is not served by the same gNodeB as cell A, the gNodeB updates the NCL based on measurement reports from UEs. For details, see [NCL Update Triggered by Measurement Reports from UEs](#).

In the measurement report sent from the UE, the NR cell identity (NCI) of the target cell is 36 bits long and is a combination of gNodeB ID and cell ID. The gNodeB ID is 22 to 32 bits long. The cell ID is 4 to 14 bits long.

- When the **gNBSharingMode.MultipleNCISwitch** parameter is set to **ON**:
 - If the PLMN corresponding to the NCI is configured through a **gNBOperator** or **gNBExternalPlmn** MO, the gNodeB takes the leftmost bits in the NCI as the gNodeB ID of the target cell based on the setting of the **gNBOperator.gNBIdLength** or **gNBExternalPlmn.gNBIdLength** parameter corresponding to the PLMN.
 - If the PLMN corresponding to the NCI is not configured through a **gNBOperator** or **gNBExternalPlmn** MO, the gNodeB takes the leftmost bits in the NCI as the gNodeB ID of the target cell based on the setting of the **gNodeBFunction.gNBIdLength** parameter.

NOTE

If the PLMN corresponding to the NCI is not configured through a **gNBOperator** or **gNBExternalPlmn** MO and the **gNodeBFunction.gNBIdLength** parameter value for the primary operator of the missing neighboring cell is different from the **gNodeBFunction.gNBIdLength** parameter value of the local gNodeB, then the gNodeB ID parsed from the NCI is incorrect after this neighboring cell is added. Therefore, you are advised to configure the PLMN corresponding to the NCI through the **gNBOperator** or **gNBExternalPlmn** MO before the neighboring cell is added.

- When the **gNBSharingMode.MultipleNCISwitch** parameter is set to **OFF**:

- If the PLMN corresponding to the NCI is configured through a **gNBExternalPlmn** MO, the gNodeB takes the leftmost bits in the NCI as the gNodeB ID of the target cell based on the setting of the **gNBExternalPlmn.gNBIdLength** parameter corresponding to the PLMN.
- If the PLMN corresponding to the NCI is not configured through a **gNBExternalPlmn** MO, the gNodeB takes the leftmost bits in the NCI as the gNodeB ID of the target cell based on the setting of the **gNodeBFunction.gNBIdLength** parameter.

For further details about the **gNBSharingMode.MultipleNCISwitch** parameter, see *Multi-Operator Sharing*.

An NRT is configured through an **NRCellRelation** MO, and an NCL is configured through an **NRExternalNCell** MO. If the number of neighbor relationships in an NRT reaches the maximum value, new neighbor relationships cannot be added to the NRT. Similarly, if the number of external cells in an NCL reaches the maximum value, new external cells cannot be added to the NCL.

Neighboring cell class label management

In the preceding procedure, assume that cell A has two adjacent cells, cell B and cell D. Cell B and cell D work on the same frequency and are configured with the same PCI, and cell B is in the NRT of cell A. When subsequent event-triggered measurement reports contain the same PCI as cell B and cell D, the gNodeB queries the NRT and selects cell B in the NRT as the target cell for the handover. When the UE moves, the measured cell is either of the two adjacent cells corresponding to the PCI. Therefore, if the gNodeB selects an incorrect target cell, the UE handover fails. In this scenario, the gNodeB automatically sets **NRCellRelation.NCellClassLabel** (this parameter cannot be set by running MML commands) to **FORMAL** or **EXTENDED**.

The **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable for automatic neighboring cell class label adjustment.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

When the number of handovers from cell A to cell B is less than the **NRCellAnr.NrNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over the next measurement period until the number is greater than or equal to this parameter value. When the number of handovers from cell A to cell B is greater than or equal to the **NRCellAnr.NrNCellHoNumThld** parameter value, the gNodeB automatically sets the neighboring cell class label (specified by the **NRCellRelation.NCellClassLabel** parameter) for cell B based on the neighboring cell handover success rate threshold (specified by the **NRCellAnr.NCellHoSuccRateThld** parameter).

- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to **0**, the gNodeB automatically sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.

- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is less than this value, the gNodeB automatically sets the **NRCellRelation.NCellClassLabel** parameter to **EXTENDED** for cell B. When the PCI of cell B and cell D is reported again in subsequent event-based measurements:
 - If the reported CGI of cell D is different from that of cell B corresponding to the PCI in the NRT, the gNodeB adds cell D to the NRT and sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.
 - If the number of consecutive measurements with the report type of *reportCGI* performed by the gNodeB on cell B is greater than or equal to the value of the **NRCellAnr.NrNCellRptCgiNumThld** parameter, the gNodeB sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.

 NOTE

After the gNodeB sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B based on measurements of report type *reportCGI*, the handover success rate threshold is determined as follows in the next measurement:

- If the **NRCellAnr.NCellHoSuccRateThld** parameter value remains unchanged in the next measurement, the handover success rate threshold in the next measurement is equal to that in the current measurement multiplied by **NRCellAnr.NCellHoSuccRateThld**.
- If there is a change in the **NRCellAnr.NCellHoSuccRateThld** parameter value in the next measurement, the handover success rate threshold in the next measurement is specified by this parameter.
- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is greater than or equal to this value, the value of the **NRCellRelation.NCellClassLabel** parameter remains unchanged.

Specifically:

- The measurement of the number of handovers is controlled by the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.
- Handover success rate = Number of successful handovers/Number of handovers x 100%

The gNodeB can detect and add two intra-frequency cells with the same PCI as neighboring cells through the preceding extended neighbor relationship processing. The two neighboring cells have a PCI confusion issue. PCI confusion can be detected and resolved by using the PCI conflict detection and self-optimization function. For details, see *PCI Conflict Detection and Self-Optimization*.

PLMN ID management

The Automatic Neighbour Relation (ANR) feature also supports PLMN ID management. If the PLMN attribute of an operator is configured in a **gNBNCellPlmnList** MO, the gNodeB performs differentiated management based on the list type of PLMN ID when obtaining the PLMN list from *CGI-Info*. For details, see [4.2 Intra-RAT PLMN ID Management](#).

PLMN list management in multi-operator sharing scenarios

On the gNodeB side, multi-operator sharing incorporates RAN sharing with common carrier and RAN sharing with dedicated carrier. For details, see *Multi-Operator Sharing*. If a neighboring NG-RAN cell works in multi-operator sharing mode, the gNodeB adds secondary operator information about a missing neighboring cell to the external-cell PLMN list when adding the missing neighboring cell to the NCL. For details, see [4.1.1.7.1 Automatic External-Cell PLMN List Addition](#).

UE blacklist management

To address UE compatibility issues, the gNodeB supports UE blacklist management in intra-RAT ANR. With this function, the gNodeB can take differentiated measures for UEs. This function is controlled by the **NR_NR_ANR_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.

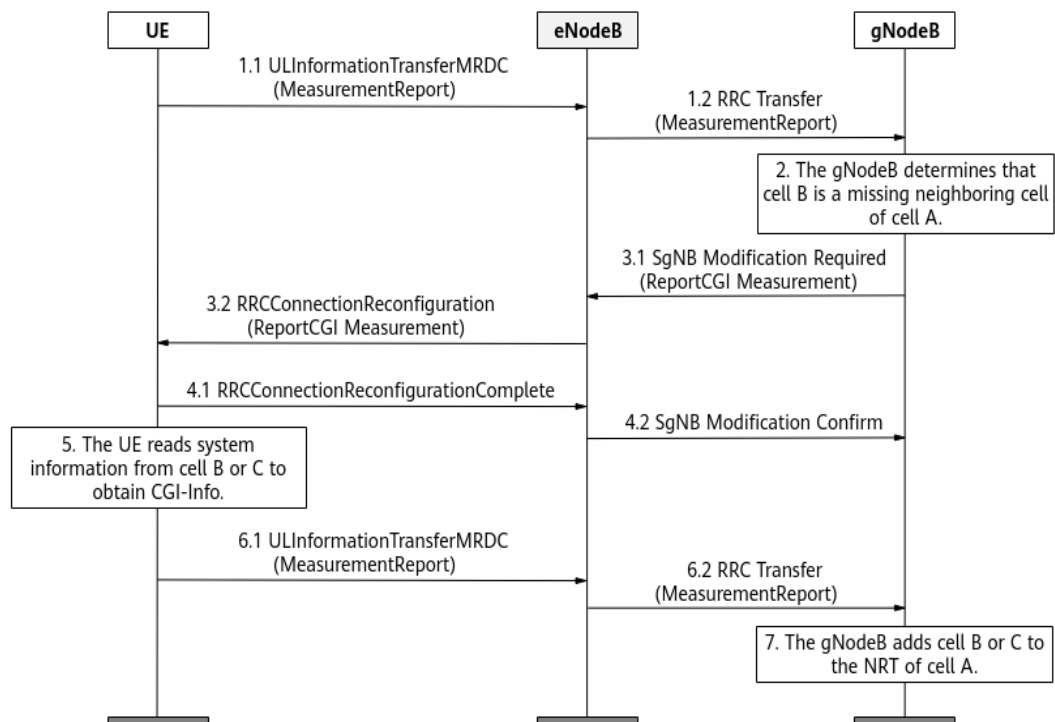
- If this option is selected, the gNodeB disables intra-RAT ANR for specified UEs and does not deliver reportCGI measurement configurations to these UEs.
- If this option is deselected, the gNodeB does not take differentiated measures for UEs and delivers reportCGI measurement configurations to all UEs.

NSA Networking

In NSA networking, the eNodeB works as the master eNodeB, which is connected to the EPC. The gNodeB works as the secondary gNodeB, which is connected to the eNodeB over the X2 interface. The PSCell will change when a UE moves between NG-RAN cells, requiring existing neighbor relationships. If no neighbor relationship exists, neighboring cells will be automatically configured. There is no direct signaling bearer between the gNodeB and the UE. Therefore, the eNodeB is responsible for forwarding the RRC messages between the gNodeB and the UE.

Assume that a UE is located in the coverage area of cell A, which provides NR services for the UE. Cell B and cell C are NG-RAN cells adjacent to cell A. The gNodeB sends measurement configurations to the UE. Based on the measurement report sent from the UE, the gNodeB detects and adds cell B, as shown in [Figure 4-4](#).

Figure 4-4 Automatically detecting and adding missing intra-RAT neighboring cells when PSCell changes occur in NSA networking



1. The UE sends a measurement report.
 - a. The UE measures cells on the frequency specified by the gNodeB and sends the eNodeB a ULInformationTransferMRDC message carrying a measurement report. The measurement report contains the PCIs of cells (cell B for example) that meet signal quality requirements.
 - If the UE sends an event-triggered measurement report, cell B has the best signal quality among all cells in the report. In this case, the procedure goes to 2.
 - If the UE sends a periodic measurement report, the procedure varies as follows:
 - If only fast ANR is enabled or both fast ANR and PCI-specific ANR are enabled, cell B has the best signal quality among all cells in the report. In this case, the procedure goes to 2.
 - If only PCI-specific ANR is enabled, the intersection of the cells contained in the measurement report and the cells each with the PCI specified by the **NRCellNcgiMeas.TargetPhysicalCellId** parameter is used. If there is an intersection, cell B has the best signal quality among all cells in the intersection. In addition, if its RSRP value is greater than or equal to the value of **NRCellAnr.IntraRatAnrRSRPThld**, the procedure goes to 3. Otherwise, the procedure ends.
 - b. The eNodeB sends the gNodeB an RRC Transfer message carrying the measurement report.

 NOTE

A UE can directly send a measurement report to the gNodeB. For details, see *EPC-based NSA Basics*.

2. The gNodeB obtains the PCI of cell B from the measurement report, queries the NCGIs of all neighboring cells in the NRT of cell A, and then searches its NCL and served cells for the PCIs and frequencies of cells indicated by these NCGIs. The gNodeB then checks whether a cell in the NCL or a served cell has the same PCI and frequency as cell B.
If it does not, cell B is not in the NRT of cell A. That is, cell B is a missing neighboring cell of cell A. In this case, the procedure goes to 3. If it does, cell B is in the NRT of cell A. That is, cell B is a neighboring cell of cell A.
 - If the measurement report is a periodic measurement report and only fast ANR is enabled, or if the measurement report is an event-triggered report, the procedure ends.
 - If the measurement report is a periodic measurement report and both fast ANR and PCI-specific ANR are enabled, the intersection of the cells contained in the measurement report and the cells each with the PCI specified by the **NRCCellNcgiMeas.TargetPhysicalCellId** parameter is used. The procedure goes to 3 in any of the following scenarios. In other scenarios, the procedure ends.
 - If there is an intersection, cell B is in the intersection and its RSRP value must be greater than or equal to the value of **NRCCellAnr.IntraRatAnrRSRPThld**.
 - If there is an intersection but cell B is not in the intersection, the gNodeB selects another cell (for example, cell C) with the best signal quality from the intersection. The RSRP value of cell C must be greater than or equal to the value of **NRCCellAnr.IntraRatAnrRSRPThld**.
3. The gNodeB sends measurement configurations to the UE, with the measurement report type of *reportCGI* and DRX configuration. With the configuration of this type of report, the UE performs measurements and reports *CGI-Info* of the cell with the specified PCI (*CGI-Info* of cell B or C). DRX configuration reduces power consumption of the UE.
 - a. The gNodeB sends the eNodeB an SgNB Modification Required message carrying the measurement configuration.
 - b. The eNodeB sends the UE an RRCConnectionReconfiguration message carrying the measurement configuration.
4. The UE responds to indicate that RRC reconfiguration is completed.
 - a. The UE responds to the eNodeB with an RRCConnectionReconfigurationComplete message.
 - b. The eNodeB responds to the gNodeB with an SgNB Modification Confirm message.
5. The UE reads SIB1 (system information) from cell B or C to obtain *CGI-Info* of cell B or C, respectively.
In NSA networking, SIB1 transmissions are optional. To ensure that missing neighboring cells can be automatically detected and added, cell B or C needs to send SIB1 so that the UE obtains *CGI-Info*.

6. The UE sends the gNodeB a measurement report carrying the *CGI-Info*.
 - a. The UE sends the eNodeB a ULInformationTransferMRDC message carrying a measurement report.
 - b. The eNodeB sends the gNodeB an RRC Transfer message carrying the measurement report.
7. The gNodeB obtains the NCGI, TAC, PCI, PLMN list, and other information about cell B or C from the *CGI-Info* IE in the measurement report. In addition, the gNodeB adds cell B or C to the NCL and the neighbor relationship with cell B or C to the NRT or updates its NCL based on the information. The gNodeB supports automatic addition of missing neighboring cells based on the number of CGI-Info reporting times. It adds cell B or cell C to NCL and the neighbor relationship with cell B or C to the NRT only when the number of times *CGI-Info* of cell B or cell C is reported is greater than or equal to the **NRCellAnr.NCellAddCgiRptCountThld** parameter value. If this parameter is set to 1, this function does not take effect.
 - If cell B or C is not in the NRT of cell A and is served by the same gNodeB as cell A, the gNodeB adds cell B or C to the NRT of cell A. If cell A is not a neighboring cell of cell B or C, the gNodeB also adds cell A to the NRT of cell B or C.
 - If cell B or C is not in the NRT of cell A and is not served by the same gNodeB as cell A, the gNodeB checks whether cell B or C is in its NCL.
 - If it is, the gNodeB adds cell B or C to the NRT of cell A.
 - If it is not, the gNodeB adds cell B or C to its NCL and then adds cell B or C to the NRT of cell A.

If the *CGI-Info* IE contains the TAC information about neighboring cells, **NRExternalNCell.NrNetworkingOption** is automatically set to **UNLIMITED** (default value) for cell B or C when the cell is being added to the NCL. The gNodeB queries the MAE-Access for the networking option of cell B or C and updates this option in the NCL.

If the *CGI-Info* IE does not contain the TAC information about neighboring cells, **NRExternalNCell.NrNetworkingOption** and **NRExternalNCell.Tac** are automatically set to **NSA** and **4294967295** (invalid value) respectively for cell B or C when the cell is being added to the NCL.
 - If cell B or C is in the NRT of cell A and is served by the same gNodeB as cell A, the procedure ends.
 - If cell B or C is in the NRT of cell A and is not served by the same gNodeB as cell A, the gNodeB updates the NCL based on measurement reports from UEs. For details, see **NCL Update Triggered by Measurement Reports from UEs**.

In NSA networking, the gNodeB determines the gNodeB ID of the target cell from the NCI in the same way as in SA networking.

An NRT is configured through an **NRCellRelation** MO, and an NCL is configured through an **NRExternalNCell** MO. If the number of neighbor relationships in an NRT reaches the maximum value, new neighbor relationships cannot be added to the NRT. Similarly, if the number of external cells in an NCL reaches the maximum value, new external cells cannot be added to the NCL.

Neighboring cell class label management

In the preceding procedure, assume that cell A has two adjacent cells, cell B and cell D. Cell B and cell D work on the same frequency and are configured with the same PCI, and cell B is in the NRT of cell A. When subsequent event-triggered measurement reports contain the same PCI as cell B and cell D, the gNodeB queries the NRT and selects cell B in the NRT as the target cell for the handover. When the UE moves, the measured cell is either of the two adjacent cells corresponding to the PCI. Therefore, if the gNodeB selects an incorrect target cell, the UE handover fails.

In this scenario, the gNodeB can automatically adjust the neighboring cell class label in NSA networking. That is, the **NRCellRelation.NCellClassLabel** parameter can automatically be set to **FORMAL** or **EXTENDED**. This function can take effect only if the **NSA_NCELL_CLASS_LABEL_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected on the NR side and the **PSCELL_RA_SUCCESS_MSG_FWD_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter is selected on the LTE side.

NOTE

The **PSCELL_RA_SUCCESS_MSG_FWD_SW** option of the **EnodebAlgoExtSwitch.NsaDcAlgoSwitch** parameter on the LTE side specifies whether to forward the message indicating the successful random access of UEs to the target PSCell. When this option is selected, the MeNB forwards the message indicating the successful random access of UEs to the target PSCell to the base station serving the source PSCell after receiving the message.

The **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable for automatic neighboring cell class label adjustment.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

When the number of PSCell change attempts from cell A to cell B is less than the **NRCellAnr.NrNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over the next measurement period until the number is greater than or equal to this parameter value. When the number of PSCell change attempts from cell A to cell B is greater than or equal to the **NRCellAnr.NrNCellHoNumThld** parameter value, the gNodeB automatically sets the neighboring cell class label (specified by the **NRCellRelation.NCellClassLabel** parameter) for cell B based on the neighboring cell handover success rate threshold (specified by the **NRCellAnr.NCellHoSuccRateThld** parameter).

- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to **0**, the gNodeB automatically sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.
- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is less than this value, the gNodeB automatically sets the **NRCellRelation.NCellClassLabel** parameter to **EXTENDED** for cell B. When the PCI of cell B and cell D is reported again in subsequent event-based measurements:

- If the reported CGI of cell D is different from that of cell B corresponding to the PCI in the NRT, the gNodeB adds cell D to the NRT and sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.
- If the number of consecutive measurements with the report type of *reportCGI* performed by the gNodeB on cell B is greater than or equal to the value of the **NRCellAnr.NrNCellRptCgiNumThld** parameter, the gNodeB sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B.

NOTE

After the gNodeB sets the **NRCellRelation.NCellClassLabel** parameter to **FORMAL** for cell B based on measurements of report type *reportCGI*, the handover success rate threshold is determined as follows in the next measurement:

- If the **NRCellAnr.NCellHoSuccRateThld** parameter value remains unchanged in the next measurement, the handover success rate threshold in the next measurement is equal to that in the current measurement multiplied by **NRCellAnr.NCellHoSuccRateThld**.
- If there is a change in the **NRCellAnr.NCellHoSuccRateThld** parameter value in the next measurement, the handover success rate threshold in the next measurement is specified by this parameter.
- When the **NRCellAnr.NCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is greater than or equal to this value, the value of the **NRCellRelation.NCellClassLabel** parameter remains unchanged.

Specifically, the handover success rate is equal to the result of the following:
Number of successful handovers/Number of PSCell change attempts x 100%.

The gNodeB can detect and add two intra-frequency cells with the same PCI as neighboring cells through the preceding extended neighbor relationship processing. The two neighboring cells have a PCI confusion issue. PCI confusion can be detected and resolved by using the PCI conflict detection and self-optimization function. For details, see *PCI Conflict Detection and Self-Optimization*.

PLMN ID management

The Automatic Neighbour Relation (ANR) feature also supports PLMN ID management. If the PLMN attribute of an operator is configured in a **gNBNCCellPlmnList** MO, the gNodeB performs differentiated management based on the list type of PLMN ID when obtaining the PLMN list from *CGI-Info*. For details, see [4.2 Intra-RAT PLMN ID Management](#).

PLMN list management in multi-operator sharing scenarios

On the gNodeB side, multi-operator sharing incorporates RAN sharing with common carrier and RAN sharing with dedicated carrier. For details, see *Multi-Operator Sharing*. If a neighboring NG-RAN cell works in multi-operator sharing mode, the gNodeB adds secondary operator information about a missing neighboring cell to the external-cell PLMN list when adding the missing neighboring cell to the NCL. For details, see [4.1.1.7.1 Automatic External-Cell PLMN List Addition](#).

UE blacklist management

To address UE compatibility issues, the gNodeB supports UE blacklist management in intra-RAT ANR. With this function, the gNodeB can take differentiated measures for UEs. This function is controlled by the **NR_NR_ANR_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.

- If this option is selected, the gNodeB disables intra-RAT ANR for specified UEs and does not deliver reportCGI measurement configurations to these UEs.
- If this option is deselected, the gNodeB does not take differentiated measures for UEs and delivers reportCGI measurement configurations to all UEs.

4.1.1.3 Automatic Removal of Neighboring Cells

Automatic removal of neighboring cells includes automatic removal of neighbor relationships and automatic removal of external cells. The gNodeB performs automatic removal of neighboring cells at the cell and frequency levels, which are controlled by cell- and frequency-specific switches, respectively.

- Cell-specific switches: To enable automatic removal of neighboring cells for a specific cell, operators need to turn on the corresponding cell-specific switches, that is, the **NR_NR_ANR_SW** and **NR_NR_ANR_DEL_SW** options of the **NRCCellAlgoSwitch.AnrSwitch** parameter.
- Frequency-specific switches: To enable automatic removal of neighboring cells on a specific neighboring frequency of a cell after the cell-specific switches are turned on for the cell, operators need to turn on the frequency-specific switch, that is, the **NRCCellFreqRelation.AnrInd** parameter set to **ALLOWED**. Frequency-specific switches are not designed for the operating frequency of the serving cell.

Automatic Removal of Neighbor Relationships

Automatic removal of neighbor relationships includes specification-based removal of neighbor relationships and removal of redundant neighbor relationships.

- Specification-based removal of neighbor relationships
The maximum number of neighboring cells that can be configured in an NRT depends on the maximum number of neighboring cells that can be configured for a cell and that for a gNodeB. For details about the maximum number of neighboring cells that can be configured for a cell and that for a gNodeB, see *NSA Mobility Management* and *SA Mobility Management in Connected Mode*.
 - When the number of neighboring cells of a cell reaches its maximum value, the gNodeB automatically removes the neighbor relationships that are least frequently used from the NRT of the cell so that new neighbor relationships can be added.
 - When the number of neighboring cells of the gNodeB reaches its maximum value but that of a cell does not, the gNodeB automatically removes the neighbor relationships that are least frequently used from the NRTs of served cells so that new neighbor relationships can be added.
- Removal of redundant neighbor relationships
The **NRCCellAnr.PeriodForDelNrt** parameter specifies the period for automatic identification and removal of redundant neighbor relationships. At the end of a period, if a neighbor relationship meets all of the following conditions, the

gNodeB identifies it as a redundant neighbor relationship and removes it from the NRT.

- In the period, the total number of handover executions from the local cell to all intra-RAT neighboring cells is greater than or equal to 50.
- The number of intra-RAT neighboring cells (including intra-frequency and inter-frequency neighboring cells) of the local cell is greater than or equal to 32.
- The number of times the neighbor relationship is used is 0 every day for **NRCellAnr.PeriodForDelNrt** consecutive days (the gNodeB periodically checks at night every day).

The measurement of the number of times a neighbor relationship is used is controlled by the **NR_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the gNodeB supports optimized measurement for automatic removal of neighboring cells. In this case, the measured number of times a neighbor relationship with a cell is used is the sum of the numbers of the following attempts performed by the gNodeB using this neighbor relationship: handover attempts, SCC addition/change attempts, and SUL carrier addition/change attempts. However, if the **NRCellRelation.NoHoFlag** parameter is set to a value other than **PERMIT_HO** for the neighbor relationship, the gNodeB cannot remove it.
- If this option is deselected, the measured number of times the neighbor relationship is used is the number of times the gNodeB detects the corresponding neighboring cell based on the PCI contained in measurement reports sent by UEs.

NOTE

The gNodeB measures the number of times the neighboring cells are queried every night and these cells must be activated during the measurement.

A neighbor relationship cannot be automatically removed if any of the following conditions is met:

- The **NRCellRelation.BlindScellConfigFlag** parameter is set to **TRUE**.
- The **COV_BASED_BLIND_HO_FLAG** option of the **NRCellRelation.BlindHoFlag** parameter is selected.
- The **NRCellRelation.NoRmvFlag** parameter is set to **FORBID_ANR_RMV**.
- The **INTRA_BS_NCELL_RMV_PROHIBIT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, and the neighbor relationship is an intra-base-station neighbor relationship.

When a neighbor relationship is automatically removed by ANR, the gNodeB imposes a penalty on the neighbor relationship to prevent it from being added back to the NRT within a short period and stores it in a gNodeB-maintained temporary list. Whether the neighbor relationship can be added back to the NRT through ANR is described as follows:

- The neighbor relationship can be added back to the NRT through ANR only when the number of requests to add the neighbor relationship is greater than the value of the **NRCellAnr.AutoDelNCellPenaltyNum** parameter within the penalty period specified by the **NRCellAnr.AutoDelNCellPenaltyPeriod** parameter.

- After the penalty period specified by the **NRCellAnr.AutoDelNCellPenaltyPeriod** parameter expires, the gNodeB can add the neighbor relationship back to the NRT through ANR.

 NOTE

Removed neighbor relationships under penalty are stored in a temporary list on the base station. The list is cleared if the base station resets and can be regenerated after the penalty mechanism evaluates neighbor relationships again. Therefore, the number of neighbor relationships may increase after the base station restarts. The number can be restored by using automatic removal of neighboring cells.

Automatic Removal of External Cells

Automatic removal of external cells includes specification-based removal of external cells and removal of redundant external cells.

- Specification-based removal of external cells

The maximum number of external cells that can be configured in an NCL depends on the maximum number of external cells that can be configured for a gNodeB. For details about the maximum number of external cells that can be configured for a gNodeB, see *NSA Mobility Management* and *SA Mobility Management in Connected Mode*.

If the number of external cells in the NCL reaches its maximum value, the gNodeB automatically removes the external cells that are least frequently used from the NCL so that new external cells can be added.

- Removal of redundant external cells

The base station automatically removes redundant external cells from the NCL in either of the following scenarios:

- The external cell was a neighboring cell removed from an NRT and no cells served by the gNodeB have neighbor relationships with this cell.
- The external cell is not used for a period of seven days and no cells served by the gNodeB have neighbor relationships with this cell.

If a neighboring NG-RAN cell works in multi-operator sharing mode and the corresponding external cell is automatically removed from the NCL, all records of the external cell in the external-cell PLMN list are also removed.

The measurement of the number of times an external cell is used is controlled by the **NR_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the gNodeB supports optimized measurement for automatic removal of neighboring cells. In this case, the measured number of times the external cell is used is the sum of the numbers of the following attempts performed by the gNodeB using this neighbor relationship: handover attempts, SCC addition/change attempts, and SUL carrier addition/change attempts.
- If this option is deselected, the measured number of times the external cell is used is the number of times the gNodeB detects this cell based on the PCI contained in measurement reports sent by UEs.

 NOTE

The gNodeB measures the number of times the neighboring cells are queried every night and these cells must be activated during the measurement.

4.1.1.4 Automatic NCL Update

This function is enabled when the **NR_NR_ANR_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected. NCL update can be triggered by measurement reports from UEs, messages transmitted over the Xn interface between gNodeBs, or queries from the MAE-Access. The following provides the details.

NCL Update Triggered by Measurement Reports from UEs

NCL update can be triggered by measurement reports from UEs in SA and NSA networking.

After receiving an NCGI, TAC, frequencyBandList, and cellReservedForOperatorUse carried in the *CGI-Info* IE in a measurement report from a UE, the gNodeB searches its NCL for the external cell corresponding to the NCGI and performs the following operations:

- The gNodeB checks whether the SSB frequency and PCI of the external cell in the NCL are the same as those carried in the *CGI-Info* IE in the measurement report.
 - If they are the same, the gNodeB performs no further operation.
 - If they are different, the gNodeB updates the NCL using the SSB frequency or PCI carried in the *CGI-Info* IE.
- The gNodeB checks whether the TAC, FrequencyBand, AdditionalFrequencyBand, and PlmnReservedFlag of the external cell in the NCL are the same as those in the measurement report.
 - If they are the same, the gNodeB performs no further operation.
 - If they are different, the gNodeB updates the NCL using the information carried in the *CGI-Info* IE.

 NOTE

When the NCL is updated based on the *CGI-Info* IE:

- If the *CGI-Info* IE contains the TAC information of an external cell but the TAC is different from the configured value of **NRExternalNCell.Tac** for the external cell, the corresponding **NRExternalNCell.NrNetworkingOption** is automatically set to **UNLIMITED** (default value).
- If the *CGI-Info* IE does not contain the TAC information of an external cell, **NRExternalNCell.NrNetworkingOption** and **NRExternalNCell.Tac** of the cell are automatically set to **NSA** and **4294967295** (invalid value), respectively.

In the preceding information:

- The NCGI is determined by the **NRExternalNCell.Mcc**, **NRExternalNCell.Mnc**, **NRExternalNCell.gNBId**, and **NRExternalNCell.CellId** parameters.
- The SSB frequency is determined by the **NRExternalNCell.SsbDescMethod** and **NRExternalNCell.SsbFreqPos** parameters.

- The PCI is specified by the **NRExternalNCell.PhysicalCellId** parameter.
- The TAC is specified by the **NRExternalNCell.Tac** parameter.
- The frequencyBandList is specified by the **NRExternalNCell.FrequencyBand** and **NRExternalNCell.AdditionalFrequencyBand** parameters.
- The cellReservedForOperatorUse is specified by the **NRExternalNCell.PlmnReservedFlag** parameter.

NCL Update Triggered by Messages Transmitted over the Xn Interface Between gNodeBs

The Xn interface between gNodeBs does not exist in NSA networking. Therefore, NCL update cannot be triggered by messages transmitted over the Xn interface between gNodeBs in this networking. NCL update can be triggered by messages transmitted over the Xn interface between gNodeBs only in SA networking. NCL update can happen in the case of Xn setup between gNodeBs or gNodeB configuration change. This function is controlled by the **NREXTERNALCELL_SELF_CONFIG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter. In addition, this function takes effect only when the **NR_NR_ANR_SW** option of the **NRCCellAlgoSwitch.AnrSwitch** parameter is selected for any cell served by the gNodeB.

- Xn setup between gNodeBs
The local gNodeB exchanges information with the peer gNodeB through the Xn setup procedure.
After receiving an NCGI, SSB frequency, PCI, TAC, and frequencyBandList carried in the *Served Cell Information NR IE* in an XN SETUP REQUEST or XN SETUP RESPONSE message, the gNodeB updates the NCL in the same way as it does in NCL update triggered by measurement reports from UEs.
The networking option in the NCL is determined based on the actual capabilities (affected by factors such as the NG link status and the value of the **NRCCellAlgoSwitch.NsaDcSwitch** parameter). Therefore, the networking option in the NCL may be inconsistent with the value of the **gNBOperator.NrNetworkingOption** or **NRDUCellOp.NrNetworkingOption** parameter.
- gNodeB configuration change
The local gNodeB exchanges information with the peer gNodeB through the NG-RAN node configuration update procedure.
After receiving an NCGI, SSB frequency, PCI, TAC, and frequencyBandList carried in the *Served Cell Information NR IE* in an NG-RAN NODE CONFIGURATION UPDATE or NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE message, the gNodeB updates the NCL in the same way as it does in NCL update triggered by measurement reports from UEs.
The networking option in the NCL is determined based on the actual capabilities (affected by factors such as the NG link status and the value of the **NRCCellAlgoSwitch.NsaDcSwitch** parameter). Therefore, the networking option in the NCL may be inconsistent with the value of the **gNBOperator.NrNetworkingOption** or **NRDUCellOp.NrNetworkingOption** parameter. In addition, the gNodeB can directly update the networking option in the NCL based on Xn messages, which is controlled by the **NRNTWOPTN_SELF_CONFIG_POL_SW** option of the

gNodeBAlgo.SelfConfigSwitch parameter. When this option is selected and the networking option in the NCL is set to **UNLIMITED** based on the actual capabilities, the gNodeB can directly update the networking option in the NCL based on the networking option carried in the *Private Cell Extension Info* IE in the NG-RAN NODE CONFIGURATION UPDATE message.

 NOTE

Cell information must be carried during NG-RAN configuration update. As there are a large number of NG-RAN cells and Xn interfaces, a large number of configuration update messages will be sent if cell information is frequently modified. To prevent this, the system checks the base station configuration every 90 seconds and sends a configuration update message when a change occurs. Therefore, the modified configuration will take effect in at most 90s.

In addition, the gNodeB supports the update of the networking option in the NCL over the Xn interface in early morning. The gNodeB can directly update the networking option in the NCL based on the networking option carried in the *Private Cell Extension Info* IE in the NG-RAN NODE CONFIGURATION UPDATE message only when the networking option in the NCL is set to **UNLIMITED**.

NCL Update Triggered by Queries from the MAE-Access

Assume that a UE is located in the coverage area of cell A, which provides NR services for the UE. Cell B is an NG-RAN cell adjacent to cell A.

If the **NRExternalNCell.NrNetworkingOption** parameter for cell B is set to the default value (**UNLIMITED**), the gNodeB queries the MAE-Access and updates the NCL entry corresponding to cell B based on its actual networking option.

- When the MAE-Access successfully returns the query result, the gNodeB changes the **NRExternalNCell.NrNetworkingOption** parameter value of the external NR cell to **SA**, **NSA**, or **SA_NSA** based on the query result.
- When the MAE-Access does not return any query result or the query result is empty, the gNodeB cannot update the **NRExternalNCell.NrNetworkingOption** parameter value of the corresponding external NG-RAN cell. In this case, the gNodeB supports fast NR networking option query. This function is controlled by the **NR_NTW_OPTN_FAST_QRY_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.
 - If this option is selected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every day.
 - If this option is deselected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every seven days.

4.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate

This function automatically identifies neighbor relationships with cells of which the handover statistics are abnormal and processes the abnormal neighbor relationships.

The **NRCellRelation.NoHoFlag** parameter specifies whether to allow handovers or redirections for UEs based on neighbor relationships.

- If this parameter is set to **PERMIT_HO**, the gNodeB allows handovers or redirections for UEs based on neighbor relationships.
- If this parameter is set to **FORBID_HO**, the gNodeB prohibits handovers or redirections for UEs based on neighbor relationships.
- If this parameter is set to **SA_PERMIT_NSA_FORBID_HO**, the gNodeB allows handovers or redirections for UEs based on neighbor relationships in SA networking while the gNodeB prohibits handovers for UEs based on neighbor relationships in NSA networking.
- If this parameter is set to **SA_FORBID_NSA_PERMIT_HO**, the gNodeB prohibits handovers or redirections for UEs based on neighbor relationships in SA networking while the gNodeB allows handovers for UEs based on neighbor relationships in NSA networking.

Prohibition of Handovers to Neighboring Cells Based on the Handover Success Rate

With this function, the gNodeB automatically identifies neighboring cells with abnormal handover statistics and prohibits UEs from being handed over to these cells. This prevents handover or redirection failures caused by abnormal neighbor relationships, thereby guaranteeing user experience. This function is controlled by the **NR_NR_ANR_AUTO_NO_HO_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter.

If this option is selected, the gNodeB optimizes the setting of the **NRCellRelation.NoHoFlag** parameter for neighbor relationships based on the handover success rate within a measurement period. Whether the measurement period is configurable is specified by the **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

If the number of handovers (in SA networking) or that of PSCell change attempts (in NSA networking) from the local cell to a neighboring cell is less than the **NRCellAnr.NrNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over and includes it in the counting process during the next measurement period until the number is greater than or equal to this parameter value, and starts a decision on the handover success rate.

The gNodeB automatically sets the **NRCellRelation.NoHoFlag** parameter for the neighboring cell based on whether the success rate of handovers from the local cell to the neighboring cell is less than or equal to the **NRCellAnr.N2NNoHoSuccRateThld** parameter value for consecutive decision times whose number is specified by the **NRCellAnr.N2NNoHoConsecPrdThld** parameter.

- The gNodeB automatically sets the **NRCellRelation.NoHoFlag** parameter for the neighboring cell to **SA_FORBID_NSA_PERMIT_HO** if the success rate of

handovers from the local cell to the neighboring cell meets the conditions in each decision in SA networking rather than NSA networking.

- The gNodeB automatically sets the **NRCellRelation.NoHoFlag** parameter for the neighboring cell to **SA_PERMIT_NSA_FORBID_HO** if the success rate of handovers from the local cell to the neighboring cell meets the conditions in each decision in NSA networking rather than SA networking.
- The gNodeB automatically sets the **NRCellRelation.NoHoFlag** parameter for the neighboring cell to **FORBID_HO** if the success rate of handovers from the local cell to the neighboring cell meets the conditions in each decision in both SA networking and NSA networking.

Specifically:

- The measurement of the number of handovers in SA networking is controlled by the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.
- Handover success rate in SA networking = Number of successful handovers / Number of handovers x 100%
- Handover success rate in NSA networking = Number of successful handovers / Number of PSCell change attempts x 100%

Automatic optimization of neighbor relationship attributes can only change the value of the **NRCellRelation.NoHoFlag** parameter from a value indicating that handovers or redirections are allowed to another value indicating that handovers or redirections are prohibited. For example, this function cannot change the parameter value from **FORBID_HO** to **PERMIT_HO**, **SA_PERMIT_NSA_FORBID_HO**, or **SA_FORBID_NSA_PERMIT_HO**. Therefore, after this function is enabled, check whether the **NRCellRelation.NoHoFlag** parameter is automatically set to a value indicating that handovers or redirections are prohibited for a neighboring cell. If such a neighboring cell is still required as a target cell for handovers or redirections, this parameter value needs to be manually changed to another value indicating that handovers or redirections are allowed; alternatively, restoration of handover-prohibited neighboring cells based on the handover success rate needs to be manually enabled.

Restoration of Handover-Prohibited Neighboring Cells Based on the Handover Success Rate

With this function, the gNodeB periodically restores handover- or redirection-prohibited neighboring cells to the handover- or redirection-allowed state based on the handover statistics. This resolves the issue that a large number of cells on a network are in the handover-prohibited state and the cells must be restored manually. This function is controlled by the **NR_NR_ANR_NO_HO_AUTO_RES_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter.

If this option is selected, the gNodeB temporarily changes the value of the **NRCellRelation.NoHoFlag** parameter to a value indicating that handovers or redirections are allowed at the interval specified by the **NRCellAnr.NrNCellNoHoResIntvl** parameter for neighboring cells for which this parameter is set to a value indicating that handovers or redirections are prohibited. The gNodeB monitors these neighboring cells for a period of time. At

the end of the period, if the success rate of handovers from the local cell to the neighboring cells is less than or equal to 80%, the gNodeB automatically changes the value of the **NRCellRelation.NoHoFlag** parameter from a value indicating that handovers or redirections are allowed to another value indicating that handovers or redirections are prohibited.

4.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells

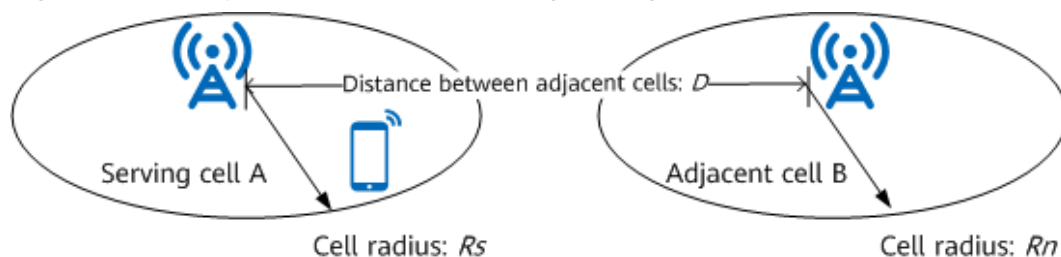
With this function, the gNodeB automatically selects neighboring cells having low handover success rates based on handover statistics, identifies over-distant neighboring cells among these neighboring cells, and handles the corresponding neighbor relationships with the over-distant neighboring cells. This function can prevent handover or redirection failures caused by a long distance between a serving cell and a target cell, thereby ensuring user experience.

An over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the sum of the access radii of the two cells.

- The distance between adjacent cells is the linear distance between the antennas serving the two cells.
- The access radius refers to the maximum distance where a UE can successfully access a cell. The distance is determined by the PRACH configurations of the cell. The PRACH configurations include the PRACH subcarrier spacing, cell radius, and PRACH configuration index.

The example shown in [Figure 4-5](#) indicates that D is greater than the sum of R_s and R_n . In conclusion, adjacent cell B is an over-distant neighboring cell of serving cell A.

Figure 4-5 Example of an over-distant neighboring cell



This function is enabled when the **NR_NR_ANR_OVER_DIST_DETECT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected.

The **NRCellRelation.NoHoFlag** parameter specifies whether to allow handovers or redirections for UEs based on neighbor relationships.

- If this parameter is set to **PERMIT_HO**, the gNodeB allows handovers or redirections for UEs based on neighbor relationships.
- If this parameter is set to **FORBID_HO**, the gNodeB prohibits handovers or redirections for UEs based on neighbor relationships.
- If this parameter is set to **SA_PERMIT_NSA_FORBID_HO**, the gNodeB allows handovers or redirections for UEs based on neighbor relationships in SA networking while the gNodeB prohibits handovers for UEs based on neighbor relationships in NSA networking.

- If this parameter is set to **SA_FORBID_NSA_PERMIT_HO**, the gNodeB prohibits handovers or redirections for UEs based on neighbor relationships in SA networking while the gNodeB allows handovers for UEs based on neighbor relationships in NSA networking.

If the **NR_NR_ANR_OVER_DIST_DETECT_SW** option of the **NRCCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB optimizes the setting of the **NRCCellRelation.NoHoFlag** parameter for neighbor relationships based on identification of over-distant neighboring cells within a measurement period. Whether the measurement period is configurable is specified by the **NR_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.NrHoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

If the number of handovers from the local cell to a neighboring cell is less than the **NRCCellAnr.NrNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over and includes it in the counting process during the next measurement period until the number is greater than or equal to this parameter value, and starts a decision. The gNodeB changes the **NRCCellRelation.NoHoFlag** parameter value from **PERMIT_HO** or **SA_PERMIT_NSA_FORBID_HO** to **FORBID_HO** if the neighboring cell meets all of the following conditions:

- The **NR Networking Option** parameter is set to **SA** or **SA_NSA** for the neighboring cell.
- The **NRCCellRelation.NoHoFlag** parameter is set to **PERMIT_HO** or **SA_PERMIT_NSA_FORBID_HO** for the neighboring cell.
- The number of successful handovers from the local cell to the neighboring cell is less than 95%.
- The distance between the local cell and the neighboring cell is greater than the sum of the access radii of the two cells.

Specifically:

- The measurement of the number of handovers in SA networking is controlled by the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.
- Handover success rate in SA networking = Number of successful handovers / Number of handovers x 100%
- Handover success rate in NSA networking = Number of successful handovers / Number of PSCell change attempts x 100%

After the **NR_NR_ANR_OVER_DIST_DETECT_SW** option of the **NRCCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB changes the **NRCCellRelation.NoHoFlag** parameter value from **PERMIT_HO** or **SA_FORBID_NSA_PERMIT_HO** to **FORBID_HO** for a neighboring cell that meets all of the following conditions:

- The **NR Networking Option** parameter is set to **SA_NSA** or **NSA** for the neighboring cell.
- The **NRCellRelation.NoHoFlag** parameter is set to **SA_FORBID_NSA_PERMIT_HO** or **PERMIT_HO** for the neighboring cell.
- The distance between the local cell and the neighboring cell is greater than the sum of the access radii of the two cells.

 NOTE

- The gNodeB queries the MAE-Access about whether the distance between a local cell and a neighboring cell is greater than the sum of the access radii of the two cells. The MAE-Access calculates the distance between the local cell and the neighboring cell based on the longitude and latitude information contained in base station engineering parameters.
- If a multi-RRU cell has multiple latitudes and longitudes, the minimum distance is used for determination.
- The gNodeB queries over-distant neighboring cells on the MAE-Access again at night every day.
- The MAE-Access allows each gNodeB to query a maximum of 256 over-distant neighboring cells every day.

Automatic optimization of neighbor relationship attributes can only change the value of the **NRCellRelation.NoHoFlag** parameter from a value indicating that handovers or redirections are allowed to another value indicating that handovers or redirections are prohibited. For example, this function cannot change the parameter value from **FORBID_HO** to **PERMIT_HO**, **SA_PERMIT_NSA_FORBID_HO**, or **SA_FORBID_NSA_PERMIT_HO**. Therefore, after this function is enabled, check the value of the **NRCellRelation.NoHoFlag** parameter. If the parameter is automatically set to a value indicating that handovers or redirections are prohibited for a neighboring cell but this cell is required as a target cell for handovers or redirections, this parameter value needs to be manually changed to another value indicating that handovers or redirections are allowed.

4.1.1.7 Automatic External-Cell PLMN List Configuration

If neighboring NG-RAN cells work in multiple-operator sharing mode, the gNodeB automatically configures NG-RAN external-cell PLMN lists (external-cell PLMN list for short) in addition to NCL and NRT configuration. An external-cell PLMN list records secondary operator information about an external NG-RAN cell. Such information includes settings of the parameters in the **NRExternalNCellPlmn** MO. The parameters include:

- **NRExternalNCellPlmn.Mcc**
- **NRExternalNCellPlmn.Mnc**
- **NRExternalNCellPlmn.gNBId**
- **NRExternalNCellPlmn.CellId**
- **NRExternalNCellPlmn.SharedMcc**
- **NRExternalNCellPlmn.SharedMnc**
- **NRExternalNCellPlmn.Tac**
- **NRExternalNCellPlmn.NrNetworkingOption**
- **NRExternalNCellPlmn.SharedPlmnCellId**

- **NRExternalNCellPlmn.SharedPlmngNBId**

This function is enabled when the **NR_NR_ANR_SW** option of the **NRCelAlgoSwitch.AnrSwitch** parameter is selected. Automatic external-cell PLMN list configuration includes automatic external-cell PLMN list addition and automatic external-cell PLMN list update. Details are provided in the following sections. For details of multiple-operator sharing, see *Multi-Operator Sharing*.

4.1.1.7.1 Automatic External-Cell PLMN List Addition

The **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCelAnr.AnrModeSwitch** parameter determines whether the gNodeB can query the PLMN list of an external NG-RAN cell from the MAE-Access. If neighboring NG-RAN cells work in multi-operator sharing mode, the following apply:

- When the **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCelAnr.AnrModeSwitch** parameter is selected, the gNodeB can query the list of PLMN IDs of the primary and secondary operators to which an external NG-RAN cell belongs from the MAE-Access.
- When the **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCelAnr.AnrModeSwitch** parameter is deselected, the gNodeB cannot query the list of PLMN IDs of the primary and secondary operators to which an external NG-RAN cell belongs from the MAE-Access.

If PLMN ID rotation mode is enabled for the neighboring NG-RAN cell, the sequence of PLMN IDs broadcast in system information periodically changes. Therefore, the primary operator is not always listed at the top of the PLMN list. The principles of PLMN configuration for the external NG-RAN cell vary depending on whether PLMN ID rotation mode is enabled for the neighboring NG-RAN cell. The following provides the details.

For details of PLMN ID rotation mode, see *Multi-Operator Sharing*.

PLMN ID Rotation Mode Disabled for Neighboring NG-RAN Cells

In this scenario, the gNodeB can add neighboring NG-RAN cells based on measurement reports from UEs. The **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCelAnr.AnrModeSwitch** parameter does not need to be selected.

When a missing neighboring cell is detected, the gNodeB receives a measurement report containing the NCGI from the UE and checks whether the reported *CGI-Info* IE contains only one PLMN ID.

- If yes, the neighboring NG-RAN cell does not work in multi-operator sharing mode. In this case, the gNodeB adds only the missing neighboring cell to the NCL and NRT.
- If no, the neighboring NG-RAN cell works in multi-operator sharing mode. In this case, the gNodeB adds the missing neighboring cell to the NCL/NRT and adds secondary operator information about the missing neighboring cell to the external-cell PLMN list.

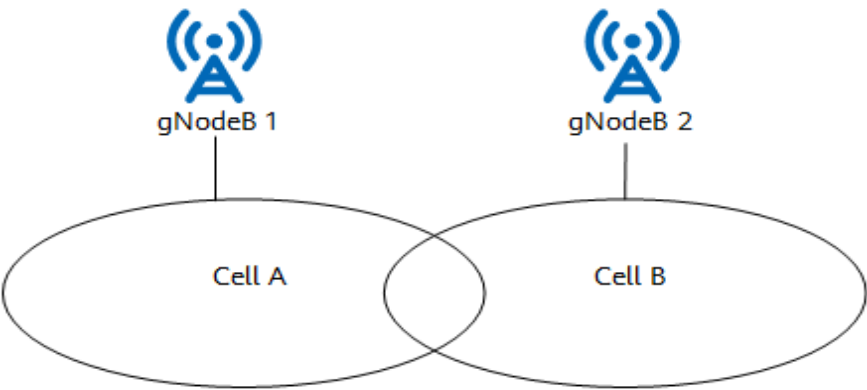
For example, cell A and cell B are respectively served by gNodeB 1 and gNodeB 2. Cell B is shared among three operators and is a missing neighboring cell of cell A. gNodeB 1 adds cell B to its NCL and NRT of cell A. In addition, gNodeB 1 adds secondary operator information about cell B to its

external-cell PLMN list. **Figure 4-6** provides the information recorded in the external-cell PLMN list.

The secondary operator PLMN selected for cell B is specified by the **ANR_PLMN_SEL_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- When this option is selected, PLMNs (configured by using the **gNBOperator** MO) supported by the local gNodeB in measurement reports from UEs are preferentially selected with the corresponding information added to the external-cell PLMN list.
- When this option is deselected, the gNodeB selects PLMNs according to their sequence in measurement reports from UEs and adds the corresponding information to the external-cell PLMN list.

Figure 4-6 External-cell PLMN list



Cell B:
Primary operator 1 (PLMN ID: 46001)
Secondary operator 2 (PLMN ID: 46003)
Secondary operator 3 (PLMN ID: 46000)

External-cell PLMN list of gNodeB 1

Parameter	Entry 1	Entry 2	...
Mcc	460	460	...
Mnc	01	01	...
gNBId	2	2	...
CellId	1	1	...
SharedMcc	460	460	...
SharedMnc	03	00	...
Tac	8	8	...
NrNetworkingOption	SA	SA	...
SharedPlmnCellId	2	3	...
SharedPlmngNBId	3	4	...

When the secondary operator information of a missing neighboring cell is added to the external-cell PLMN list:

- If the PLMN information in the *CGI-Info* IE contains the TAC corresponding to the PLMN, **NRExternalNCellPlmn.NrNetworkingOption** of the secondary operator is automatically set to **UNLIMITED** (default value) in the external-cell PLMN list. The gNodeB then queries the MAE-Access before updating the option in the external-cell PLMN list to the actual networking option. For details about the procedure of external-cell PLMN list update triggered by queries from the MAE-Access, see [External-Cell PLMN List Update Triggered by Queries from the MAE-Access](#).
- If the PLMN information in the *CGI-Info* IE does not contain the TAC corresponding to the PLMN, **NRExternalNCellPlmn.NrNetworkingOption** and **NRExternalNCellPlmn.Tac** of the secondary operator are automatically set to **NSA** and **4294967295** (invalid value) respectively in the external-cell PLMN list.
- The principles for setting the **NRExternalNCellPlmn.SharedPlmnCellId** and **NRExternalNCellPlmn.SharedPlmnNbId** parameters are as follows:
 - When the **gNBSharingMode.MultipleNCISwitch** parameter is turned on:
 - If the PLMN corresponding to the NCI is configured through a **gNBOperator** or **gNBExternalPlmn** MO, the information in the NCI is filled into the **NRExternalNCellPlmn.SharedPlmnCellId** and **NRExternalNCellPlmn.SharedPlmnNbId** parameters based on the setting of the **gNBOperator.gNBIdLength** or **gNBExternalPlmn.gNBIdLength** parameter corresponding to the PLMN.
 - If the PLMN corresponding to the NCI is not configured through a **gNBOperator** or **gNBExternalPlmn** MO, the information in the NCI is filled into the **NRExternalNCellPlmn.SharedPlmnCellId** and **NRExternalNCellPlmn.SharedPlmnNbId** parameters based on the setting of the **gNodeBFunction.gNBIdLength** parameter.

 NOTE

If the PLMN corresponding to the NCI is not configured through a **gNBOperator** or **gNBExternalPlmn** MO and the **gNodeBFunction.gNBIdLength** parameter value for the primary operator of the missing neighboring cell is different from the **gNodeBFunction.gNBIdLength** parameter value of the local gNodeB, then the gNodeB ID parsed from the NCI is incorrect after this neighboring cell is added. Therefore, you are advised to configure the PLMN corresponding to the NCI through the **gNBOperator** or **gNBExternalPlmn** MO before the neighboring cell is added.

- When the **gNBSharingMode.MultipleNCISwitch** parameter is turned off:
 - If the PLMN corresponding to the NCI is configured through a **gNBExternalPlmn** MO, the information in the NCI is filled into the **NRExternalNCellPlmn.SharedPlmnCellId** and **NRExternalNCellPlmn.SharedPlmnNbId** parameters based on the setting of the **gNBExternalPlmn.gNBIdLength** parameter corresponding to the PLMN.
 - If the PLMN corresponding to the NCI is not configured through a **gNBExternalPlmn** MO, the information in the NCI is filled into the

NRExternalNCellPlmn.SharedPlmnCellId and **NRExternalNCellPlmn.SharedPlmnngNBId** parameters based on the setting of the **gNodeBFunction.gNBIdLength** parameter.

For further details about the **gNBSharingMode.MultipleNCISwitch** parameter, see *Multi-Operator Sharing*.

PLMN ID Rotation Mode Enabled for Neighboring NG-RAN Cells

In this scenario, the primary operator is not always listed at the top of the PLMN list and UEs cannot obtain the neighboring NG-RAN cell's correct PLMN IDs of the primary and secondary operators. As a result, the gNodeB cannot add neighboring NG-RAN cells based on measurement reports from UEs. For this reason, the **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter needs to be selected. When this option is selected, the gNodeB can obtain the PLMN list of the neighboring NG-RAN cell from the MAE-Access.

When a missing neighboring cell is detected, the gNodeB receives a measurement report containing the NCGI from a UE and manages the external-cell PLMN list as follows:

1. The gNodeB checks whether the reported *CGI-Info* IE contains only one PLMN ID.
 - If yes, the neighboring NG-RAN cell does not work in multi-operator sharing mode and no further action is required.
 - If no, the neighboring NG-RAN cell works in multi-operator sharing mode. The process goes to step 2.
2. The gNodeB checks whether one of the reported PLMN IDs is marked as Reserved.
 - If yes, the PLMN marked as Reserved belongs to the primary operator, and other PLMNs belong to secondary operators.
 - If no, the process goes to step 3.
3. The gNodeB checks whether the reported neighboring cell is served by the gNodeB.
 - If yes, an external-cell PLMN list does not need to be configured.
 - If no, the process goes to step 4.
4. The gNodeB checks whether an external-cell PLMN list has been configured for the reported neighboring cell.
 - If yes, the eNodeB identifies the primary and secondary operators based on the external-cell PLMN list.
 - If no, the process goes to step 5.
5. The gNodeB reports the PCI and NCGI sent from the UE to the MAE-Access.
6. The MAE-Access checks the PLMN list of the indicated neighboring NG-RAN cell based on the PCI, gNodeB ID, and cell ID.
7. The MAE-Access sends the check result to the gNodeB.
8. The gNodeB adds the check result to the external-cell PLMN list. PLMN selection is controlled by the **ANR_PLMN_SEL_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.
 - When this option is selected, PLMNs (configured by using the **gNBOperator** MO) supported by the local gNodeB in the query result

from the MAE-Access are preferentially selected with the corresponding information added to the external-cell PLMN list.

- When this option is deselected, the gNodeB selects PLMNs according to their sequence in the query result from the MAE-Access and adds the corresponding information to the external-cell PLMN list.

4.1.1.7.2 Automatic External-Cell PLMN List Update

External-cell PLMN list update includes:

- Addition: The PLMN ID of an external cell is added.
- Modification: The values of PLMN-ID-specific **Tac**, **SharedPlmnCellId**, **SharedPlmngNBId**, and **NrNetworkingOption** of an external cell are changed.

The value of **NrNetworkingOption** is changed only when the value of **Tac** changes. The value of **SharedPlmnCellId** or **SharedPlmngNBId** does not affect the value of **NrNetworkingOption**. When the external-cell PLMN list update is triggered based on the CGI, then:

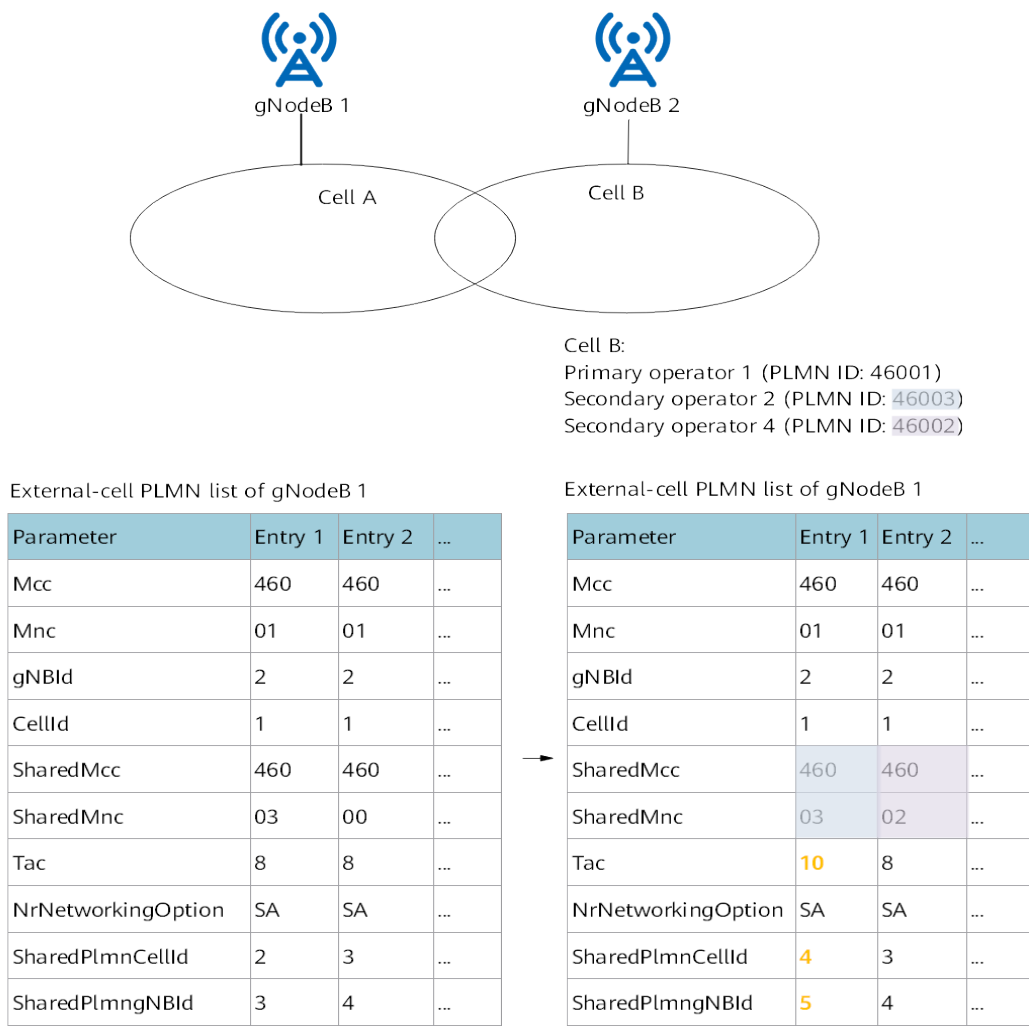
- If the current TAC is valid and the value of **NrNetworkingOption** is changed to **UNLIMITED**, the gNodeB queries the MAE-Access and then updates the value of **NrNetworkingOption** in the external-cell PLMN list using the actual architecture option of the neighboring cell.
- If the TAC is invalid, the value of **NrNetworkingOption** is changed to **NSA**.
- Removal: The PLMN ID of an external cell is removed.

For example, cell A and cell B are respectively served by gNodeB 1 and gNodeB 2. Cell B is shared among three operators. The secondary operator information about cell B is recorded in the external-cell PLMN list of gNodeB 1. [Figure 4-6](#) shows this example. The following describes the different outcomes when the operators to which cell B belongs change:

- Addition: The cell belongs to a new operator (operator 4).
- Modification: The **Tac**, **SharedPlmnCellId**, and **SharedPlmngNBId** of operator 2 are changed.
- Removal: The cell no longer belongs to operator 3.

[Figure 4-7](#) shows an example where the information recorded in the external-cell PLMN list is updated.

Figure 4-7 External-cell PLMN list



External-cell PLMN list update can be triggered by measurement reports from UEs, messages transmitted over the Xn interface between gNodeBs, or queries from the MAE-Access. The following provides the details.

External-Cell PLMN List Update Triggered by UE Measurement Reports

External-cell PLMN list update can be triggered by measurement reports from UEs in SA and NSA networking.

External-cell PLMN list update can be triggered by measurement reports from UEs. After receiving the NCGI and PLMN list carried in the *CGI-Info* IE in the measurement report from a UE, the gNodeB queries the external-cell PLMN list for the external cell indicated by the NCGI. The gNodeB then checks whether the PLMN list and TAC of the external cell are same as those in the measurement report.

- If they are the same, the gNodeB performs no further operation.
- If they are different, the gNodeB updates the external-cell PLMN list using the information carried in the IE.

 NOTE

When the external-cell PLMN list is updated based on the *CGI-Info* IE:

- If the PLMN information in the *CGI-Info* IE contains the TAC corresponding to the PLMN but the TAC is different from the configured value of **NRExternalNCellPlmn.Tac** for the PLMN, the corresponding **NRExternalNCellPlmn.NrNetworkingOption** is automatically set to **UNLIMITED** (default value).
- If the PLMN information in the *CGI-Info* IE does not contain the TAC corresponding to the PLMN, **NRExternalNCellPlmn.NrNetworkingOption** and **NRExternalNCellPlmn.Tac** are automatically set to **NSA** and **4294967295** (invalid value), respectively.

External-Cell PLMN List Update Triggered by Messages over the Xn Interface Between gNodeBs

As the Xn interface between gNodeBs does not exist in NSA networking, external-cell PLMN list update cannot be triggered by messages transmitted over the Xn interface between gNodeBs in this networking. External-cell PLMN list update can be triggered by messages transmitted over the Xn interface between gNodeBs only in SA networking. The triggering scenarios include:

- Xn setup between gNodeBs
The gNodeB exchanges information with the peer gNodeB through the Xn setup procedure.
After receiving the NCGI and PLMN list carried in the *Served Cell Information NR* IE in the XN SETUP REQUEST or XN SETUP RESPONSE message, the gNodeB queries the external-cell PLMN list for the external cell indicated by the NCGI and checks whether the PLMN list and TAC of the external cell are same as those in the measurement report.
 - If they are the same, the gNodeB performs no further operation.
 - If they are different, the gNodeB updates the external-cell PLMN list using the information carried in the IE.
- gNodeB configuration change
The gNodeB exchanges information with the peer gNodeB through the NG-RAN node configuration update procedure.
After receiving the NCGI and PLMN list carried in the *Served Cell Information NR* IE in the NG-RAN NODE CONFIGURATION UPDATE or NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE message, the gNodeB queries the external-cell PLMN list for the external cell indicated by the NCGI. The gNodeB then checks whether the PLMN list and TAC of the external cell are same as those in the measurement report.
 - If they are the same, the gNodeB performs no further operation.
 - If they are different, the gNodeB updates the external-cell PLMN list using the information carried in the IE.

The networking option in the external-cell PLMN list information is determined based on actual capabilities (affected by factors such as the NG link status and the value of the **NRCellAlgoSwitch.NsaDcSwitch** parameter). Therefore, the networking option in the external-cell PLMN list information may be inconsistent with the value of the **gNBOperator.NrNetworkingOption** or **NRDUCellOp.NrNetworkingOption** parameter. In addition, the gNodeB can directly update the networking option

in the external-cell PLMN list information based on Xn messages, which is controlled by the **NRNTWOPTN_SELF_CONFIG_POL_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter. When this option is selected and the networking option in the external-cell PLMN list information is set to **UNLIMITED** based on the actual capabilities, the gNodeB can directly update the networking option in the external-cell PLMN list information based on the networking option carried in the *Private Cell Extension Info* IE in the NG-RAN NODE CONFIGURATION UPDATE message.

In addition, the gNodeB supports the update of the networking option in the external-cell PLMN list information over the Xn interface in early morning. The gNodeB can directly update the networking option in the external-cell PLMN list information based on the networking option carried in the *Private Cell Extension Info* IE in the NG-RAN NODE CONFIGURATION UPDATE message only when the networking option in the external-cell PLMN list information is set to **UNLIMITED**.

External-Cell PLMN List Update Triggered by Queries from the MAE-Access

Assume that a UE is located in the coverage area of cell A, which provides NR services for the UE. Cell B is an NG-RAN cell adjacent to cell A.

If the **NRExternalNCellPlmn.NrNetworkingOption** parameter for cell B is set to the default value (**UNLIMITED**), the gNodeB queries the MAE-Access and updates the external-cell PLMN list corresponding to cell B based on its actual networking option.

- When the MAE-Access successfully returns the query result, the gNodeB changes the **NRExternalNCell.NrNetworkingOption** parameter value to **SA**, **NSA**, or **SA_NSA** in the NR external-cell PLMN list based on the query result.
- When the MAE-Access does not return any query result or the query result is empty, the gNodeB cannot update the **NRExternalNCell.NrNetworkingOption** parameter value in the NR external-cell PLMN list. In this case, the gNodeB supports fast NR networking option query. This function is controlled by the **NR_NTW_OPTN_FAST_QRY_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.
 - If this option is selected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every day.
 - If this option is deselected, the gNodeB queries the **NRExternalNCell.NrNetworkingOption** parameter values of external NG-RAN cells from the MAE-Access at night every seven days.

4.1.1.8 Automatic Neighboring Frequency Configuration

When multiple intra-RAT neighboring frequencies are configured for a cell and the gNodeB initiates an inter-frequency handover for a UE, the gNodeB can automatically identify the frequencies to be measured based on the coverage of the neighboring frequencies of the cell. The gNodeB delivers only frequencies whose coverage areas cover the UE's location for measurements. Automatic neighboring frequency configuration is enabled when the **AUTO_NR_FREQ_RELATION_SW** option of the **NRCCellAlgoSwitch.AnrSwitch** parameter is selected.

- If this option is selected, the gNodeB automatically adjusts the value of **NRCellFreqRelation.MeasActvFlag** based on the inter-frequency measurement report sent by the UE.
- If this option is deselected, automatic neighboring frequency configuration does not take effect.

The **NRCellFreqRelation.MeasActvFlag** parameter specifies whether a neighboring frequency of a cell can be measured. This parameter can be set to **ACTIVE** or **INACTIVE**.

- If this parameter is set to **ACTIVE**, the frequency can be delivered in scenarios such as inter-frequency measurements.
- If this parameter is set to **INACTIVE**, the frequency cannot be delivered in scenarios such as inter-frequency measurements.

For details about inter-frequency measurements, see *NSA Mobility Management* and *SA Mobility Management in Connected Mode*.

Automatic neighboring frequency configuration provides the following two sub-functions which are different in the mechanism for delivering inter-frequency measurement configurations:

- Basic automatic neighboring frequency configuration: The gNodeB cannot proactively deliver measurement configurations through this function, but instead it selects UEs when other inter-frequency measurements are initiated. Then, the gNodeB automatically adjusts the value of **NRCellFreqRelation.MeasActvFlag** based on the measurement reports.
- Enhanced automatic neighboring frequency configuration: With this function, the gNodeB proactively selects UEs and delivers periodic measurement configurations to them. Then, the gNodeB automatically adjusts the value of **NRCellFreqRelation.MeasActvFlag** based on the measurement reports.

Additionally, forbidding automatic neighboring frequency configuration for a specific neighboring frequency is supported. This function is controlled by the **AUTO_NR_FREQ_REL_FORBID_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter.

- If this option is selected, the gNodeB cannot automatically adjust the **NRCellFreqRelation.MeasActvFlag** parameter value for the corresponding neighboring frequency based on measurement reports.
- If this option is deselected, forbidding automatic neighboring frequency configuration does not take effect.

Basic Automatic Neighboring Frequency Configuration

After the **AUTO_NR_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB automatically adjusts the value of **NRCellFreqRelation.MeasActvFlag** as follows:

- For a frequency with the **NRCellFreqRelation.MeasActvFlag** parameter set to **INACTIVE**:
The gNodeB temporarily allows measurements on this frequency every hour. During this period, the gNodeB automatically selects UEs to which this frequency is delivered for measurements when other inter-frequency measurement events are triggered. When receiving a valid measurement report of the frequency, the gNodeB changes the value of the

NRCellFreqRelation.MeasActvFlag parameter to **ACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in system information block type 4 (SIB4).

- For a frequency with the **NRCellFreqRelation.MeasActvFlag** parameter set to **ACTIVE**:

If the number of times this frequency is delivered for measurements reaches 1000, the accumulative measurement period is longer than or equal to seven days, and the gNodeB has not received a valid measurement report of the frequency, then the gNodeB changes the value of the **NRCellFreqRelation.MeasActvFlag** parameter to **INACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB4.

Enhanced Automatic Neighboring Frequency Configuration

After the **AUTO_NR_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, enhanced automatic neighboring frequency configuration enables the gNodeB to deliver a frequency with the **NRCellFreqRelation.MeasActvFlag** parameter set to **INACTIVE** or **ACTIVE** for periodic measurements and determine the coverage of the neighboring frequency based on the RSRP threshold. This function is controlled by the **AUTO_NR_FREQ_RELATION_MEAS_SW** option of the **NRCellAnr.AnrModeSwitch** parameter.

- If this option is selected, the gNodeB automatically adjusts the **NRCellFreqRelation.MeasActvFlag** parameter value based on the inter-frequency measurement report sent by the UE and the RSRP threshold.
- If this option is deselected, enhanced automatic neighboring frequency configuration does not take effect.

After the **AUTO_NR_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter and the **AUTO_NR_FREQ_RELATION_MEAS_SW** option of the **NRCellAnr.AnrModeSwitch** parameter are both selected, the gNodeB automatically adjusts the value of the **NRCellFreqRelation.MeasActvFlag** parameter as follows:

- For a frequency with the **NRCellFreqRelation.MeasActvFlag** parameter set to **INACTIVE**:

The gNodeB temporarily allows measurements on this frequency every 61 minutes. During this period, the gNodeB selects three UEs in each cell to which this frequency is delivered for measurements. When the gNodeB receives a valid measurement report of the frequency and the RSRP value of a neighboring cell on the frequency is greater than the RSRP threshold specified by the **NRCellFreqRelation.AutoNrFreqRelationRsrpThld** parameter, the gNodeB changes the value of the **NRCellFreqRelation.MeasActvFlag** parameter to **ACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB4.

- For a frequency with the **NRCellFreqRelation.MeasActvFlag** parameter set to **ACTIVE**:

The base station delivers this frequency for measurements to three UEs in each cell every 10 minutes. Assume that the number of times this frequency is delivered for measurements reaches 1000 and the accumulative measurement

period is longer than or equal to seven days. If the gNodeB has not received a valid measurement report of the frequency or the RSRP values of neighboring cells on the frequency are less than or equal to the RSRP threshold specified by the **NRCellFreqRelation.AutoNrFreqRelationRsrpThld** parameter, then the gNodeB changes the value of the **NRCellFreqRelation.MeasActvFlag** parameter to **INACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB4.

 NOTE

When delivering frequency measurement configurations to a UE, the gNodeB is not allowed to select the frequency on which an SUL cell operates.

4.1.1.9 NCL/NRT Management Based on the gNodeB Blacklist

With NCL/NRT management based on the gNodeB blacklist, the gNodeB can identify blacklisted gNodeBs and perform differentiated management on NR cells served by these gNodeBs.

The **gNBNRBlacklist** MO is used to set a gNodeB blacklist. A maximum of 128 gNodeBs can be blacklisted for each gNodeB. The MCC, MNC, and gNodeB ID are specified by the following parameters:

- MCC: **gNBNRBlacklist.Mcc**
- MNC: **gNBNRBlacklist.Mnc**
- gNodeB ID: **gNBNRBlacklist.gNodeBId**

The gNodeB performs the following operations on NR cells served by another gNodeB after adding this gNodeB to the blacklist using the **gNBNRBlacklist** MO:

- If these NR cells are identified as unknown neighboring cells by automatic detection and addition of missing neighboring cells, the gNodeB does not add these cells to the NCL or add the neighbor relationships with these cells to NRTs.
- The gNodeB does not instruct UEs to initiate handovers or redirections to these NR cells.

4.1.2 Network Analysis

4.1.2.1 Benefits

Intra-RAT ANR ensures the integrity, validity, and correctness of intra-RAT neighbor relationships with NG-RAN cells. By automatically optimizing the attributes of the neighboring cells, it reduces manual intervention to decrease the costs of network planning, network optimization, and O&M.

- When the gNodeB automatically adds neighboring cells based on measurement results reported by UEs, it may add over-distant neighboring cells or some neighboring cells with low signal strength to NRTs. These neighboring cells will decrease the handover success rate.
 - In SA networking, ANR automatically maintains the validity of neighboring cells based on the handover success rate every day and sets the **NRCellRelation.NoHoFlag** parameter to **FORBID_HO** or **SA_FORBID_NSA_PERMIT_HO** for neighboring cells with low handover success rates. Then, the handover success rate will gradually recover.

- In NSA networking, there may be no X2 link between the base station serving the added neighboring cell and the LTE anchor base station. As a result, handover failures occur and the handover success rate decreases. ANR automatically maintains the validity of neighboring cells based on the handover success rate every day and sets the **NRCellRelation.NoHoFlag** parameter to **FORBID_HO** or **SA_PERMIT_NSA_FORBID_HO** for neighboring cells to which the number of successful handovers is 0 and the number of handovers exceeds 100. Then, the handover success rate will gradually recover. Therefore, the controlled mode is recommended. In this mode, manual intervention is required to select and add appropriate neighboring cells based on the measurement results reported by UEs.

The number and distribution of ANR-capable UEs affect how fast ANR can detect missing neighboring cells. Therefore, the decrease in the service drop rate and increase in the intra-RAT handover success rate produced by intra-RAT ANR are impossible to quantify.

- Automatic addition of missing neighboring cells based on the number of CGI-Info reporting times reduces the probability of adding an over-distant neighboring cell to the NCL and the neighbor relationship with this cell to the NRT.
- If neighboring cell class label management is enabled in NSA networking, neighboring NR cells with the handover success rate lower than the threshold specified by the **NRCellAnr.NCellHoSuccRateThld** parameter are configured as extended neighboring cells. The gNodeB can detect and add neighboring cells with PCI confusions, preventing the **Inter-SgNB PSCell Change Success Rate (CU)** from decreasing due to PCI confusions.
- If the **SA_NR_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter is selected, a neighboring NR cell is identified as an abnormal neighboring cell in SA networking if preparations of a handover to this NR cell fail, increasing the handover success rate. Specifically:
 - Intra-frequency handover success rate =
$$\frac{(N.HO.IntraFreq.IntragNB.ExecSuccOut + N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut + N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut)}{(N.HO.IntraFreq.IntragNB.PrepAttOut + N.HO.IntraFreq.Ng.IntergNB.PrepAttOut + N.HO.IntraFreq.Xn.IntergNB.PrepAttOut)} \times 100\%$$
 - Inter-frequency handover success rate =
$$\frac{(N.HO.InterFreq.IntragNB.ExecSuccOut + N.HO.InterFreq.Ng.IntergNB.ExecSuccOut + N.HO.InterFreq.Xn.IntergNB.ExecSuccOut)}{(N.HO.InterFreq.IntragNB.PrepAttOut + N.HO.InterFreq.Ng.IntergNB.PrepAttOut + N.HO.InterFreq.Xn.IntergNB.PrepAttOut)} \times 100\%$$
- With automatic removal of neighboring cells enabled, if the **NR_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, unused neighboring NR cells are more likely to be removed.
- In NSA and SA hybrid networking scenarios, after the function of directly updating the networking option in the NCL based on Xn messages or the

function of directly updating the networking option in the external-cell PLMN list information based on Xn messages is enabled, **Inter-SgNB PSCell Change Success Rate (CU)** may increase and **SgNB Abnormal Release Rate (CU)** may decrease.

- NCL/NRT management based on the gNodeB blacklist prevents UEs from being handed over or redirected to a cell served by a fake base station. As a result, the NG-based intra-frequency and inter-frequency inter-gNodeB handover preparation success rates may increase. It is recommended that this function be enabled when a fake gNodeB is near a gNodeB. For details about how to identify a fake base station, contact Huawei technical support. Specifically:
 - NG-based inter-gNodeB intra-frequency handover preparation success rate = $\frac{\text{N.HO.IntraFreq.Ng.IntergNB.ExecAttOut}}{\text{N.HO.IntraFreq.Ng.IntergNB.PrepAttOut}} \times 100\%$
 - NG-based inter-gNodeB inter-frequency handover preparation success rate = $\frac{\text{N.HO.InterFreq.Ng.IntergNB.ExecAttOut}}{\text{N.HO.InterFreq.Ng.IntergNB.PrepAttOut}} \times 100\%$
- In the case of automatic external-cell PLMN list addition, with PLMN selection optimization enabled, PLMNs supported by the local gNodeB are preferentially selected. This reduces the impact of PLMN addition failure due to insufficient capacity of the external-cell PLMN list.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

- After the value of **NRExternalNCell.CtrlMode** is changed to **MANUAL_MODE**, the gNodeB cannot add or modify the external-cell PLMN list in a timely manner with ANR. As a result, UEs cannot initiate handovers or handovers fail, which may increase the **Service Call Drop Rate (CU)** or may decrease the **Intra-Frequency Handover Out Success Rate (CU)** and **Inter-Frequency Handover Out Success Rate (CU)**.
- To detect an unknown neighboring cell, the gNodeB instructs the UEs that meet specific conditions to perform reportCGI measurements. This process prolongs the delay in a handover from the source cell to the target cell. After the neighbor relationship with the cell is added, measurements of report type *reportCGI* are not triggered during subsequent handovers to the neighboring cell and therefore the delay in handovers to the cell is no longer prolonged.
- The average downlink UE throughput (**User Downlink Average Throughput (DU)**) decreases during DRX in which the UEs read the NCGI of the target cell. This is because the gNodeB limits scheduling for downlink transmission for the UEs. After a neighbor relationship with the cell is added, DRX configuration will not be triggered for reportCGI measurements, and therefore UEs are not subject to such changes in the average downlink UE throughput during subsequent handovers to the cell.
- Uplink data transmission causes ongoing measurements of report type *reportCGI* to fail. UEs read NCGIs in sleep time during DRX. The sleep time can be interrupted by transmission of uplink data packets (such as uplink data from UEs or RLC ACKs from UEs in RLC AM mode). This decreases the NCGI reading success rate. After an NCGI is acquired, the identified cell is

automatically configured as a neighboring cell and UEs no longer read the NCGI.

- Automatic addition of missing neighboring cells based on the number of CGI-Info reporting times increases the number of reportCGI measurements, which may cause decreases in **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**. A smaller value of the **NRCellAnr.NCellAddCgiRptCountThld** parameter results in a shorter time required for adding a missing neighboring cell but a higher risk that an addition is performed by mistake. A larger value of this parameter produces the opposite effects.
- During automatic neighboring cell class label adjustment in NSA networking, the number of messages transmitted over the X2 interface between the MeNB and the base station serving the source PSCell and between the MeNB and the base station serving the target PSCell increases, which may cause message congestion.
- Penalty on reportCGI measurements may change the number of handovers. As a result, the **Intra-Frequency Handover Out Success Rate (CU)** and **Inter-Frequency Handover Out Success Rate (CU)** may fluctuate.
- After forbidding automatic neighboring frequency configuration is enabled, the **NRCellFreqRelation.MeasActvFlag** parameter value will not be automatically adjusted for the corresponding neighboring frequency based on measurement reports. This may increase the number of unnecessary measurements on this frequency.
- After UE blacklist management in intra-RAT ANR is enabled, intra-RAT ANR cannot take effect for specific UEs, which may increase the service drop rate of these UEs.
- The NG-RAN cell needs to broadcast SIB1 to support intra-RAT ANR for NG-RAN or inter-RAT ANR with NG-RAN for E-UTRAN. The **NR_CELL_SIB1_SW** option of the **NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt** parameter specifies whether an NG-RAN cell broadcasts SIB1.
 - If the **NR_CELL_SIB1_SW** option of the **NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt** parameter is selected for an NG-RAN cell, then the cell can broadcast SIB1, and intra-RAT ANR for NG-RAN or inter-RAT ANR with NG-RAN for E-UTRAN can take effect. This occupies RB resources in the cell and slightly decreases the peak rate of the cell.
 - If the **NR_CELL_SIB1_SW** option of the **NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt** parameter is deselected for an NG-RAN cell, then the cell cannot broadcast SIB1. This does not occupy RB resources in the cell or affect the peak rate of the cell. However, intra-RAT ANR for NG-RAN and inter-RAT ANR with NG-RAN for E-UTRAN do not take effect.

NOTE

In low-frequency TDD, the **NR_CELL_SIB1_SW** option of the **NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt** parameter applies only to NSA networking.

- After fast NR networking option query is enabled, the number of NR networking option queries increases. As a result, signaling resources are consumed due to repeated invalid requests.

- After restoration of handover-prohibited neighboring cells based on the handover success rate is enabled, the gNodeB periodically changes the value of **NRCellRelation.NoHoFlag** from a value indicating that handovers or redirections are prohibited to another value indicating that handovers or redirections are allowed. As a result, **Inter-SgNB PSCell Change Success Rate (CU)** in NSA networking as well as **Intra-Frequency Handover Out Success Rate (CU)** and **Inter-Frequency Handover Out Success Rate (CU)** in SA networking fluctuate.
- After enhanced automatic neighboring frequency configuration is enabled, the gNodeB delivers neighboring frequency measurements for selected UEs. In this case, RRC connection reconfiguration may increase the **Service Call Drop Rate (CU)**. In addition, measurement gaps may decrease the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** of the UEs selected for measurement.

The setting of the **NRCellFreqRelation.AutoNrFreqRelationRsrpThld** parameter has the following impacts:

- A larger value of this parameter results in a higher probability that enhanced automatic neighboring frequency configuration determines that the measured frequency provides weak or no coverage. This decreases the number of unnecessary measurements on the intra-RAT neighboring frequency, but increases the **Service Call Drop Rate (CU)** if the neighboring frequency provides weak coverage.
- A smaller value of this parameter results in a lower probability that enhanced automatic neighboring frequency configuration determines that the measured frequency provides weak or no coverage. This does not decrease the number of unnecessary measurements on the intra-RAT neighboring frequency.

4.1.3 Requirements

4.1.3.1 Licenses

NCL/NRT management based on the gNodeB blacklist is a basic function and has no license requirements. The following table lists the license requirements for other functions.

Feature ID	Feature Name	Model	Sales Unit
FOFD-021204	Automatic Neighbour Relation (ANR)	NR0S0AUTNE00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

None

4.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive PCI conflict detection based on intra-RAT ANR	INTRA_RAT_ANR_PCI_CONFL_DET_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>PCI Conflict Detection and Self-Optimization</i>	Proactive PCI conflict detection based on intra-RAT ANR is mutually exclusive with any of the following intra-RAT ANR functions: • Fast ANR • PCI-specific ANR

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	• The UE can perform reportCGI measurements only after entering the DRX sleep time. • The gNodeB automatically configures a long DRX cycle for UEs to read the NCGIs. The setting of the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter does not affect the configuration of the long DRX cycle initiated by intra-RAT ANR.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Not configuring DRX for UEs with specific RFSP indexes	NOT_CONFIG_DRX_SW option of the gNB RfSP Config.RfSPAlgoSwitch parameter	<i>Flexible User Steering</i>	- The UE can perform reportCGI measurements only after entering the DRX sleep time. - When the gNodeB triggers a UE to read the NCGIs, the UE automatically uses a long DRX cycle. The setting of the NOT_CONFIG_DRX_SW option of the gNB RfSP Config.RfSPAlgoSwitch parameter does not affect the configuration of the long DRX cycle initiated by intra-RAT ANR.
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTICARRIER_SHUTDOWN_SW or INTER_GNB_MULTICARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When automatic removal of neighboring cells in ANR takes effect, intelligent carrier shutdown does not take effect if the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	Capacity-layer cells: INTRA_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter (for intra-base-station multi-carrier scenarios)/ INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter (for inter-base-station multi-carrier scenarios) FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter Basic-layer cells: FAST_CARR_SHUT_SERV_ASSUR_SW option of the NRDUCellAlgoSwitch.PowerS	<i>Energy Conservation and Emission Reduction</i>	When automatic removal of neighboring cells in ANR takes effect, fast carrier shutdown does not take effect if the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells.

RAT	Function Name	Function Switch	Reference	Description
		<i>avingExtSwitc h</i> parameter		
Low- frequenc y TDD	Operator- specific NCI configura tion	gNBSharingM ode.MultipleN CISwitch	<i>Multi-Operator Sharing</i>	When an Xn interface has been set up between a gNodeB shared by multiple operators in a version earlier than V100R017C10 and a gNodeB in V100R017C10 with the gNBSharingMode.MultipleNciEnhanceSw parameter set to ON, the "NCL update triggered by messages transmitted over the Xn interface between gNodeBs" function of the former gNodeB requires that the Xn interface of the primary operator work properly. When an Xn interface has been set up between a non-shared gNodeB in a version earlier than V100R017C10 and a gNodeB in V100R017C10 with the gNBSharingMode.MultipleNciEnhanceSw parameter set to ON and the operator that owns the former gNodeB is a secondary operator of the latter gNodeB, the "NCL update triggered by messages transmitted over the Xn interface between gNodeBs" function of the former gNodeB does not take effect. NOTE It is recommended that the gNBSharingMode.MultipleNciEnhanceSw parameter be set to ON for all gNodeBs each shared by multiple operators only after the gNodeBs are upgraded to V100R017C10. This avoids impacts on the operator-specific NCI configuration function.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter NRCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the NRCellRelation.MlbHoFlag parameter is set to TRUE for the neighboring cell.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter NRCellMlb.MlbTriggerMode set to SPEC_EFFECT_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the NRCellRelation.MlbHoFlag parameter is set to TRUE for the neighboring cell.
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SWITCH option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter NRCellMlb.MlbTriggerMode set to SPECIFIED_5QI_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the NRCellRelation.MlbHoFlag parameter is set to TRUE for the neighboring cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQUENCY_CONNECTED_MLB_SWITCH option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter NRCellMlb.MlbTriggerMode set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	A neighboring cell added by ANR cannot be a candidate cell for MLB unless the NRCellRelation.MlbHoFlag parameter is set to TRUE for the neighboring cell.
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	NR_IF_HO_MEAS_QTY_PRED_FUN_SW option of the NRCellSmartMultiCarrier.NrMultiCarrierAlgoSwitch parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	After intra-RAT ANR is enabled, neighbor relationships are automatically added and removed. The base station periodically monitors the change in the proportion of CGIs of neighboring cells in a grid and determines whether to rebuild the grid model.
Low-frequency TDD	Intra-RAT PLMN ID management	None	<i>ANR</i>	The gNodeB checks whether the PLMN to which a neighboring cell belongs is blacklisted before adding the neighboring cell through intra-RAT ANR. If the PLMN is blacklisted, the gNodeB does not add the neighboring cell. If the PLMN is not blacklisted, the gNodeB adds the neighboring cell.
Low-frequency TDD	Cell-level MRO	MRO_SCENARIO_IDENT_SW option of the gNBMr.ScenarioIdentifySwitch parameter and the INTRA_FREQUENCY_MRO_SW option of the NRCellMrroSwitch parameter	<i>MRO</i>	If the NRCellRelation.CtrlMode parameter is set to MANUAL_MODE for a neighboring cell, cell-level intra-frequency MRO does not take effect for the neighboring cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	COVERAGE_BASSED_HO and REDIRECTION options of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i> <i>NSA Mobility Management</i>	If measurement-based inter-frequency handover is enabled and the NRCellFreqRelation.FreqIntrfrFlag parameter is set to INTRF or INTRF_SINR, the gNodeB delivers measurement configurations with the report type of reportCGI to the UE involving only the neighboring cells whose measured signal quality is higher than the value of the NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2 or NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2 parameter. If redirection is enabled and the NRCellFreqRelation.FreqIntrfrFlag parameter is set to INTRF or INTRF_SINR, the gNodeB delivers measurement configurations with the report type of reportCGI to the UE involving only the neighboring cells whose measured signal quality is higher than the value of the NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2 or NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2 parameter.
Low-frequency TDD	RF module deep dormancy	RRU.DORMANCYSW set to ON or ON_DEEP_SUPP	<i>Multi-RAT Coordinated Energy Saving</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, load-based RF module deep dormancy does not take effect.
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, RF module fast deep dormancy does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module super dormancy	RRU.DORMANCYSW set to ON_SUPER or ON_DEEP_SUPER	<i>Multi-RAT Coordinated Energy Saving</i>	When automatic removal of neighboring cells in ANR takes effect and the neighbor relationships with the basic-layer cells are automatically removed for capacity-layer cells, load-based RF module super dormancy does not take effect.
Low-frequency TDD	Inter-cell interference avoidance	INTER_CELL_INTERF_AVOID_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When UEs measure cell global identifiers (CGIs), inter-cell interference avoidance may not take effect due to failures to determine correlations. If this occurs, the benefits of inter-cell interference avoidance change.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.1.3.4 Others

In NSA networking, intra-RAT ANR requires that UEs support neighboring NG-RAN cell measurements of report type *reportCGI* (nr-CGI-Reporting-ENDC) in E-UTRA-NR dual connectivity (EN-DC). In SA networking, intra-RAT ANR requires that UEs support neighboring NG-RAN cell measurements of report type *reportCGI* (nr-CGI-

Reporting). For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V15.5.0.

3GPP TS 38.331 V15.4.0 and earlier versions do not define "nr-CGI-Reporting-ENDC". In NSA networking, UEs complying with 3GPP TS 38.331 V15.4.0 or earlier versions do not include "nr-CGI-Reporting-ENDC" in reports. Therefore, the gNodeB considers that the UEs do not support measurements of report type *reportCGI* and does not deliver measurement configurations carrying the report type of *reportCGI* to the UEs. As a result, intra-RAT ANR does not take effect. UEs complying with 3GPP TS 38.331 V15.5.0 or later versions are required to support intra-RAT ANR in NSA networking. Such UEs can include "nr-CGI-Reporting-ENDC" in reports.

4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

4.1.4.1.1 Data Preparation

[Table 4-5](#) and [Table 4-6](#) describe the parameters used for function activation and optimization, respectively.

Table 4-5 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
ANR Switch	NRCellAlgoSwitch.AnrSwitch	NR_NR_ANR_SW	Select this option to enable automatic detection and addition of missing neighboring cells, automatic NCL update, and automatic external-cell PLMN list configuration.
ANR Switch	NRCellAlgoSwitch.AnrSwitch	NR_NR_ANR_DEL_SW	Select this option to enable automatic removal of neighboring cells. Before enabling this function, ensure that the NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter is selected.
Self-Configuration on Switch	gNodeBAlgo.SelfConfigSwitch	INTRA_BS_NCELL_RMVB_PROHIBIT_SW	Select this option to enable base-station-level prohibition on intra-base-station neighbor relationship removal.

Parameter Name	Parameter ID	Option	Setting Notes
Self-Configurati on Switch	gNodeBAlgo.SelfConfigSwitch	NREXTERNALN CELL_SELF_CO NFIG_SW	Select this option to enable NCL update triggered by messages transmitted over the Xn interface between gNodeBs. Before enabling this function, ensure that the NR_NR_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter is selected for any cell served by the gNodeB.
Self-Configurati on Switch	gNodeBAlgo.SelfConfigSwitch	NRNTWOPNTN_ SELF_CONFIG_ POL_SW	Select this option to enable NCL update triggered by messages transmitted over the Xn interface between gNodeBs or external-cell PLMN list information update triggered by messages transmitted over the Xn interface between gNodeBs.
Self-Configurati on Switch	gNodeBAlgo.SelfConfigSwitch	NR_NTW_OPT N_FAST_QRY_S W	Select this option to enable fast NR networking option query when NCL update triggered by queries from the MAE-Access or external-cell PLMN list update triggered by queries from the MAE-Access is enabled.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_NR_ANR_A UTO_NO_HO_S W	Select this option to enable prohibition of handovers to neighboring cells based on the handover success rate.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_NR_ANR_O VER_DIST_DET ECT_SW	Select this option to enable automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_NR_FAST_A NR_SW	Select this option to enable fast ANR. Before enabling this function, ensure that the NR_NR_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter is selected.

Parameter Name	Parameter ID	Option	Setting Notes
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	PCI_SPECIFIC_N R_ANR_SW	Select this option to enable PCI-specific ANR. Before enabling this function, ensure that the NR_NR_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter is selected.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	BACK_TO_HPL MN_HO_ANR_SW	Select this option to enable compatibility of ANR with RFSP-specific inter-frequency handover back to the HPLMN.
ANR Indication	NRCCellFreqRelation.AnrInd	None	Use the recommended value. This parameter takes effect only when the cell-specific switches for automatic detection and addition of missing neighboring cells or automatic removal of neighboring cells are turned on.
Target Physical Cell ID	NRCCellNcgiMeasure.TargetPhysicalCellId	None	Configure a specific PCI as required to enable PCI-specific ANR.
Target SSB Freq Position Describe Method	NRCCellNcgiMeasure.TargetSsbDescribeMethod	None	Configure a frequency associated with a specific PCI as required to enable PCI-specific ANR.
Target SSB Frequency Position	NRCCellNcgiMeasure.TargetSsbFreqPos	None	
Self-Configuration Switch	gNodeBAlgoSelfConfigSwitch	NR_HO_SUCC_ RATE_INTVL_C FG_SW	Select this option in the case of measurement period modification when neighboring cell class label management, prohibition of handovers to neighboring cells based on the handover success rate, or automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
NCell Addition CGI Report Count Thld	NRCellAnr.NCellAddCgiRptCountThld	None	Use the recommended value to enable automatic addition of missing neighboring cells based on the number of CGI-Info reporting times. This function takes effect only when this parameter is set to a value other than 1 and the NR_NR_ANR_CTRL_MODE_HO_SW option of the NRCellAnr.AnrModeSwitch parameter is deselected.
ANR Algorithm Switch	NRCellAnr.AnrAlgoSwitch	SA_NR_NCELL_SUCC_RT_STAT_SW	Select this option when neighboring cell class label management is required in SA networking. Select this option when identification of scenarios with a handover preparation failure is required in prohibition of handovers to neighboring cells based on the handover success rate or automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells in SA networking.
Self-Configuration Switch	gNodeBAlgo.SelfConfigSwitch	NSA_NCELL_CLASS_LABEL_SW	Use the recommended value.
NSA DC Algorithm Switch	EnodebAlgoExtSwitch.NsaDcAlgoSwitch	PSCELL_RA_SUCCESS_MSG_FWD_SW	Use the recommended value.
Blacklist Control Extension Switch	gNBUECompatSw.BlacklistCtrlExtSwitch	NR_NR_ANR_SW_OFF	Select this option if intra-RAT ANR is required to be disabled for specific UEs.
Self-Configuration Switch	gNodeBAlgo.SelfConfigSwitch	NR_NCELL_DELETE_STAT_OPT_SW	Select this option if automatic removal of neighboring cells is enabled and measurement based on the actual number of times a neighbor cell is used is required.

Parameter Name	Parameter ID	Option	Setting Notes
ANR Algorithm Switch	NRCellAnr.AnrAlgoSwitch	NR_NR_ANR_NO_HO_AUTO_REES_SW	Select this option to enable restoration of handover-prohibited neighboring cells based on the handover success rate.
ANR Mode Switch	NRCellAnr.AnrModeSwitch	NR_NCELL_PLMN_RR_MODE_SW	Select this option if neighboring NG-RAN cells work in multi-operator sharing mode and broadcast PLMN IDs in rotation mode.
Self-Configuration Switch	gNodeBAlgo.SelfConfigSwitch	ANR_PLMN_SELECTION_OPT_SW	Select this option to enable PLMN selection optimization in automatic external-cell PLMN list addition.
ANR Switch	NRCellAlgoSwitch.AnrSwitch	AUTO_NR_FREQ_RELATION_SW	Select this option to enable automatic neighboring frequency configuration.
ANR Mode Switch	NRCellAnr.AnrModeSwitch	AUTO_NR_FREQ_RELATION_MEAS_SW	Select this option to enable enhanced automatic neighboring frequency configuration. Before enabling this function, ensure that the AUTO_NR_FREQ_RELATION_SW option of the NRCellAlgoSwitch.AnrSwitch parameter is selected.
Frequency Control Switch	NRCellFreqRelation.FrequencyCtrlSwitch	AUTO_NR_FREQ_REL_FORBID_SW	Select this option to enable forbidding automatic neighboring frequency configuration for a specific neighboring frequency.
Algorithm Compatibility Switch Extension	NRDUCellAlgoSwitch.AlgoCompatibilitySwitchExt	NR_CELL_SIB1_SW	Select this option to enable intra-RAT ANR for NG-RAN or inter-RAT ANR with NG-RAN for E-UTRAN in NSA networking.
Mobile Country Code	gNBBlacklist.Mcc	None	If a fake gNodeB is near a gNodeB, set this parameter based on the fake gNodeB information.

Parameter Name	Parameter ID	Option	Setting Notes
Mobile Network Code	gNBNRBlacklist.<i>Mnc</i>	None	If a fake gNodeB is near a gNodeB, set this parameter based on the fake gNodeB information.
gNodeB ID	gNBNRBlacklist.<i>gNodeBId</i>	None	If a fake gNodeB is near a gNodeB, set this parameter based on the fake gNodeB information.

Table 4-6 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Intra-RAT ANR RSRP Threshold	NRCellAnr.<i>IntraRatAnrRsrpThld</i>	None	Use the default value.
ANR Mode Switch	NRCellAnr.<i>AnrModeSwitch</i>	NR_NR_ANR_CTRL_MODE_SW	Use the default value. Alternatively, set this option based on the operator's policy.
ANR Mode Switch	NRCellAnr.<i>AnrModeSwitch</i>	NR_NR_ANR_CTRL_MODE_HO_SW	Use the default value. Alternatively, set this option based on the operator's policy.
Control Mode	NRCellRelation.<i>CtrlMode</i>	None	Use the default value.
Control Mode	NRExternalNCell.<i>CtrlMode</i>	None	Use the default value.
NR Handover Success Rate Statistic Interval	gNodeBAlgo.<i>NrHoSuccRateStatInterval</i>	None	Use the default value.
NR Neighbor Cell ReportCGI Number Thld	NRCellAnr.<i>NrNCellRptCgiNumThld</i>	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Over Distant Neighbor Cell Threshold	NRCellAnr.OverDistantNCellThld	None	Use the default value. This parameter applies only to the controlled mode.
Neighbor Cell Handover Success Rate Thld	NRCellAnr.NCellHoSuccRateThld	None	Use the default value.
Period for Delete NRT	NRCellAnr.PeriodForDelNrt	None	Use the default value. If you want redundant neighbor relationships to be removed quickly, set this parameter to a small value.
Auto Deleted Neighbor Cell Penalty Period	NRCellAnr.AutoDelNCellPenaltyPeriod	None	Use the default value.
Auto Deleted Neighbor Cell Penalty Number	NRCellAnr.AutoDelNCellPenaltyNum	None	Use the recommended value.
NR Neighbor Cell Handover Number Thld	NRCellAnr.NrNCellHoNumThld	None	Use the default value.
NR to NR No Handover Success Rate Thld	NRCellAnr.N2NNoHoSuccRateThld	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
NR to NR No Handover Consecutive Period Thld	NRCellAnr.N2N NoHoConsecPr dThld	None	Use the default value. It is recommended that this parameter be set to a value greater than or equal to 2 when the NR_NR_ANR_SW and NR_NR_ANR_AUTO_NO_HO_SW options of the NRCellAlgoSwitch.AnrSwitch parameter, the NSA_NCELL_CLASS_LABEL_SW option of the gNodeBAlgo.SelfConfigSwitch parameter (only in NSA networking), and the PSCell_RA_SUCCESS_MSG_FWD_SW option of the EnodebAlgoExtSwitch.NsaDcAlgoSwitch parameter (only in NSA networking) on the LTE side are all selected.
NR Neighbor Cell No HO Restore Interval	NRCellAnr.NrN CellNoHoResIntvl	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
No Handover Flag	NRCellRelation.NoHoFlag	None	If the neighboring cell has high handover success rates for both SA UEs and NSA UEs, setting the NRCellRelation.NoHoFlag parameter to PERMIT_HO reduces the number of service drops caused by handover or redirection failures; whereas setting this parameter to FORBID_HO increases the service drop rate caused by handover or redirection failures. If the neighboring cell has a high handover success rate for SA UEs but a low handover success rate for NSA UEs, setting the NRCellRelation.NoHoFlag parameter to SA_PERMIT_NSA_FORBID_HO reduces the number of service drops caused by handover failures of SA UEs. In addition, this setting prevents handover attempts and failures regarding this neighboring cell for NSA UEs, increasing the handover success rate of NSA UEs in the local cell. If the neighboring cell has a low handover success rate for SA UEs but a high handover success rate for NSA UEs, setting the NRCellRelation.NoHoFlag parameter to SA_FORBID_NSA_PERMIT_HO prevents handover attempts and failures regarding the neighboring cell for SA UEs, increasing the handover success rate of SA UEs in a cell. In addition, this setting reduces the number of service drops caused by

Parameter Name	Parameter ID	Option	Setting Notes
			handover failures of NSA UEs. If the neighboring cell has low handover success rates for both SA UEs and NSA UEs, setting the NRCellRelation.NoHoFlag parameter to PERMIT_HO reduces the handover success rate of the local cell; whereas setting this parameter to FORBID_HO prevents handover attempts and failures regarding this neighboring cell, increasing the handover success rate of the local cell.
No Removal Flag	NRCellRelation.NoRmvFlag	None	Use the default value.
Measurement Activation Flag	NRCellFreqRelation.MeasActvFlag	None	Use the default value.
Auto NR Frequency Relation RSRP Threshold	NRCellFreqRelation.AutoNrFreqRelationRsrpThld	None	Use the default value if enhanced automatic neighboring frequency configuration is enabled.

NOTE

Assume that both SA networking and NSA networking are implemented on the network and base stations in the two networking options are managed by different MAE-Access systems. Inter-networking-option handovers (handovers from an SA cell to an NSA cell or from the NSA cell to the SA cell) may fail. To prevent this, the following must be performed after ANR is enabled: (1) Run the **LST NREXTERNALNCELL** and **LST NREXTERNALNCELLPLMN** commands to check information about external NG-RAN cells. (2) For external NG-RAN cells with the **NRExternalNCell.NrNetworkingOption** and **NRExternalNCellPlmn.NrNetworkingOption** parameters set to **UNLIMITED**, check the actual networking options of these cells. (3) Run the **MOD NREXTERNALNCELL** and **MOD NREXTERNALNCELLPLMN** commands to change the settings of the **NRExternalNCell.NrNetworkingOption** and **NRExternalNCellPlmn.NrNetworkingOption** parameters to **SA**, **NSA**, or **SA_NSA** accordingly.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling automatic detection and addition of missing neighboring cells, automatic NCL update, and
automatic external-cell PLMN list configuration for a specific cell
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1;
//Enabling fast ANR for a specific cell
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1&NR_NR_FAST_ANR_SW-1;
//Enabling PCI-specific ANR for a specific cell
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1&PCI_SPECIFIC_NR_ANR_SW-1;
//Configuring a specified PCI when PCI-specific ANR is enabled
ADD NRCELLNCGIMEAS: NrCellId=0, TargetSsbDescMethod=SSB_DESC_TYPE_GSCN,
TargetSsbFreqPos=6600, TargetPhysicalCellId=1;
//Enabling automatic neighboring cell class label adjustment on both the NR and LTE sides in NSA
networking
MOD GNODEBALGO: SelfConfigSwitch=NSA_NCELL_CLASS_LABEL_SW-1;
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=PCCELL_RA_SUCCESS_MSG_FWD_SW-1;
//Enabling UE blacklist management in intra-RAT ANR when intra-RAT ANR is required to be disabled for
specific UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=NR_NR_ANR_SW_OFF-1;
//Enabling automatic removal of neighboring cells for a specific cell
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-1&NR_NR_ANR_DEL_SW-1;
//Enabling automatic detection and addition of missing neighboring cells and automatic removal of
neighboring cells for a specific frequency
MOD NRCELLFREQUELATION: NrCellId=0, SsbFreqPos=6600, AnrInd=ALLOWED;
//Enabling compatibility of ANR with RFSP-specific inter-frequency handover back to the HPLMN
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=BACK_TO_HPLMN_HO_ANR_SW-1;
//Enabling automatic addition of missing neighboring cells based on the number of CGI-Info reporting
times when uncontrolled mode is used
MOD NRCELLANR: NrCellId=0, NCellAddCgiRptCountThld=3;
//Selecting the NR_HO_SUCC_RATE_INTVL_CFG_SW option in the case of measurement period modification
when neighboring cell class label management, prohibition of handovers to neighboring cells based on the
handover success rate, or automatic optimization of neighbor relationship attributes based on identification
of over-distant neighboring cells is enabled
MOD GNODEBALGO: SelfConfigSwitch=NR_HO_SUCC_RATE_INTVL_CFG_SW-1;
//((SA networking) Enabling optimized measurement of the success rate of handovers to neighboring NR
cells when neighboring cell class label management is required, and identification of scenarios with a
handover preparation failure is required in prohibition of handovers to neighboring cells based on the
handover success rate or automatic optimization of neighbor relationship attributes based on identification
of over-distant neighboring cells
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=SA_NR_NCELL_SUCC_RT_STAT_SW-1;
//Enabling optimized measurement for automatic removal of neighboring NR cells when automatic removal
of neighboring cells is enabled and measurement based on the actual number of times a neighbor cell is
used is required
MOD GNODEBALGO: SelfConfigSwitch=NR_NCELL_DEL_STAT_OPT_SW-1;
//Enabling base-station-level prohibition on intra-base-station neighbor relationship removal
MOD GNODEBALGO: SelfConfigSwitch=INTRA_BS_NCELL_RMV_PROHIBIT_SW-1;
//Enabling NCL update triggered by messages transmitted over the Xn interface between gNodeBs
MOD GNODEBALGO: SelfConfigSwitch=NREXTERNALNCELL_SELF_CONFIG_SW-1;
//Enabling the function of directly updating the networking option in the NCL based on Xn messages or the
function of directly updating the networking option in the external-cell PLMN list information based on Xn
messages
MOD GNODEBALGO: SelfConfigSwitch=NRNTWOPTN_SELF_CONFIG_POL_SW-1;
//Enabling fast NR networking option query as required when NCL update triggered by queries from the
MAE-Access or external-cell PLMN list update triggered by queries from the MAE-Access is enabled
MOD GNODEBALGO: SelfConfigSwitch=NR_NTW_OPTN_FAST_QRY_SW-1;
//Enabling prohibition of handovers to neighboring cells based on the handover success rate
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_AUTO_NO_HO_SW-1;
```

```
//Enabling restoration of handover-prohibited neighboring cells based on the handover success rate
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NR_NR_ANR_NO_HO_AUTO_RES_SW-1;
//Enabling automatic optimization of neighbor relationship attributes based on identification of over-
distant neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_OVER_DIST_DETECT_SW-1;
//Selecting the NR_NCELL_PLMN_RR_MODE_SW option if the neighboring NG-RAN cells work in multi-
operator sharing mode and broadcast PLMN IDs in rotation mode
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_NCELL_PLMN_RR_MODE_SW-1;
//Enabling PLMN selection optimization as required when automatic external-cell PLMN list addition is
enabled
MOD GNODEBALGO: SelfConfigSwitch=ANR_PLMN_SEL_OPT_SW-1;
//Enabling automatic neighboring frequency configuration
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=AUTO_NR_FREQ_RELATION_SW-1;
//Enabling enhanced automatic neighboring frequency configuration
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=AUTO_NR_FREQ_RELATION_MEAS_SW-1;
//Selecting the AUTO_NR_FREQ_REL_FORBID_SW option when forbidding automatic neighboring frequency
configuration for a specific neighboring frequency is required
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6600,
FrequencyCtrlSwitch=AUTO_NR_FREQ_REL_FORBID_SW-1;
//Turning on NR_CELL_SIB1_SW to support intra-RAT ANR for NG-RAN or inter-RAT ANR with NG-RAN for
E-UTRAN in NSA networking
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AlgoCompatibilitySwitchExt=NR_CELL_SIB1_SW-1;
//Adding a fake gNodeB to the blacklist if it is near the local gNodeB
ADD GNBNRBLACKLIST: gNodeBId=1, Mcc="302", Mnc="220";
```

Optimization Command Examples

```
//Configuring an RSRP threshold when PCI-specific ANR is enabled
MOD NRCELLANR: NrCellId=0, IntraRatAnrRsrpThld=-102;
//Configuring the control mode of neighbor relationships
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CtrlMode=AUTO_MODE;
//Configuring the control mode of external cells
MOD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, Tac=1, SsbFreqPos=6600,
CtrlMode=AUTO_MODE;
//((Recommended) Disabling the controlled mode for intra-RAT ANR based on operator's policies
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_NR_ANR_CTRL_MODE_SW-0;
//((Recommended) Prohibiting handovers to unconfirmed neighboring NG-RAN cells for intra-RAT ANR
based on operator's policies when controlled mode is used
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_NR_ANR_CTRL_MODE_HO_SW-0;
//Modifying the threshold for determining over-distant neighboring cells in controlled mode as required
MOD NRCELLANR: NrCellId=0, OverDistantNCellThld=5000;
//Modifying the neighboring cell handover success rate threshold and the threshold expressed as the
number of reportCGI measurements on neighboring NR cells for neighboring cell class label management
as required
MOD NRCELLANR: NrCellId=0, NCellHoSuccRateThld=80, NrNCellRptCgiNumThld=100;
//Modifying the measurement period for neighboring cell class label management, prohibition of handovers
to neighboring cells based on the handover success rate, or automatic optimization of neighbor relationship
attributes based on identification of over-distant neighboring cells as required
MOD GNODEBALGO: NrHoSuccRateStatInterval=24;
//Modifying the threshold expressed as the number of handovers to a neighboring NR cell and the
thresholds expressed as the success rate and number of consecutive periods for deciding whether to enable
NR-to-NR handover prohibition as required
MOD NRCELLANR: NrCellId=0, NrNCellHoNumThld=100, N2NNoHoSuccRateThld=0,
N2NNoHoConsecPrdThld=1;
//Allowing handovers or redirections to optimized neighboring cells
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, NoHoFlag=PERMIT_HO;
//Enabling removal of neighbor relationships through ANR
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
NoRmvFlag=PERMIT_ANR_RMV;
//Configuring the period for automatic removal of redundant neighbor relationships as required
MOD NRCELLANR: NrCellId=0, PeriodForDelNrt=7;
//Modifying the penalty on automatically removed neighbor relationships as required
MOD NRCELLANR: NrCellId=0, AutoDelNCellPenaltyPeriod=10080, AutoDelNCellPenaltyNum=100;
//Setting the NrNCellNoHoResIntvl parameter when the interval at which the neighboring NR cells marked
as handover-prohibited are restored needs to be changed
MOD NRCELLANR: NrCellId=0, NrNCellNoHoResIntvl=30;
//Configuring the measurement activation flag of an intra-RAT neighboring frequency when automatic
neighboring frequency configuration is enabled
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6600, MeasActvFlag=ACTIVE;
```

```
//Modifying the RSRP threshold of enhanced automatic neighboring frequency configuration as required  
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6600, AutoNrFreqRelationRsrpThld=-140;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling PLMN selection optimization  
MOD GNODEBALGO: SelfConfigSwitch=ANR_PLMN_SEL_OPT_SW-0;  
//Deselecting the NR_NCELL_PLMN_RR_MODE_SW option  
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_NCELL_PLMN_RR_MODE_SW-0;  
//Disabling fast NR networking option query  
MOD GNODEBALGO: SelfConfigSwitch=NR_NTW_OPTN_FAST_QRY_SW-0;  
//Disabling the function of directly updating the networking option in the NCL based on Xn messages or  
the function of directly updating the networking option in the external-cell PLMN list information based on  
Xn messages  
MOD GNODEBALGO: SelfConfigSwitch=NRNTWOPTN_SELF_CONFIG_POL_SW-0;  
//Disabling NCL update triggered by messages transmitted over the Xn interface between gNodeBs  
MOD GNODEBALGO: SelfConfigSwitch=NREXTERNALNCELL_SELF_CONFIG_SW-0;  
//Deselecting the NR_HO_SUCC_RATE_INTVL_CFG_SW option  
MOD GNODEBALGO: SelfConfigSwitch=NR_HO_SUCC_RATE_INTVL_CFG_SW-0;  
//Disabling automatic neighboring cell class label adjustment on both the NR and LTE sides in NSA  
networking  
MOD GNODEBALGO: SelfConfigSwitch=NSA_NCELL_CLASS_LABEL_SW-0;  
MOD ENODEBALGOEXTSWITCH: NsaDcAlgoSwitch=PSCELL_RA_SUCCESS_MSG_FWD_SW-0;  
//Disabling UE blacklist management in intra-RAT ANR  
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,  
BlacklistCtrlExtSwitch=NR_NR_ANR_SW_OFF-0;  
//Disabling optimized measurement for automatic removal of neighboring NR cells  
MOD GNODEBALGO: SelfConfigSwitch=NR_NCELL_DEL_STAT_OPT_SW-0;  
//Disabling optimized measurement of the success rate of handovers to neighboring NR cells in SA  
networking  
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=SA_NR_NCELL_SUCC_RT_STAT_SW-0;  
//Disabling automatic addition of missing neighboring cells based on the number of CGI-Info reporting  
times when uncontrolled mode is used  
MOD NRCELLANR: NrCellId=0, NCellAddCgiRptCountThld=1;  
//Disabling fast ANR for a specific cell  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_FAST_ANR_SW-0;  
//Disabling PCI-specific ANR for a specific cell  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=PCI_SPECIFIC_NR_ANR_SW-0;  
//Disabling base-station-level prohibition on intra-base-station neighbor relationship removal  
MOD GNODEBALGO: SelfConfigSwitch=INTRA_BS_NCELL_RMV_PROHIBIT_SW-0;  
//Disabling automatic removal of neighboring cells for a specific cell  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_DEL_SW-0;  
//Disabling compatibility of ANR with RFSP-specific inter-frequency handover back to the HPLMN  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=BACK_TO_HPLMN_HO_ANR_SW-0;  
//Disabling automatic detection and addition of missing neighboring cells, automatic NCL update, and  
automatic external-cell PLMN list configuration for a specific cell  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_SW-0;  
//Disabling automatic detection and addition of missing neighboring cells or automatic removal of  
neighboring cells for a specific frequency  
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6600, AnrInd=NOT_ALLOWED;  
//Disabling restoration of handover-prohibited neighboring cells based on the handover success rate  
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=NR_NR_ANR_NO_HO_AUTO_RES_SW-0;  
//Disabling prohibition of handovers to neighboring cells based on the handover success rate  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_AUTO_NO_HO_SW-0;  
//Disabling automatic optimization of neighbor relationship attributes based on identification of over-  
distant neighboring cells  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_NR_ANR_OVER_DIST_DETECT_SW-0;  
//Disabling forbidding automatic neighboring frequency configuration  
MOD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=6600,  
FrequencyCtrlSwitch=AUTO_NR_FREQ_REL_FORBID_SW-0;  
//Disabling enhanced automatic neighboring frequency configuration  
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=AUTO_NR_FREQ_RELATION_MEAS_SW-0;  
//Disabling automatic neighboring frequency configuration  
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=AUTO_NR_FREQ_RELATION_SW-0;
```

```
//Turning off NR_CELL_SIB1_SW in NSA networking  
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AlgoCompatibilitySwitchExt=NR_CELL_SIB1_SW-0;  
//Removing a gNodeB from the blacklist  
RMV GNBNRBLACKLIST: gNodeBId=1, Mcc="302", Mnc="220";
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

Automatic Detection and Addition of Missing Neighboring Cells/Automatic NCL Update/Automatic External-Cell PLMN List Configuration/Automatic Neighboring Frequency Configuration

To check whether the functions have taken effect, query self-organizing network (SON) logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Add Neighboring Cell**, **Add External Cell**, **Update External Cell**, **Add External Cell PLMN List**, **Delete External Cell PLMN List**, **Update External Cell PLMN List**, or **Update Neighboring Frequency Measurement Configuration**. If any of the events is logged, these functions have taken effect.

----End

Automatic Removal of Neighboring Cells

To check whether the function has taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Delete Neighboring Cell** or **Delete External Cell**. If either event is logged, automatic removal of neighboring cells has taken effect.

----End

Prohibition of Handovers to Neighboring Cells Based on the Handover Success Rate

To check whether the functions have taken effect, query self-organizing network (SON) logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Set NoHo Flag**. If the event is logged and the value of **SaHoSuccTimes/SaHoTimes** or **NsaHoSuccTimes/NsaHoTimes** is less than or equal to that of **NRCellAnr.N2NNoHoSuccRateThld** in **Event Description**, prohibition of handovers to neighboring cells based on the handover success rate has taken effect.

----End

Restoration of Handover-Prohibited Neighboring Cells Based on the Handover Success Rate

To check whether the functions have taken effect, query self-organizing network (SON) logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Auto Restore NoHo Flag**. If the event is logged, restoration of handover-prohibited neighboring cells based on the handover success rate has taken effect.

----End

Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells

To check whether the function has taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.

Step 3 After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.

Step 4 In the **Event Name** area, select **Set NoHo Flag**. If the event is logged and **CellRachRadiusSum** and **InterCellDistance** are not **N/A** in **Event Description**, automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells has taken effect.

----End

Optimization Mode

The following uses neighboring cell addition as an example to describe activation verification.

- Prerequisites:
 - a. Cell A does not exist in the NRT of cell B.
 - b. Automatic detection and addition of missing neighboring cells is enabled.
- Expected result:
 - The uncontrolled mode is used if the **LST NRCELLRELATION** command output shows that cell A exists in the NRT of cell B.
 - The controlled mode is used if the **LST NRCELLRELATION** command output shows that cell A still does not exist in the NRT of cell B and the **Neighboring Cell Relationship to Be Handled** area on the **Optimization Advice** tab page of the MAE-Access shows that cell A needs to be added as a neighboring cell of cell B. (To open the **Optimization Advice** tab page, choose **SON > NR ANR**.)

4.1.4.3 Network Monitoring

To monitor the running status about intra-RAT ANR, query SON logs on the MAE-Access. For details about how to query SON logs on the MAE-Access, see [4.1.4.2 Activation Verification](#).

Use the KPIs in [4.1.2.2 Impacts](#) to monitor changes in the service drop rate and average UE throughput after enhanced automatic neighboring frequency configuration is enabled.

4.2 Intra-RAT PLMN ID Management

4.2.1 Principles

With PLMN ID management, the gNodeB checks the PLMN IDs carried in the *CGI-Info* IE reported from UEs and performs differentiated management.

PLMN ID management applies in either of the following scenarios:

- The gNodeB is on the inter-operator network borders.
In this scenario, the gNodeB is located adjacent to areas covered by networks of different operators and UEs served by the gNodeB cannot be handed over

or redirected to adjacent cells owned by other operators. If such cells are measured by UEs and the gNodeB does not add these cells to the NCL or NRT, UEs repeatedly read the CGIs of these cells, affecting data transmission. However, if the gNodeB adds these cells to the NCL and NRT, handovers or redirections will fail.

This function enables the gNodeB to add these cells to the NCL and NRT, and prohibits UEs served by the gNodeB from being handed over or redirected to these cells.

- Abnormal UEs report invalid PLMN IDs of neighboring cells.

In this scenario, the measurement results reported from UEs are incorrect. If the gNodeB adds incorrect neighboring cell information to the NCL and NRT, the NCL and NRT contain incorrect configurations. If the gNodeB selects the target cell that is incorrectly configured in the NCL and NRT, handovers or redirections will fail.

This function prohibits incorrect neighboring cell information from being added to the NCL and NRT, ensuring correct configurations in the NCL and NRT.

The **gNBCellPlmnList** MO is used to configure PLMN ID attributes. For each PLMN ID, the following parameters need to be configured:

gNBCellPlmnList.Mcc, **gNBCellPlmnList.Mnc**, **gNBCellPlmnList.RatType**, and **gNBCellPlmnList.PlmnListType**. The parameters are defined as follows:

- The **gNBCellPlmnList.Mcc** and **gNBCellPlmnList.Mnc** parameters indicate the MCC and MNC, respectively. The MCC and MNC form a PLMN ID.
- The **gNBCellPlmnList.RatType** parameter indicates the RAT of the PLMN to which a neighboring cell belongs. It can be set to **NR** or **EUTRAN**.
- The **gNBCellPlmnList.PlmnListType** parameter indicates the type of a PLMN. It can be set to **BLACKLIST**, **GREYLIST**, or **WHITELIST**. Up to two PLMN types can be configured for neighboring cells of a given RAT.
 - If this parameter is set to **BLACKLIST** for a PLMN, neighboring cells or external cells of the PLMN cannot be added through ANR.
 - If this parameter is set to **GREYLIST** for a PLMN, neighboring cells or external cells of the PLMN can be added through ANR but UEs served by the gNodeB cannot be handed over or redirected to cells of the greylisted PLMN.
 - If this parameter is set to **WHITELIST** for a PLMN, neighboring cells or external cells of the PLMN can be added through ANR and UEs served by the gNodeB can be handed over or redirected to cells of the whitelisted PLMN.

For the PLMN to which the gNodeB belongs (configured in the **gNBOperator** MO), the following principles apply:

- If the **ADD GNBCELLPLMNLIST** command is not executed to add the PLMN to the whitelist for NR, the PLMN is considered to be in the whitelist for NR by default.
- If the **ADD GNBCELLPLMNLIST** command is executed to add the PLMN to the whitelist for NR, the PLMN can only be whitelisted. Specifically, if **gNBCellPlmnList.RatType** is set to **NR**, **gNBCellPlmnList.PlmnListType** can only be set to **WHITELIST**.

Therefore, the PLMN configured in the **gNBOperator** MO is considered to be in the whitelist for NR.

 NOTE

The preceding principles are inapplicable to inter-RAT PLMN ID management. For details about how inter-RAT PLMN ID management works, see [5.2.1 Principles](#).

Based on the preceding principles, intra-RAT PLMN ID management works as follows:

- If two PLMN types are configured:
 - If no type is configured for the PLMN to which the gNodeB belongs, the PLMN is considered whitelisted by default.
 - If no type is configured for a PLMN to which the gNodeB does not belong, the PLMN is considered to have the third type by default.
- If only one PLMN type is configured:
 - If only a PLMN blacklist or greylist is configured, the PLMN for which no type is configured is considered whitelisted by default, regardless of whether the gNodeB belongs to the PLMN.
 - If only a PLMN whitelist is configured:
 - If no type is configured for the PLMN to which the gNodeB belongs, the PLMN is considered whitelisted by default.
 - If no type is configured for a PLMN to which the gNodeB does not belong, the PLMN is considered greylisted by default.
- If no PLMN types are configured, the PLMN for which no type is configured is considered whitelisted by default, regardless of whether the gNodeB belongs to the PLMN.

In multi-operator sharing scenarios, a cell is configured with multiple PLMN IDs. If a reported cell is configured with a blacklisted PLMN ID, the gNodeB regards that the measurement report is invalid. It will not add the cell to the NCL or NRT, or initiate handovers, redirections, or an NR SCell change to this cell.

4.2.2 Network Analysis

4.2.2.1 Benefits

This function delivers the following scenario-specific gains:

- Scenario where the gNodeB is on the border of different operators
In this scenario, the gNodeB is located adjacent to areas covered by networks of different operators and UEs served by the gNodeB cannot be handed over or redirected to adjacent cells owned by the operators. This function enables the gNodeB to add these cells to the NCL and NRT, and prohibits UEs served by the gNodeB from being handed over or redirected to these cells. In this way, data transmissions are not affected by repeated reading of the CGIs of these cells.
- Scenario where abnormal UEs report invalid PLMN IDs of neighboring cells
In this scenario, the measurement results reported from UEs are incorrect. This function prohibits incorrect neighboring cell information from being

added to the NCL and NRT, ensuring correct configurations in the NCL and NRT. In this way, this function prevents handover failures caused by selection of the target cells that are incorrectly configured in the NCL and NRT.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.2.3 Requirements

4.2.3.1 Licenses

None

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

None

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	The gNodeB checks whether the PLMN to which a neighboring cell belongs is blacklisted before adding the neighboring cell through intra-RAT ANR. If the PLMN is blacklisted, the gNodeB does not add the neighboring cell. If the PLMN is not blacklisted, the gNodeB adds the neighboring cell.

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Others

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

[Table 4-7](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 4-7 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBNCeIlPlmnList.Mcc	Set this parameter to the correct MCC.
Mobile Network Code	gNBNCeIlPlmnList.Mnc	Set this parameter to the correct MNC.
RAT Type	gNBNCeIlPlmnList.RatType	Set this parameter to NR .

Parameter Name	Parameter ID	Setting Notes
PLMN List Type	gNBNCeLLPlmnList.PlmnListType	If the gNodeB is on the border of different operators, either of the following configuration is applied: - Set this parameter to GREYLIST for the PLMNs of the operators that own the neighboring cells to which UEs cannot be handed over or redirected. - Set this parameter to WHITELIST for the PLMNs of the operators that own the neighboring cells to which UEs can be handed over or redirected. With this setting, the PLMNs of the operators that own the neighboring cells to which UEs cannot be handed over or redirected are considered greylisted. If abnormal UEs report invalid PLMN IDs of neighboring cells, set this parameter to BLACKLIST for the invalid PLMNs.

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to [4.2.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling intra-RAT PLMN ID management  
ADD GNBNCeLLPLMNLIST: RatType=NR, Mcc="302", Mnc="220", PlmnListType=WHITELIST;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling intra-RAT PLMN ID management  
RMV GNBNCeLLPLMNLIST: RatType=NR, Mcc="302", Mnc="220";
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Prerequisites

The **ADD GNBCELLPLMNLIST** command is executed with **gNBCellPlmnList.PlmnListType** set to **BLACKLIST** to blacklist the PLMN to which a neighboring cell belongs.

Expected Result

When the *CGI-Info* IE in the measurement report with type of *reportCGI* sent from a UE contains the information of the cell for which **gNBCellPlmnList.PlmnListType** is set to **BLACKLIST**, the gNodeB does not add the cell to the NCL or NRT.

4.2.4.3 Network Monitoring

None

5 Inter-RAT ANR Management

5.1 Inter-RAT ANR

5.1.1 Principles

Inter-RAT ANR enables automatic management of inter-RAT (between NR and E-UTRAN) NCLs and inter-RAT NRTs (NCL and NRT for short). [Table 5-1](#) lists support for subfunctions of inter-RAT ANR depending on whether neighboring E-UTRAN cells are shared among multiple operators.

Table 5-1 Subfunctions of inter-RAT ANR

Subfunction	Scenarios with Multi-Operator Sharing	Scenarios Without Multi-Operator Sharing
5.1.1.1 Optimization Mode	Supported	Supported
5.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells	Supported	Supported
5.1.1.3 Automatic Removal of Neighboring Cells	Supported	Supported
5.1.1.4 Automatic NCL Update	Supported	Supported
5.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate	Supported	Supported

Subfunction	Scenarios with Multi-Operator Sharing	Scenarios Without Multi-Operator Sharing
5.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells	Supported	Supported
5.1.1.7 Automatic External-Cell PLMN List Configuration	Supported	N/A
5.1.1.8 Automatic Neighboring Frequency Configuration	Supported	Supported
5.1.1.9 NCL/NRT Management Based on the eNodeB Blacklist	Supported	Supported

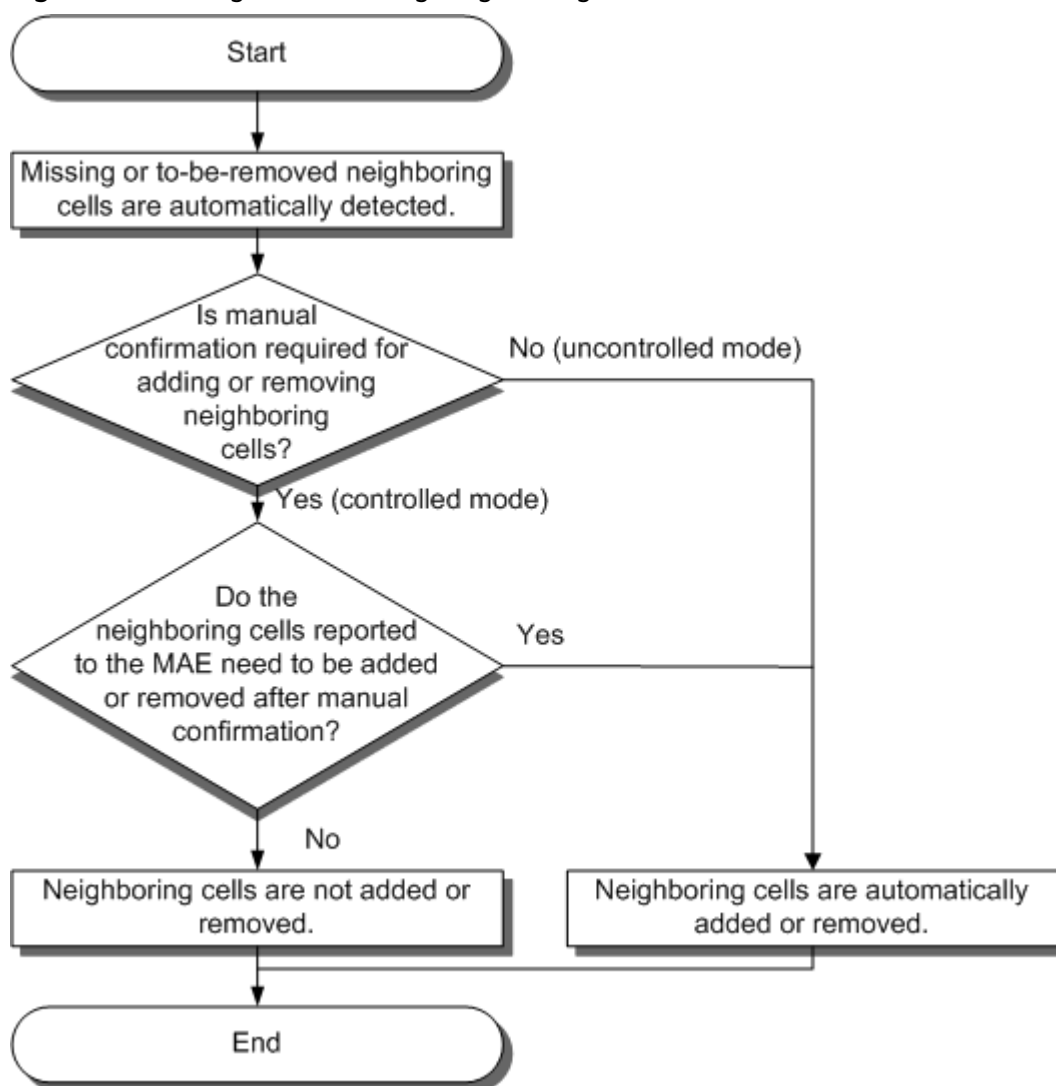
In NSA networking, when an NG-RAN cell serves as a PSCell, it involves only PSCell change between NG-RAN cells and does not involve mobility management from the NG-RAN cell to E-UTRAN cells. Therefore, inter-RAT ANR is involved only in SA networking.

Inter-RAT ANR is controlled by cell-level parameters and takes effect on a per cell basis. The following describes the principles.

5.1.1.1 Optimization Mode

The optimization mode is classified into uncontrolled mode and controlled mode, depending on whether manual confirmation is required in neighboring cell addition and removal. [Figure 5-1](#) shows how neighboring cells are added and removed in each mode. The **NR_EUTRAN_ANR_CTRL_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter specifies the optimization mode. When this option is selected, controlled mode is used. When this option is deselected, uncontrolled mode is used.

Figure 5-1 Adding and removing neighboring cells



Uncontrolled Mode

In uncontrolled mode, the gNodeB automatically adds or removes neighboring E-UTRAN cells once missing or to-be-removed neighboring E-UTRAN cells are detected.

Controlled Mode

In controlled mode, the gNodeB reports optimization advice on neighboring cells to the MAE-Access once missing or to-be-removed neighboring E-UTRAN cells are detected. After manual confirmation on the MAE-Access, the gNodeB adds or removes neighboring E-UTRAN cells. For details about operations on the MAE-Access, see descriptions of NR ANR in *SON Management in MAE Product Documentation (EulerOS, TaiShan)*.

The gNodeB periodically reports optimization advice on neighboring cells to the MAE-Access every night. Before the optimization advice is manually confirmed on the MAE-Access, the gNodeB temporarily stores the detected missing neighboring cells in its memory to reduce reportCGI measurements on the missing neighboring

cells. When detecting the PCIs of the missing neighboring cells again, the gNodeB does not deliver reportCGI measurement configurations of these cells to UEs. Instead, the number of times that these cells are measured is incremented by one in the gNodeB memory. Therefore, if the E-UTRAN cell global identifiers (ECGIs) of the missing neighboring cells change but their PCIs remain unchanged, the ECGIs of the missing neighboring cells in the optimization advice reported by the gNodeB are still the original ones. The ECGI consists of the PLMN ID, eNodeB ID, and cell ID. The PLMN ID consists of the MCC and MNC.

The **NR_EUTRAN_ANR_CTRL_MODE_HO_SW** option of the **NRCellAnr.AnrModeSwitch** parameter determines whether to allow handovers or redirections to unconfirmed neighboring E-UTRAN cells in controlled mode.

- When this option is selected, the gNodeB can initiate handovers or redirections to neighboring E-UTRAN cells unconfirmed on the MAE-Access based on the ECGIs of the neighboring cells reported from UEs.
- When this option is deselected, handovers and redirections to neighboring E-UTRAN cells unconfirmed on the MAE-Access are prohibited.

 NOTE

EPS fallback is of high priority and is not controlled by this option. Handovers and redirections to the unconfirmed neighboring E-UTRAN cells are allowed in this scenario.

In controlled mode, when the MAE-Access displays the optimization advice for adding a neighbor relationship, it also displays a message indicating whether the neighboring cell is an over-distant neighboring cell. The **NRCellAnr.OverDistantNCellThld** parameter is used to determine whether the neighboring cell is an over-distant neighboring cell in the preceding scenario.

- When the **NRCellAnr.OverDistantNCellThld** parameter is not set to **0**, an over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the parameter value.
- When the **NRCellAnr.OverDistantNCellThld** parameter is set to **0**, an over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the sum of the access radii of the two cells.

 NOTE

The MAE-Access calculates the distance between the local cell and a neighboring cell based on the longitude and latitude information contained in base station engineering parameters. For details about engineering parameters, see the description of setting NR-LTE overshoot query in *SON Management in MAE Product Documentation (EulerOS, TaiShan)*. If a multi-RRU cell has multiple latitudes and longitudes, the minimum distance is used for determination.

5.1.1.2 Automatic Detection and Addition of Missing Neighboring Cells

Automatic detection and addition of missing neighboring cells can be triggered by event-triggered measurements or periodic measurements.

Automatic detection and addition of missing neighboring cells triggered by event-triggered measurements is enabled when the **NR_EUTRAN_ANR_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected.

Automatic detection and addition of missing neighboring cells can be triggered by coverage-based inter-RAT handover measurements. The gNodeB detects missing

neighboring cells based on information included in inter-RAT measurement reports. Coverage-based inter-RAT handover measurements are measurements related to inter-RAT interoperability. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Automatic detection and addition of missing neighboring cells triggered by periodic measurements can detect and add missing neighboring cells fast. This function is enabled when both the **NR_EUTRAN_ANR_SW** and **NR_EUTRAN_FAST_ANR_SW** options of the **NRCellAlgoSwitch.AnrSwitch** parameter are selected. After fast ANR is enabled, the gNodeB selects some UEs in RRC_CONNECTED mode in a serving cell and sends inter-RAT measurement configurations to the UEs. The gNodeB detects missing neighboring cells based on the periodic measurement reports sent from these UEs. The UEs perform the measurements for four times and send measurement reports at an interval of 5120 ms.

After fast ANR is enabled, the gNodeB randomly selects UEs that meet the following conditions and sends measurement configurations to them at an interval of 30s. A maximum of five UEs in the serving cell can be selected for measurements at a time.

- The UE is in RRC_CONNECTED mode and a PDU session has been set up for the UE.
- The UE supports E-UTRAN neighboring cell measurements with the report type of *reportCGI* (eutra-CGI-Reporting).
- The UE is only running one or more of the following types of non-GBR services, except for FWA Internet, FWA video, and FWA private line services (services for which the **gNBQciBearer.ServiceType** parameter is set to **FWA_INTERNET**, **FWA_VIDEO**, and **FWA_PRIVATE_LINE**, respectively).
 - In NSA networking, services with a QCI of 5, 6, 8, or 9
 - In SA networking, services with a 5QI of 5, 6, 8, or 9 or services with a mapped QCI of 5, 6, 8, or 9
- The UE has not been selected for periodic measurements within one hour while the UE context is set up and retained.

NOTE

CA UEs that meet the preceding conditions but treat the local cell as their SCell cannot be selected for periodic measurements.

The gNodeB instructs the UEs to measure all inter-RAT neighboring frequencies configured for the serving cell. If the number of frequencies to be measured is limited and not all the frequencies can therefore be delivered for measurements, the gNodeB selects frequencies based on their priorities. For details about configuring the candidate frequency list based on frequency priorities, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

The maximum number of periodic measurements of each frequency is 100. The rules for calculating the number of periodic measurements are as follows:

- Once fast ANR is enabled, the gNodeB sets the number of periodic measurements of every inter-RAT neighboring frequency of a cell to zero.
- After a periodic measurement of a frequency is complete, the gNodeB increases the number of periodic measurements for this frequency by one.

- If the neighbor relationship with a cell on a frequency is successfully added by ANR, the gNodeB sets the number of periodic measurements of the frequency to zero. If this neighbor relationship is added by running MML commands, the gNodeB does not set this number to zero.
- The number of periodic measurements of a new inter-RAT neighboring frequency is counted from zero.

The gNodeB delivers frequency measurement configurations based on the following rules:

- If the number of consecutive periodic measurements of the frequency is greater than or equal to 100 but no neighbor relationship with a cell on this frequency is added, this frequency will no longer be delivered for periodic measurements.
- If the number of inter-RAT neighbor relationships configured for a cell or the gNodeB reaches its maximum value, all inter-RAT neighboring frequencies of the cell will no longer be delivered for measurements. When the number of inter-RAT neighbor relationships configured for a cell or the gNodeB decreases from the maximum value, the counting of periodic measurements of these frequencies continues from the previous numbers.
- In controlled mode, if the sum of the number of inter-RAT neighbor relationships configured for a cell/gNodeB and the number of neighbor relationships configured for the cell/gNodeB to be confirmed on the MAE-Access reaches the maximum number of neighbor relationships that can be configured, all inter-RAT neighboring frequencies of the cell will no longer be delivered for measurements. When the sum decreases from the maximum value, the counting of periodic measurements of these frequencies continues from the previous numbers.
- In the inter-RAT candidate frequency list, frequencies for which the **NRCeUtraNFreq.MeasActvFlag** parameter is set to **INACTIVE** are not delivered for measurements.

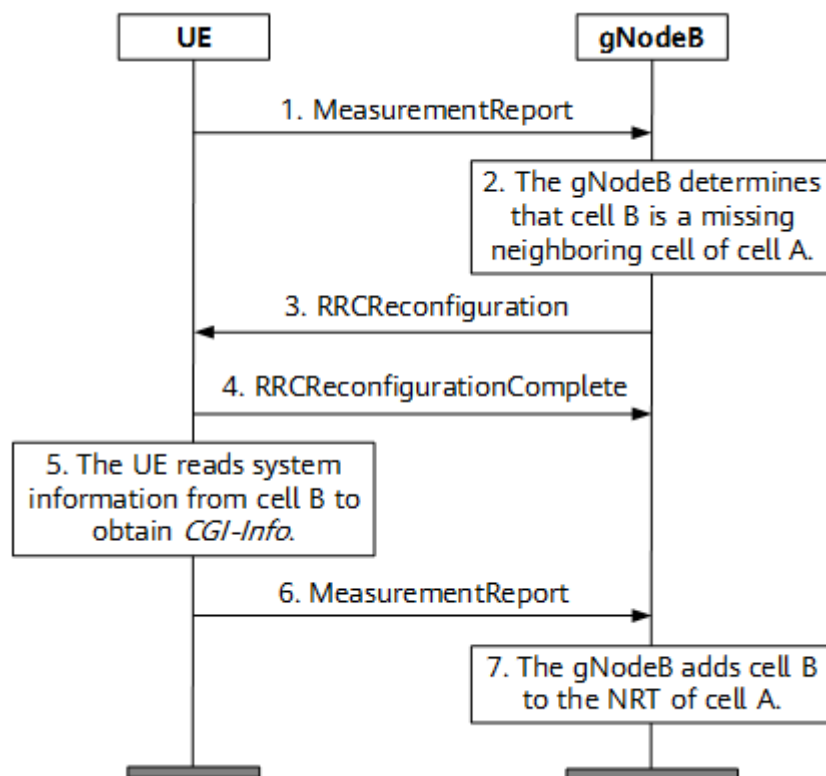
NOTE

- Neighbor relationship addition triggered by fast ANR is not service-triggered. It does not trigger specification-based neighboring relationship replacement or specification-based external cell replacement. In controlled mode, no optimization advice is reported when the number of neighbor relationships reaches its maximum.
- NRT query triggered by fast ANR is not service-triggered. It is not counted in the number of times a neighbor relationship is used and does not affect specification-based neighbor relationship and external cell replacement and identification of redundant neighbor relationships.
- If a UE starts periodic measurements but the measurements conflict with other functions (such as cell handover, CA, and DC) that require measurement resources, the periodic measurements triggered by fast ANR automatically pauses to meet the measurement requirements of the functions.
- For details about the maximum allowed number of neighbor relationships in an inter-RAT NRT, see the descriptions of the **NRCeUtraNRelat** MO in *3900 & 5900 Series Base Station MO and Parameter Reference* in *3900 & 5900 Series Base Station Product Documentation*.

The gNodeB automatically detects and adds missing neighboring cells based on event-triggered or periodic measurement reports sent by UEs. The following provides the details.

Assume that a UE is located in the coverage area of cell A, which works as the serving cell for the UE. Cell B is an E-UTRAN cell adjacent to cell A. The gNodeB sends measurement configurations to the UE. Based on the measurement report sent from the UE, the gNodeB detects and adds cell B, as shown in [Figure 5-2](#).

Figure 5-2 Automatically detecting and adding missing inter-RAT neighboring cells during movement in SA networking



1. The UE sends a measurement report.
The UE measures cells on the frequency specified by the gNodeB, and then sends the gNodeB a measurement report containing the PCIs of cells (cell B for example) that meet signal quality requirements.
2. The gNodeB obtains the PCI of cell B from the measurement report, queries the ECGIs of all neighboring cells in the NRT of cell A, and then searches its NCL for the PCIs and frequencies of external cells indicated by these ECGIs. The gNodeB then checks whether a cell in the NCL has the same PCI and frequency as cell B.
 - If a match is found, cell B is an existing neighboring cell of cell A and no further action is required.
 - If no match is found, cell B is a missing neighboring cell of cell A and the gNodeB proceeds to step 3.
3. The gNodeB sends the UE an RRCReconfiguration message carrying the measurement report type of *reportCGI* and DRX configuration. With the configuration of this type of report, the UE performs measurements and reports *CGI-Info* of the cell with the specified PCI (*CGI-Info* of cell B). DRX terminates downlink data reception in the serving cell so that the UE can receive system information of the target cell.

4. The UE responds to the gNodeB with an RRCReconfigurationComplete message.
5. The UE reads SIB1 (system information) from cell B to obtain *CGI-Info* of cell B.
6. The UE sends the gNodeB a measurement report carrying the *CGI-Info*.
7. The gNodeB obtains the ECGI, TAC, PCI, PLMN list, and other information about cell B from the *CGI-Info* IE in the measurement report and adds cell B to the NCL and the neighbor relationship with cell B to the NRT based on the information. The gNodeB supports automatic addition of missing neighboring cells based on the number of CGI-Info reporting times. It adds cell B to the NCL and the neighbor relationship with cell B to the NRT only when the number of times *CGI-Info* of cell B is reported is greater than or equal to the **NRCellAnr.NCellAddCgiRptCountThld** parameter value. If this parameter is set to 1, this function does not take effect.

The gNodeB checks whether cell B is in its NCL.

- If cell B is in the NCL, the gNodeB adds cell B to the NRT of cell A.
- If cell B is not in the NCL, the gNodeB adds cell B to its NCL and then adds cell B to the NRT of cell A.

The FOFD-021204 Automatic Neighbour Relation (ANR) feature also supports PLMN ID management. If the PLMN attribute of an operator is configured in a **gNBNCePlmnList** MO, the gNodeB performs differentiated management based on the list type of PLMN ID when obtaining the PLMN list from *CGI-Info*. For details, see [5.2 Inter-RAT PLMN ID Management](#).

An NRT is configured through an **NRCellEutraNRelation** MO, and an NCL is configured through a **gNBEutraExternalCell** MO. If the number of neighbor relationships in an NRT reaches the maximum value, new neighbor relationships cannot be added to the NRT. Similarly, if the number of external cells in an NCL reaches the maximum value, new external cells cannot be added to the NCL.

Neighboring cell class label management

In the preceding procedure, assume that cell A has two adjacent cells, cell B and cell C. Cell B and cell C work on the same frequency and are configured with the same PCI, and cell B is in the NRT of cell A. When subsequent event-triggered measurement reports contain the same PCI as cell B and cell C, the gNodeB queries the NRT and selects cell B in the NRT as the target cell for the handover. When the UE moves, the measured cell is either of the two adjacent cells corresponding to the PCI. Therefore, if the gNodeB selects an incorrect target cell, the UE handover fails.

In this scenario, the gNodeB automatically sets **NRCellEutraNRelation.EutraNCellClassLabel** (this parameter cannot be set by running MML commands) to **FORMAL** or **EXTENDED**. This function can be enabled by selecting the **EUTRAN_NCELL_CLASS_LABEL_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

The **N2L_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter specifies whether the measurement period is configurable for automatic neighboring cell class label adjustment.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.N2LhoSuccRateStatInterval** parameter.

- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

When the number of handovers from cell A to cell B is less than the **NRCellAnr.LteNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over the next measurement period until the number is greater than or equal to this parameter value. When the number of handovers from cell A to cell B is greater than or equal to the **NRCellAnr.LteNCellHoNumThld** parameter value, the gNodeB automatically sets the neighboring cell class label (specified by the **NRCellEutraNRelation.EutraNCellClassLabel** parameter) for cell B based on the neighboring cell handover success rate threshold (specified by the **NRCellAnr.EutraNCellHoSuccRateThld** parameter).

- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to **0**, the gNodeB automatically sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.
- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is less than this value, the gNodeB automatically sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **EXTENDED** for cell B. When the PCI of cell B and cell C is reported again in subsequent event-based measurements:
 - If the reported CGI of cell C is different from that of cell B corresponding to the PCI in the NRT, the gNodeB adds cell C to the NRT and sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.
 - If the number of consecutive measurements with the report type of *reportCGI* performed by the gNodeB on cell B is greater than or equal to the value of the **NRCellAnr.EutraNCellRptCgiNumThld** parameter, the gNodeB sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B.

NOTE

After the gNodeB sets the **NRCellEutraNRelation.EutraNCellClassLabel** parameter to **FORMAL** for cell B through measurements of report type *reportCGI*, the handover success rate threshold is determined as follows in the next measurement:

- If the **NRCellAnr.EutraNCellHoSuccRateThld** parameter value remains unchanged in the next measurement, the handover success rate threshold in the next measurement is equal to that in the current measurement multiplied by **NRCellAnr.EutraNCellHoSuccRateThld**.
- If there is a change in the **NRCellAnr.EutraNCellHoSuccRateThld** parameter value in the next measurement, the handover success rate threshold in the next measurement is specified by this parameter.
- When the **NRCellAnr.EutraNCellHoSuccRateThld** parameter is set to a value other than **0** and the handover success rate of cell B is greater than or equal to this value, the value of the **NRCellEutraNRelation.EutraNCellClassLabel** parameter remains unchanged.

Specifically:

- The measurement of the number of handovers is controlled by the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is

measured. If this option is deselected, the number of handover executions is measured.

- The handover success rate is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

Primary PLMN reserved flag management in multi-operator sharing scenarios

Unlike intra-RAT automatic detection and addition of missing neighboring cells, in inter-RAT automatic detection and addition of missing neighboring cells, the UE does not include the primary PLMN reserved flags of neighboring E-UTRAN cells in the measurement report. Therefore, the gNodeB needs to determine the primary PLMN reserved flag (**MASTER_PLMN_RESERVED_FLAG** option of the **gNBEutraExternalCell.AggregationAttribute** parameter) of an external E-UTRAN cell in the NCL as follows:

The **SEPARATED_FREQ_SPEC_SHR_FLAG** option of the **NRCellEutranNFreq.AggregationAttribute** parameter specifies whether a neighboring E-UTRA frequency is a secondary operator's frequency in RAN sharing with special dedicated carrier mode.

- If the **SEPARATED_FREQ_SPEC_SHR_FLAG** option is selected, the neighboring E-UTRA frequency is a secondary operator's frequency in RAN sharing with special dedicated carrier mode.
- If the **SEPARATED_FREQ_SPEC_SHR_FLAG** option is deselected, the neighboring E-UTRA frequency is not a secondary operator's frequency in RAN sharing with special dedicated carrier mode.

In step 7, the gNodeB automatically determines the configuration of the primary PLMN reserved flag of the E-UTRAN cell in the NCL based on the setting of the **SEPARATED_FREQ_SPEC_SHR_FLAG** option for the operating frequency of cell B, as shown in [Table 5-2](#).

Table 5-2 Automatic configuration of primary PLMN reserved flags of neighboring E-UTRAN cells

If...	Then...
Option A is selected	If option B is selected, which indicates that the primary PLMN of an external E-UTRAN cell is reserved, this E-UTRAN cell cannot be selected as the target cell for handovers of UEs served by this primary PLMN in the gNodeB.
Option A is deselected	If option B is deselected, which indicates that the primary PLMN of an external E-UTRAN cell is not reserved, this E-UTRAN cell can be selected as the target cell for handovers of UEs served by this primary PLMN in the gNodeB.

If...	Then...
Note: • Option A: SEPARATED_FREQ_SPEC_SHR_FLAG option of the NRCellEutranNFreq.AggregationAttribute parameter • Option B: MASTER_PLMN_RESERVED_FLAG option of the gNBEutraExternalCell.AggregationAttribute parameter When the eNodeB switches from another mode to the RAN sharing with special dedicated carrier mode or from the RAN sharing with special dedicated carrier mode to another mode, the setting of option A must be manually changed. When the setting of option A changes, the setting of option B in the NCL is not automatically updated and needs to be manually changed.	

 NOTE

The preceding mechanism applies only to the scenario where an E-UTRAN cell has two PLMN IDs in the measurement report sent by the UE. If an E-UTRAN cell in the measurement report sent by a UE has one or more than two PLMN IDs, the gNodeB considers that the E-UTRAN cell is not a cell of a secondary operator in RAN sharing with special dedicated carrier mode. When adding the E-UTRAN cell to the NCL, the gNodeB deselects the **MASTER_PLMN_RESERVED_FLAG** option of the **gNBEutraExternalCell.AggregationAttribute** parameter.

On the eNodeB side, multi-operator sharing incorporates functions such as RAN sharing with common carrier, RAN sharing with dedicated carrier, and RAN sharing with special dedicated carrier. For details, see *RAN Sharing in eRAN Feature Documentation*. If a neighboring E-UTRAN cell works in multi-operator sharing mode, the gNodeB adds secondary operator information about a missing neighboring cell to the external-cell PLMN list when adding the missing neighboring cell to the NCL. For details, see [5.1.1.7.1 Automatic External-Cell PLMN List Addition](#). If a neighboring E-UTRAN cell broadcasts its PLMN IDs in rotation mode, the sequence of PLMN IDs broadcast in system information periodically changes. Therefore, the primary operator is not always listed at the top of the PLMN list.

- If the eNodeB serving the target cell is not provided by Huawei, the gNodeB cannot obtain information about the target cell from the MAE-Access. In this case, the gNodeB does not add the target cell to the NCL or NRT.
- If the eNodeB serving the target cell is provided by Huawei, the **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter needs to be selected. With this setting, the gNodeB can obtain the PLMN list of an external E-UTRAN cell from the MAE-Access to get the target cell's correct MCC and MNC of the primary operator. The gNodeB can then add the target cell to the NCL and NRT.

For details about PLMN ID rotation mode, see *RAN Sharing in eRAN Feature Documentation*.

UE blacklist and whitelist management

To address UE compatibility issues, the gNodeB supports UE blacklist management and UE whitelist management in inter-RAT ANR. With these functions, the gNodeB can take differentiated measures for UEs.

- UE blacklist management in inter-RAT ANR is controlled by the **NR_EUTRAN_ANR_SW_OFF** option of the **gNBUECompatSw.BlacklistCtrlExtSwitch** parameter.
 - If this option is selected, the gNodeB disables inter-RAT ANR for specified UEs and does not deliver reportCGI measurement configurations to these UEs.
 - If this option is deselected, the gNodeB does not take differentiated measures for UEs and delivers reportCGI measurement configurations to all UEs.
- UE whitelist management in inter-RAT ANR is controlled by the **NR_EUTRAN_ANR_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlExtSwitch** parameter.
 - If this option is selected, the gNodeB enables inter-RAT ANR for specified UEs and delivers reportCGI measurement configurations only to these UEs.
 - If this option is deselected, the gNodeB does not take differentiated measures for UEs and delivers reportCGI measurement configurations to all UEs.

5.1.1.3 Automatic Removal of Neighboring Cells

Automatic removal of neighboring cells includes automatic removal of neighbor relationships and automatic removal of external cells. This function is enabled when both the **NR_EUTRAN_ANR_SW** and **NR_EUTRAN_ANR_DEL_SW** options of the **NRCeIIAlgoSwitch.AnrSwitch** parameter are selected.

Automatic Removal of Neighbor Relationships

Automatic removal of neighbor relationships from NRTs includes specification-based removal of neighbor relationships and removal of redundant neighbor relationships. Details are as follows:

- Specification-based removal of neighbor relationships

The maximum number of neighboring cells that can be configured in an NRT depends on the maximum number of neighboring cells that can be configured for a cell and that for a gNodeB. For details about the maximum number of neighboring cells that can be configured for a cell and that for a gNodeB, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

 - When the number of neighboring cells of a cell reaches its maximum value, the gNodeB automatically removes the neighbor relationships that are least frequently used from the NRT of the cell so that new neighbor relationships can be added.
 - When the number of neighboring cells of the gNodeB reaches its maximum value but that of a cell does not, the gNodeB automatically removes the neighbor relationships that are least frequently used from the NRTs of served cells so that new neighbor relationships can be added.
- Removal of redundant neighbor relationships

The gNodeB considers an inter-RAT neighbor relationship redundant if the neighbor relationship is not used for a period of seven days. After this period elapses, the gNodeB automatically removes the neighbor relationship from the NRT.

The measurement of the number of times a neighbor relationship is used is controlled by the **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the gNodeB supports optimized measurement for automatic removal of neighboring cells. In this case, the measured number of times a neighbor relationship with a cell is used is the number of handover attempts performed by the gNodeB using this neighbor relationship. However, if the **NRCellEutraNRelation.NoHoFlag** parameter is set to **FORBID_HO** for the neighbor relationship, the gNodeB cannot remove it.
- If this option is deselected, the measured number of times the neighbor relationship is used is the number of times the gNodeB detects the corresponding neighboring cell based on the PCI contained in measurement reports sent by UEs.

 NOTE

The gNodeB measures the number of times the neighboring cells are queried every night and these cells must be activated during the measurement.

The neighbor relationship cannot be automatically removed if any of the following conditions is met:

- The **NRCellEutraNRelation.NoRmvFlag** parameter is set to **FORBID_ANR_RMV**.
- The **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the **NRCellEutraNRelation.PowerSavingOverlapInd** parameter is selected.

When a neighbor relationship is automatically removed by ANR, the gNodeB imposes a penalty on the neighbor relationship to prevent it from being added back to the NRT within a short period and stores it in a gNodeB-maintained temporary list. Whether the neighbor relationship can be added back to the NRT through ANR is described as follows:

- The neighbor relationship can be added back to the NRT through ANR only when the number of requests to add the neighbor relationship is greater than the value of the **NRCellAnr.AutoDelNCellPenaltyNum** parameter within the penalty period specified by the **NRCellAnr.AutoDelNCellPenaltyPeriod** parameter.
- After the penalty period specified by the **NRCellAnr.AutoDelNCellPenaltyPeriod** parameter expires, the gNodeB can add the neighbor relationship back to the NRT through ANR.

 NOTE

Removed neighbor relationships under penalty are stored in a temporary list on the base station. The list is cleared if the base station resets and can be regenerated after the penalty mechanism evaluates neighbor relationships again. Therefore, the number of neighbor relationships may increase after the base station restarts. The number can be restored by using automatic removal of neighboring cells.

Automatic Removal of External Cells

Automatic removal of external cells includes specification-based removal of external cells and removal of redundant external cells.

- Specification-based removal of external cells

The maximum number of external cells that can be configured in an NCL depends on the maximum number of external cells that can be configured for a gNodeB. For details about the maximum number of external cells that can be configured for a gNodeB, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

If the number of external cells in the NCL reaches its maximum value, the gNodeB automatically removes the external cells that are least frequently used from the NCL so that new external cells can be added.

- Removal of redundant external cells

The base station automatically removes redundant external cells from the NCL in either of the following scenarios:

- The external cell was a neighboring cell removed from an NRT and no cells served by the gNodeB have neighbor relationships with this cell.
- The external cell is not used for a period of seven days and no cells served by the gNodeB have neighbor relationships with this cell.

If a neighboring E-UTRAN cell works in multi-operator sharing mode and the corresponding external cell is automatically removed from the NCL, all records of the external cell in the external-cell PLMN list are also removed.

The measurement of the number of times an external cell is used is controlled by the **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter. If this option is selected, the gNodeB supports optimized measurement for automatic removal of neighboring cells. In this case, the number of times the external cell is used is the actual number of handover attempts the gNodeB performs using this external cell. If this option is deselected, the number of times the external cell is used is the number of times the gNodeB detects this cell based on the PCI contained in measurement reports sent by UEs.

NOTE

The gNodeB measures the number of times the neighboring cells are queried every night and these cells must be activated during the measurement.

5.1.1.4 Automatic NCL Update

This function is enabled when the **NR_EUTRAN_ANR_SW** option of the **NRCelAlgoSwitch.AnrSwitch** parameter is selected. NCL update can be triggered by measurement reports from UEs. The following provides the details.

After receiving the ECGI and TAC carried in the *CGI-Info* IE in the measurement report from a UE, the gNodeB queries its NCL for the external cell indicated by the ECGI and performs the following operations:

- The gNodeB checks whether the downlink EARFCN and PCI of the external cell in the NCL are the same as those in measurement configuration.
 - If they are the same, the gNodeB performs no further operation.

- If they are different, the gNodeB updates the NCL using the downlink EARFCN or PCI carried in the measurement configuration.
- The gNodeB checks whether the TAC of the external cell in the NCL is the same as that in the measurement report.
 - If they are the same, the gNodeB performs no further operation.
 - If they are different, the gNodeB updates the NCL using the information carried in the IE.

In this context:

- The ECGI is determined by the settings of **gNBEutraExternalCell.Mcc**, **gNBEutraExternalCell.Mnc**, **gNBEutraExternalCell.EnodebId**, and **gNBEutraExternalCell.CellId**.
- The downlink EARFCN is specified by the **gNBEutraExternalCell.DLEarfcn** parameter.
- The PCI is specified by the **gNBEutraExternalCell.PhysicalCellId** parameter.
- The TAC is specified by the **gNBEutraExternalCell.Tac** parameter.

5.1.1.5 Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate

This function is used to automatically identify neighboring cells with abnormal handover statistics and handle inappropriate neighbor relationships. This prevents handover or redirection failures caused by inappropriate neighbor relationships, thereby guaranteeing user experience. This function is enabled when the **NR_EUTRAN_ANR_AUTO_NO_HO_SW** option of the **NRCellaAlgoSwitch.AnrSwitch** parameter is selected.

The **NRCellaEutraNRelation.NoHoFlag** parameter specifies whether to allow handovers or redirections for UEs based on neighbor relationships. This parameter can be set to **PERMIT_HO** or **FORBID_HO**.

- If this parameter is set to **PERMIT_HO**, handovers or redirections for UEs can be initiated based on neighbor relationships.
- If this parameter is set to **FORBID_HO**, handovers or redirections for UEs cannot be initiated based on neighbor relationships.

If the **NR_EUTRAN_ANR_AUTO_NO_HO_SW** option of the **NRCellaAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB optimizes the setting of the **NRCellaEutraNRelation.NoHoFlag** parameter for neighbor relationships based on the handover success rate within a measurement period. Whether the measurement period is configurable is specified by the **N2L_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.N2LhoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

If the number of handovers from the local cell to a neighboring cell is less than the **NRCellaAnr.LteNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over and includes it in the counting

process during the next measurement period until the number is greater than or equal to this parameter value, and starts a decision on the handover success rate.

The gNodeB automatically sets the **NRCellEutraNRelation.NoHoFlag** parameter to **FORBID_HO** for the neighboring cell if the success rate of handovers from the local cell to the neighboring cell is less than or equal to the **NRCellAnr.N2LNoHoSuccRateThld** parameter value for consecutive decision times whose number is specified by the **NRCellAnr.N2LNoHoConsecPrdThld** parameter.

Specifically:

- The measurement of the number of handovers is controlled by the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.
- The handover success rate is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

After the **NRCellEutraNRelation.NoHoFlag** parameter is automatically set to **FORBID_HO** for a neighboring cell in the NRT, this parameter cannot be set to **PERMIT_HO** through ANR. For this reason, with this function enabled, it is necessary to check whether the neighboring cells for which **NRCellEutraNRelation.NoHoFlag** is automatically set to **FORBID_HO** can be selected as target cells for handovers or redirections after their configurations are optimized.

- If they can, **NRCellEutraNRelation.NoHoFlag** needs to be manually set to **PERMIT_HO**.
- If they cannot, no further operation is required.

5.1.1.6 Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells

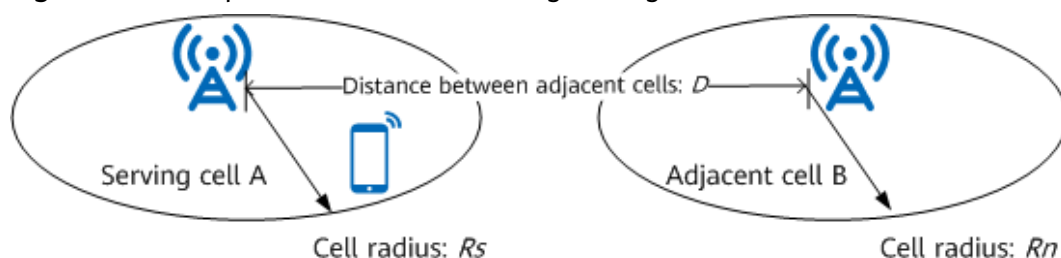
With this function, the gNodeB automatically selects neighboring cells having low handover success rates based on handover statistics, identifies over-distant neighboring cells among these neighboring cells, and handles the corresponding neighbor relationships with the over-distant neighboring cells. This function can prevent handover or redirection failures caused by a long distance between a serving cell and a target cell, thereby ensuring user experience.

An over-distant neighboring cell refers to a cell whose distance from the local cell is greater than the sum of the access radii of the two cells.

- The distance between adjacent cells is the linear distance between the antennas serving the two cells.
- The access radius refers to the maximum distance where a UE can successfully access a cell. The distance is determined by the PRACH configurations of the cell. The PRACH configurations include the PRACH subcarrier spacing, cell radius, and PRACH configuration index.

The example shown in [Figure 5-3](#) indicates that D is greater than the sum of R_s and R_n . In conclusion, adjacent cell B is an over-distant neighboring cell of serving cell A.

Figure 5-3 Example of an over-distant neighboring cell



This function is enabled when the **NR_EUTRAN_OVER_DIST_DETECT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected.

The **NRCellEutraNRelation.NoHoFlag** parameter specifies whether to allow handovers or redirections for UEs based on neighbor relationships.

- If this parameter is set to **PERMIT_HO**, the gNodeB allows handovers or redirections for UEs based on neighbor relationships.
- If this parameter is set to **FORBID_HO**, the gNodeB prohibits handovers or redirections for UEs based on neighbor relationships.

If the **NR_EUTRAN_OVER_DIST_DETECT_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB optimizes the setting of the **NRCellEutraNRelation.NoHoFlag** parameter for neighbor relationships based on identification of over-distant neighboring cells within a measurement period. Whether the measurement period is configurable is specified by the **N2L_HO_SUCC_RATE_INTVL_CFG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter.

- If this option is selected, the measurement period can be configured using the **gNodeBAlgo.N2LhoSuccRateStatInterval** parameter.
- If this option is deselected, the measurement period is fixed at 24 hours and the gNodeB performs a measurement at night every day.

If the number of handovers from the local cell to a neighboring cell is less than the **NRCellAnr.LteNCellHoNumThld** parameter value within a measurement period, the gNodeB carries the number over and includes it in the counting process during the next measurement period until the number is greater than or equal to this parameter value, and starts a decision. The gNodeB changes the **NRCellEutraNRelation.NoHoFlag** parameter value from **PERMIT_HO** to **FORBID_HO** if the neighboring cell meets all of the following conditions:

- The **NRCellEutraNRelation.NoHoFlag** parameter is set to **PERMIT_HO** for the neighboring cell.
- The number of successful handovers from the local cell to the neighboring cell is less than 95%.
- The distance between the local cell and the neighboring cell is greater than the sum of the access radii of the two cells.

Specifically:

- The measurement of the number of handovers is controlled by the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter. If this option is selected, the number of handover attempts is measured. If this option is deselected, the number of handover executions is measured.

- The handover success rate is equal to the result of the following: Number of successful handovers/Number of handovers x 100%.

 NOTE

- The gNodeB queries the MAE-Access about whether the distance between a local cell and a neighboring cell is greater than the sum of the access radii of the two cells. The MAE-Access calculates the distance between the local cell and the neighboring cell based on the longitude and latitude information contained in base station engineering parameters. For details about engineering parameters, see the description of setting NR-LTE overshoot query in *SON Management in MAE Product Documentation (EulerOS, TaiShan)*.
- If a multi-RRU cell has multiple latitudes and longitudes, the minimum distance is used for determination.
- The gNodeB queries over-distant neighboring cells on the MAE-Access again at night every day.
- The MAE-Access allows each gNodeB to query a maximum of 256 over-distant neighboring cells every day.

Automatic optimization of neighbor relationship attributes in the NRT can only change the value of the **NRCellEutraNRelation.NoHoFlag** parameter from **PERMIT_HO** to **FORBID_HO**, not from **FORBID_HO** to **PERMIT_HO**. Therefore, after this function is enabled, check the value of the **NRCellEutraNRelation.NoHoFlag** parameter. If the parameter is automatically set to **FORBID_HO** for a neighboring cell but this cell is required as a target cell for handovers or redirections, this parameter value needs to be manually changed from **FORBID_HO** to **PERMIT_HO**.

5.1.1.7 Automatic External-Cell PLMN List Configuration

If neighboring E-UTRAN cells work in multiple-operator sharing mode, the gNodeB automatically configures E-UTRAN external-cell PLMN lists (external-cell PLMN list for short) in addition to NCL and NRT configuration. An external-cell PLMN list records secondary operator information about an external E-UTRAN cell. Such information includes settings of the parameters in the **gNBEutraExtCellPlmn** MO. The parameters include:

- **gNBEutraExtCellPlmn.Mcc**
- **gNBEutraExtCellPlmn.Mnc**
- **gNBEutraExtCellPlmn.Enodebld**
- **gNBEutraExtCellPlmn.Celld**
- **gNBEutraExtCellPlmn.SharedMcc**
- **gNBEutraExtCellPlmn.SharedMnc**

This function is enabled when the **NR_EUTRAN_ANR_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected. Automatic external-cell PLMN list configuration includes automatic external-cell PLMN list addition and automatic external-cell PLMN list update. Details are provided in the following sections. For details of multiple-operator sharing, see *Multi-Operator Sharing*.

5.1.1.7.1 Automatic External-Cell PLMN List Addition

The **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter determines whether the gNodeB can

query the PLMN list of an external E-UTRAN cell from the MAE-Access. If neighboring E-UTRAN cells work in multi-operator sharing mode, the following apply:

- When the **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter is selected, the gNodeB can query the list of PLMN IDs of the primary and secondary operators to which an external E-UTRAN cell belongs from the MAE-Access.
- When the **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter is deselected, the gNodeB cannot query the list of PLMN IDs of the primary and secondary operators to which an external E-UTRAN cell belongs from the MAE-Access.

If PLMN ID rotation mode is enabled for the neighboring E-UTRAN cell, the sequence of PLMN IDs broadcast in system information periodically changes. Therefore, the primary operator is not always listed at the top of the PLMN list. The principles of PLMN configuration for the external E-UTRAN cell vary depending on whether PLMN ID rotation mode is enabled for the neighboring E-UTRAN cell. The following provides the details.

For details about PLMN ID rotation mode, see *RAN Sharing* in *eRAN Feature Documentation*.

PLMN ID Rotation Mode Disabled for Neighboring E-UTRAN Cells

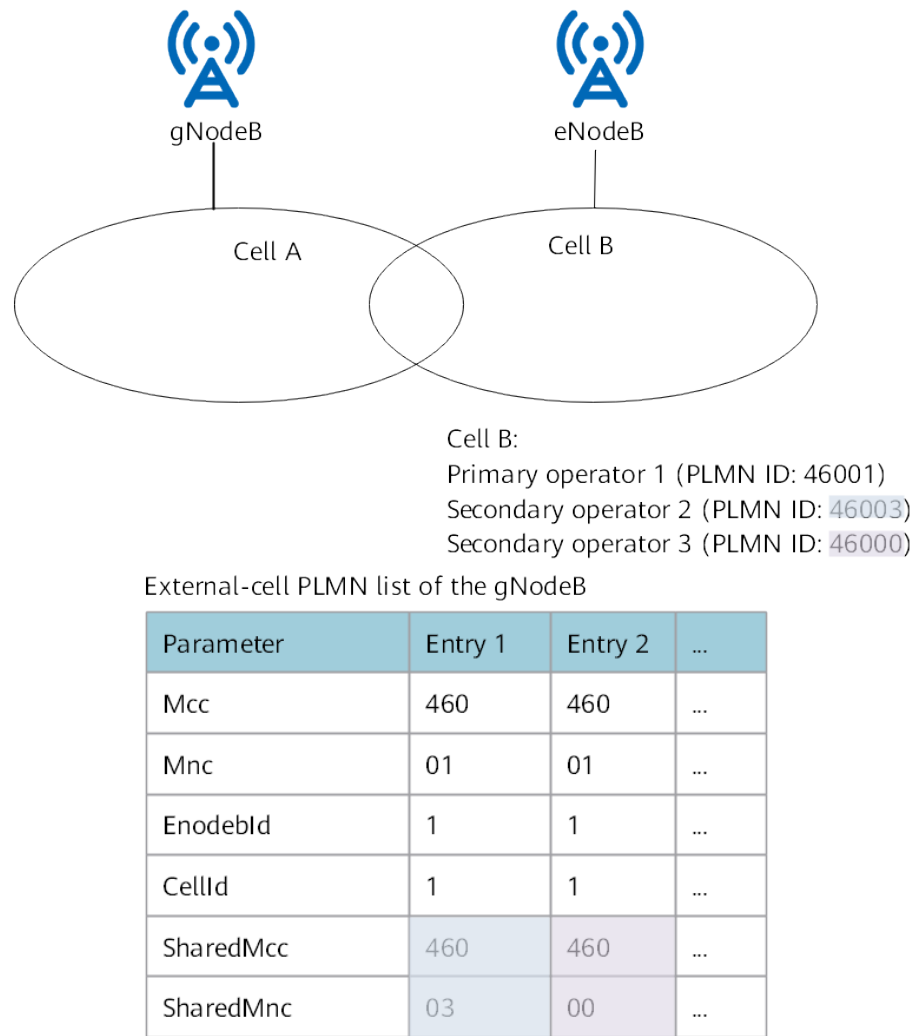
In this scenario, the gNodeB can add neighboring E-UTRAN cells based on measurement reports from UEs. The **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter does not need to be selected.

When a missing neighboring cell is detected, the gNodeB receives a measurement report containing the ECGI from the UE and checks whether the reported *CGI-Info* IE contains only one PLMN ID.

- If yes, the neighboring E-UTRAN cell does not work in multi-operator sharing mode. In this case, the gNodeB adds only the missing neighboring cell to the NCL and NRT.
- If no, the neighboring E-UTRAN cell works in multi-operator sharing mode. In this case, the gNodeB adds the missing neighboring cell to the NCL/NRT and adds secondary operator information about the missing neighboring cell to the external-cell PLMN list.

For example, cell A and cell B are respectively served by the gNodeB and eNodeB. Cell B is shared among three operators and is a missing neighboring cell of cell A. The gNodeB adds cell B to its NCL and NRT of cell A. In addition, the gNodeB adds secondary operator information about cell B to its external-cell PLMN list. [Figure 5-4](#) provides the information recorded in the external-cell PLMN list.

Figure 5-4 External-cell PLMN list



PLMN ID Rotation Mode Enabled for Neighboring E-UTRAN Cells

In this scenario, the primary operator is not always listed at the top of the PLMN list and UEs cannot obtain the neighboring E-UTRAN cell's correct PLMN IDs of the primary and secondary operators. As a result, the gNodeB cannot add neighboring E-UTRAN cells based on measurement reports from UEs. For this reason, the **EUTRAN_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter needs to be selected. When this option is selected, the gNodeB can obtain the PLMN list of the neighboring E-UTRAN cell from the MAE-Access.

When a missing neighboring cell is detected, the gNodeB receives a measurement report containing the ECGI from a UE and manages the external-cell PLMN list as follows:

1. The gNodeB checks whether the reported *CGI-Info* IE contains only one PLMN ID.
 - If yes, the neighboring E-UTRAN cell does not work in multi-operator sharing mode and no further action is required.
 - If no, the neighboring E-UTRAN cell works in multi-operator sharing mode. The gNodeB proceeds to step 2.

2. The gNodeB checks whether one of the reported PLMN IDs is marked as Reserved.
 - If yes, the PLMN marked as Reserved belongs to the primary operator, and other PLMNs belong to secondary operators.
 - If no, go to 3.
3. The gNodeB checks whether an external-cell PLMN list has been configured for the reported neighboring cell.
 - If yes, the eNodeB identifies the primary and secondary operators based on the external-cell PLMN list.
 - If no, go to 4.
4. The gNodeB reports the PCI and ECGI sent from the UE to the MAE-Access.
5. The MAE-Access checks the PLMN list of the indicated neighboring E-UTRAN cell based on the PCI, eNodeB ID, and cell ID.
6. The MAE-Access sends the check result to the gNodeB.
7. The gNodeB adds the obtained information to the external-cell PLMN list.

5.1.1.7.2 Automatic External-Cell PLMN List Update

External-cell PLMN list update includes:

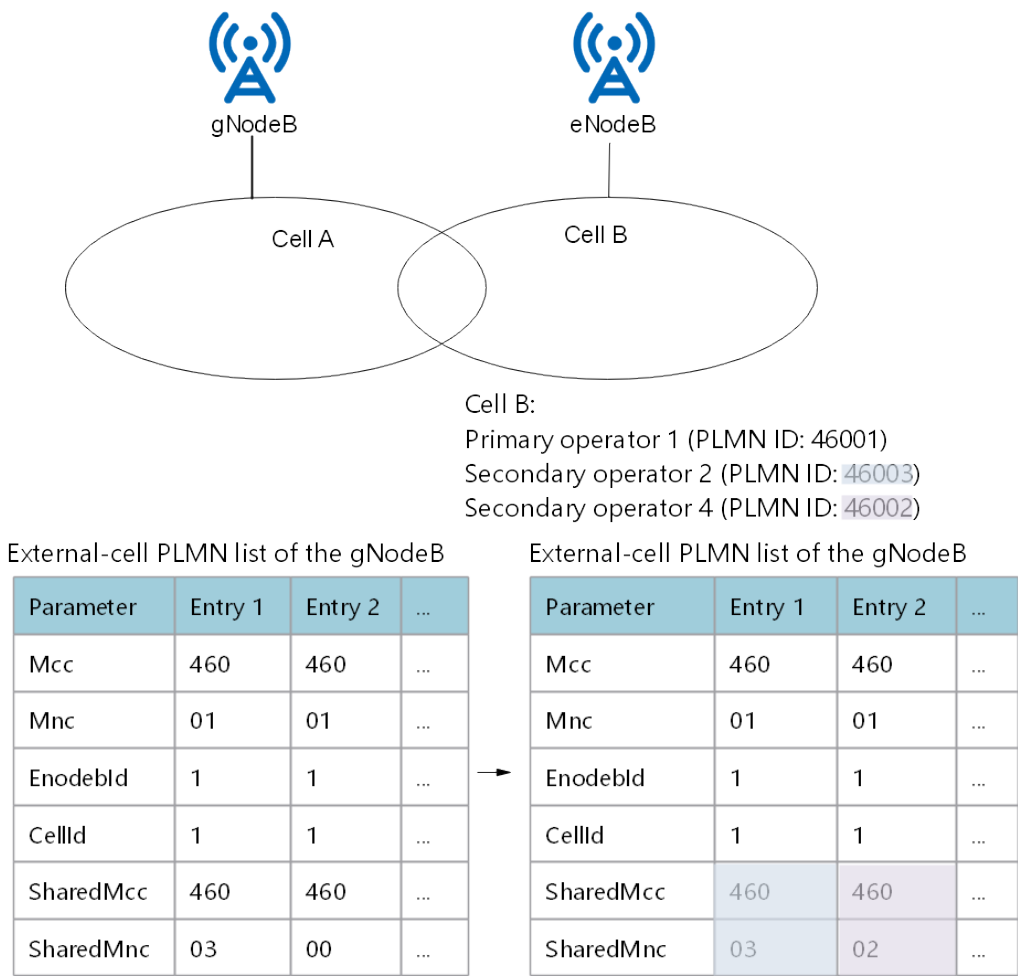
- Addition: The PLMN ID of an external cell is added.
- Removal: The PLMN ID of an external cell is removed.

For example, cell A and cell B are respectively served by the gNodeB and eNodeB. Cell B is shared among three operators. The secondary operator information about cell B is recorded in the external-cell PLMN list of the gNodeB. [Figure 5-5](#) shows this example. The following describes the different outcomes when the operators to which cell B belongs change:

- Addition: The cell belongs to a new operator (operator 4).
- Removal: The cell no longer belongs to operator 3.

[Figure 5-5](#) shows an example where the information recorded in the external-cell PLMN list is updated.

Figure 5-5 External-cell PLMN list



External-cell PLMN list update can be triggered by measurement reports from UEs. After receiving the ECGI and PLMN list carried in the *CGI-Info* IE in the measurement report from a UE, the gNodeB queries the external-cell PLMN list for the external cell indicated by the ECGI. The gNodeB then checks whether the PLMN list of the external cell is same as that in the measurement report.

- If they are the same, the gNodeB performs no further operation.
- If they are different, the gNodeB updates the external-cell PLMN list using the information carried in the IE.

5.1.1.8 Automatic Neighboring Frequency Configuration

When multiple inter-RAT neighboring frequencies are configured for a cell and the gNodeB initiates an inter-RAT handover for a UE, the gNodeB can automatically identify the frequencies to be measured based on the coverage of the neighboring frequencies of the cell. The gNodeB delivers only frequencies whose coverage areas cover the UE's location for measurements. Automatic neighboring frequency configuration is enabled when the **AUTO_EUTRAN_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected.

- If this option is selected, the gNodeB automatically adjusts the value of **NRCellEutranNFreq.MeasActvFlag** based on the inter-RAT measurement report sent by the UE.

- If this option is deselected, automatic neighboring frequency configuration does not take effect.

The **NRCellEutranNFreq.MeasActvFlag** parameter specifies whether a neighboring frequency of a cell can be measured. This parameter can be set to **ACTIVE** or **INACTIVE**.

- If this parameter is set to **ACTIVE**, the frequency can be delivered in scenarios such as inter-RAT measurements.
- If this parameter is set to **INACTIVE**, the frequency cannot be delivered in scenarios such as inter-RAT measurements.

For details about inter-RAT measurements, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

Automatic neighboring frequency configuration provides the following two sub-functions which are different in the mechanism for delivering inter-RAT measurement configurations:

- Basic automatic neighboring frequency configuration: The gNodeB cannot proactively deliver measurement configurations through this function, but instead it selects UEs when other inter-RAT measurements are initiated. Then, the gNodeB automatically adjusts the value of **NRCellEutranNFreq.MeasActvFlag** based on the measurement reports.
- Enhanced automatic neighboring frequency configuration: With this function, the gNodeB proactively selects UEs and delivers periodic measurement configurations to them. Then, the gNodeB automatically adjusts the value of **NRCellEutranNFreq.MeasActvFlag** based on the measurement reports.

Additionally, forbidding automatic neighboring frequency configuration for a specific neighboring frequency is supported. This function is controlled by the **AUTO_EUTRAN_FREQ_REL_FORBID_SW** option of the **NRCellEutranNFreq.FrequencyCtrlSwitch** parameter.

- If this option is selected, the gNodeB cannot automatically adjust the **NRCellEutranNFreq.MeasActvFlag** parameter value for the corresponding neighboring frequency based on measurement reports.
- If this option is deselected, forbidding automatic neighboring frequency configuration does not take effect.

Basic Automatic Neighboring Frequency Configuration

After the **AUTO_EUTRAN_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, the gNodeB automatically adjusts the value of **NRCellEutranNFreq.MeasActvFlag** as follows:

- For a frequency with the **NRCellEutranNFreq.MeasActvFlag** parameter set to **INACTIVE**:

The gNodeB temporarily allows measurements on this frequency every hour. During this period, the gNodeB automatically selects UEs to which this frequency is delivered for measurements when other inter-RAT measurement events are triggered. When receiving a valid measurement report of the frequency, the gNodeB changes the value of the **NRCellEutranNFreq.MeasActvFlag** parameter to **ACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB5.

- For a frequency with the **NRCellEutranNFreq.MeasActvFlag** parameter set to **ACTIVE**:
If the number of times this frequency is delivered for inter-RAT measurements reaches 1000, the accumulative measurement period is longer than or equal to seven days, and the gNodeB has not received a valid measurement report of the frequency, then the gNodeB changes the value of the **NRCellEutranNFreq.MeasActvFlag** parameter to **INACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB5.

Enhanced Automatic Neighboring Frequency Configuration

After the **AUTO_EUTRAN_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected, enhanced automatic neighboring frequency configuration enables the gNodeB to deliver a frequency with the **NRCellEutranNFreq.MeasActvFlag** parameter set to **INACTIVE** or **ACTIVE** for periodic measurements and determine the coverage of the neighboring frequency based on the RSRP threshold. Enhanced automatic neighboring frequency configuration is controlled by the **AUTO_EUTRAN_FREQ_RELN_MEAS_SW** option of the **NRCellAnr.AnrModeSwitch** parameter.

- When this option is selected, the gNodeB automatically adjusts the **NRCellEutranNFreq.MeasActvFlag** parameter value based on the inter-RAT measurement report sent by the UE and the RSRP threshold.
- If this option is deselected, enhanced automatic neighboring frequency configuration does not take effect.

After the **AUTO_EUTRAN_FREQ_RELATION_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter and the **AUTO_EUTRAN_FREQ_RELN_MEAS_SW** option of the **NRCellAnr.AnrModeSwitch** parameter are both selected, the gNodeB automatically adjusts the value of the **NRCellEutranNFreq.MeasActvFlag** parameter as follows:

- For a frequency with the **NRCellEutranNFreq.MeasActvFlag** parameter set to **INACTIVE**:
The gNodeB temporarily allows measurements on this frequency every 61 minutes. During this period, the gNodeB selects three UEs in each cell to which this frequency is delivered for measurements. When the gNodeB receives a valid measurement report of the frequency and the RSRP value of a neighboring cell on the frequency is greater than the RSRP threshold specified by the **NRCellEutranNFreq.AutoEutranFreqRelnRsrpThld** parameter, the gNodeB changes the value of the **NRCellEutranNFreq.MeasActvFlag** parameter to **ACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB5.
- For a frequency with the **NRCellEutranNFreq.MeasActvFlag** parameter set to **ACTIVE**:
The base station delivers this frequency for measurements to three UEs in each cell every 10 minutes. Assume that the number of times this frequency is delivered for measurements reaches 1000 and the accumulative measurement period is longer than or equal to seven days. If the gNodeB has not received a valid measurement report of the frequency or the RSRP values of neighboring

cells on the frequency are less than or equal to the RSRP threshold specified by the **NRCellEutranNFreq.AutoEutranFreqReInRsrpThld** parameter, then the gNodeB changes the value of the **NRCellEutranNFreq.MeasActvFlag** parameter to **INACTIVE** for the frequency and records the change in SON logs. In addition, it updates the neighboring frequency list in SIB5.

5.1.1.9 NCL/NRT Management Based on the eNodeB Blacklist

With NCL/NRT management based on the eNodeB blacklist, the gNodeB can identify blacklisted eNodeBs and perform differentiated management on E-UTRAN cells served by these eNodeBs.

The **gNBEutranBlacklist** MO is used to set an eNodeB blacklist. A maximum of 128 eNodeBs can be blacklisted for each gNodeB. The MCC, MNC, and eNodeB ID are specified by the following parameters:

- MCC: **gNBEutranBlacklist.Mcc**
- MNC: **gNBEutranBlacklist.Mnc**
- eNodeB ID: **gNBEutranBlacklist.EnodebId**

After adding an eNodeB to the blacklist using the **gNBEutranBlacklist** MO, the gNodeB performs the following operations on E-UTRAN cells served by the eNodeB:

- If these E-UTRAN cells are identified as unknown neighboring cells by automatic detection and addition of missing neighboring cells, the gNodeB does not add these cells to the NCL or NRT.
- The gNodeB does not instruct UEs to initiate handovers or redirections to these E-UTRAN cells. However, this restriction does not apply to voice fallback for emergency calls.

5.1.2 Network Analysis

5.1.2.1 Benefits

Inter-RAT ANR ensures the integrity, validity, and correctness of inter-RAT neighbor relationships with E-UTRAN cells. By automatically optimizing the attributes of the neighboring cells, it reduces manual intervention to decrease the costs of network planning, network optimization, and O&M.

NOTE

When the gNodeB automatically adds neighboring cells based on measurement results reported by UEs, it may add over-distant neighboring cells or some neighboring cells with low signal strength to NRTs. These neighboring cells will decrease the handover success rate. In SA networking, ANR automatically maintains the validity of neighboring cells based on the handover success rate every day and sets the **NRCellRelation.NoHoFlag** parameter to **FORBID_HO** for neighboring cells with low handover success rates. Then, the handover success rate will gradually recover. Inter-RAT ANR is not involved in NSA networking.

The number and distribution of ANR-capable UEs affect how fast inter-RAT ANR can detect missing neighboring cells. Therefore, the decrease in the service drop rate and increase in the inter-RAT handover success rate produced by inter-RAT ANR are impossible to quantify.

- Automatic addition of missing neighboring cells based on the number of CGI-Info reporting times reduces the probability of adding an over-distant

neighboring cell to the NCL and the neighbor relationship with this cell to the NRT.

- If neighboring cell class label management is enabled, neighboring LTE cells with the handover success rate lower than the threshold specified by the **NRCellAnr.EutraNCellHoSuccRateThld** parameter are configured as extended neighboring cells. The gNodeB can detect adjacent cells with PCI confusions and then add them as neighboring cells, preventing the handover success rate from decreasing due to PCI confusions. Specifically, the handover success rate is equal to the result of the following:
N.HO.InterRAT.N2E.ExecSuccOut/N.HO.InterRAT.N2E.PrepareAttOut × 100%.
- After UE whitelist management in inter-RAT ANR is enabled, inter-RAT ANR takes effect for specific UEs, which may decrease the service drop rate of these UEs.
- If the **SA_LTE_NCELL_SUCC_RT_STAT_SW** option of the **NRCellAnr.AnrAlgoSwitch** parameter is selected, a neighboring LTE cell is identified as an abnormal neighboring cell if preparations of a handover to this LTE cell fail, increasing the handover success rate. Specifically, the handover success rate is equal to the result of the following:
N.HO.InterRAT.N2E.ExecSuccOut/N.HO.InterRAT.N2E.PrepareAttOut × 100%.
- With automatic removal of neighboring cells enabled, if the **LTE_NCELL_DEL_STAT_OPT_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, unused neighboring LTE cells are more likely to be removed.
- NCL/NRT management based on the eNodeB blacklist prevents UEs from being handed over or redirected to a cell served by a fake base station. As a result, the inter-RAT handover preparation success rate may increase. It is recommended that this function be enabled when a fake eNodeB is near a gNodeB. For details about how to identify a fake base station, contact Huawei technical support. The inter-RAT handover preparation success rate is equal to the result of **N.HO.InterRAT.N2E.ExecAttOut/N.HO.InterRAT.N2E.PrepareAttOut** × 100%.

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

- To detect an unknown neighboring cell, the gNodeB instructs the UEs that meet specific conditions to perform reportCGI measurements. This process prolongs the delay in a handover from the source cell to the target cell. After the neighbor relationship with the cell is added, measurements of report type *reportCGI* are not triggered during subsequent handovers to the neighboring cell and therefore the delay in handovers to the cell is no longer prolonged.
- Downlink throughput decreases for UEs during DRX in which the UEs read the ECGI of the target cell. This is because the gNodeB limits scheduling for downlink transmission for the UEs. After a neighbor relationship with the cell is added, DRX configuration will not be triggered for reportCGI measurements, and therefore UEs are not subject to such changes in downlink throughput during subsequent handovers to the cell.
- Uplink data transmission causes ongoing measurements of report type *reportCGI* to fail. UEs read ECGIs in sleep time during DRX. The sleep time can

be interrupted by transmission of uplink data packets (such as uplink data from UEs or RLC ACKs from UEs in RLC AM mode). This decreases the ECGI reading success rate. After an ECGI is acquired, the identified cell is automatically configured as a neighboring cell and UEs no longer read the ECGI.

- Automatic addition of missing neighboring cells based on the number of CGI-Info reporting times increases the number of reportCGI measurements, which may cause decreases in **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)**. A smaller value of the **NRCellAnr.NCellAddCgiRptCountThld** parameter results in a shorter time required for adding a missing neighboring cell but a higher risk that an addition is performed by mistake. A larger value of this parameter produces the opposite effects.
- After forbidding automatic neighboring frequency configuration is enabled, the **NRCellEutranNFreq.MeasActvFlag** parameter value will not be automatically adjusted for the corresponding neighboring frequency based on measurement reports. This may increase the number of unnecessary measurements on this frequency.
- After UE blacklist management in inter-RAT ANR is enabled, inter-RAT ANR cannot take effect for specific UEs, which may increase the service drop rate of these UEs.
- After enhanced automatic neighboring frequency configuration is enabled, the gNodeB delivers neighboring frequency measurements to selected UEs. In this case, RRC connection reconfiguration may increase the **Service Call Drop Rate (CU)**. In addition, due to measurement gaps, the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** of the UEs selected for measurement may decrease.

The setting of the **NRCellEutranNFreq.AutoEutranFreqRelnRsrpThld** parameter has the following impacts:

- A larger value of this parameter results in a higher probability that enhanced automatic neighboring frequency configuration determines that the measured frequency provides weak or no coverage. This decreases the number of unnecessary measurements on the inter-RAT neighboring frequency, but increases the **Service Call Drop Rate (CU)** if the neighboring frequency provides weak coverage.
- A smaller value of this parameter results in a lower probability that enhanced automatic neighboring frequency configuration determines that the measured frequency provides weak or no coverage. This does not decrease the number of unnecessary measurements on the inter-RAT neighboring frequency.

5.1.3 Requirements

5.1.3.1 Licenses

NCL/NRT management based on the gNodeB blacklist is a basic function and has no license requirements. The following table lists the license requirements for other functions.

Feature ID	Feature Name	Model	Sales Unit
FOFD-021204	Automatic Neighbour Relation (ANR)	NR0S0AUTNE00	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

None

5.1.3.2.2 Mutually Exclusive Functions

None

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	- The UE can perform reportCGI measurements only after entering the DRX sleep time. - The gNodeB automatically configures a long DRX cycle for UEs to read the ECGIs. The setting of the BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter does not affect the configuration of the long DRX cycle initiated by inter-RAT ANR.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Not configuring DRX for UEs with specific RFSP indexes	NOT_CONFIG_DRX_SW option of the gNB RfspConfig.RfspAlgoSwitch parameter	<i>Flexible User Steering</i>	- The UE can perform reportCGI measurements only after entering the DRX sleep time. - When the gNodeB triggers a UE to read the ECGIs, the UE automatically uses a long DRX cycle. The setting of the NOT_CONFIG_DRX_SW option of the gNB RfspConfig.RfspAlgoSwitch parameter does not affect the configuration of the long DRX cycle initiated by inter-RAT ANR.
Low-frequency TDD	Coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_T O_EUTRAN_SW option of the NRCellAlgoSwitch.InterRatServiceMobilitySw parameter EUTRAN_H O_SWITCH or EUTRAN_REDIRECT_SWITCH option of the NRInterRatHoParam.H oModeSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	Inter-RAT ANR requires that measurement-based inter-RAT handover and measurement-based redirection take effect, and coverage-based inter-RAT mobility from NG-RAN to E-UTRAN be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink - coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_T O_EUTRAN_SW option of the NRCCellAlgoSwitch.InterRatServiceMobilitySw parameter EUTRAN_H O_SWITCH or EUTRAN_RE DIRECT_SWI TCH option of the NRInterRatHoParam.H oModeSwit ch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	If downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN is enabled and the NRCCellEutranNFreq.FreqIntrffFlag parameter is set to INTRF, CGI-based measurement configuration is delivered only when the RSRQ in the inter-frequency measurement report is greater than or equal to the following calculated result: $[NRCCellHoEutranMeaGrp.CovBasedHoB1RsrqThld - NRCCellHoEutranMeaGrp.CovBasedHoB1B2Hyst - NRCCellEutranNFreq.FreqSpecificOffset - (NRCCellEutraNRelation.EutranCellIndividualOffset + NRCCellOpPolicy.EutranCioOffset)].$
Low-frequency TDD	Coverage-based redirection to the strongest neighboring LTE cell	EUTRAN_STRO NGEST_NCELL_REDI_SW option of the NRInterRatHoParam.HoModeSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	When the gNodeB detects a missing neighboring cell through inter-RAT ANR and coverage-based redirection to the strongest neighboring LTE cell is enabled, the gNodeB does not deliver measurement configurations with report type <i>reportCGI</i> to UEs. As a result, the missing neighboring cell cannot be added to the NCL and the neighbor relationship with this cell cannot be added to the NRT. In this case, the NR_EUTRAN_COV_REDIRECT_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter can be selected so that the gNodeB can deliver measurement configurations with report type <i>reportCGI</i> to UEs and add the missing neighboring cell to the NCL and the neighbor relationship with this cell to the NRT.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-RAT PLMN ID management	None	<i>ANR</i>	The gNodeB checks whether the gNBCellPlmnList.PlmnListType parameter corresponding to the PLMN to which a cell belongs is set to BLACKLIST before adding the cell as a neighboring cell through inter-RAT ANR. The gNodeB adds the neighboring cell only if the parameter is not set to BLACKLIST .

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.1.3.4 Others

In SA networking, inter-RAT ANR requires that UEs support neighboring E-UTRAN cell measurements of report type *reportCGI* (eutra-CGI-Reporting). For details, see section 6.3.3 "UE capability information elements" in 3GPP TS 38.331 V15.5.0.

Automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells requires that the MAE-Access and MAE-Deployment versions be V100R022C10SPC010 or later.

5.1.4 Operation and Maintenance

5.1.4.1 When to Use

It is recommended that inter-RAT ANR be enabled in new deployment, capacity expansion, and network swapping scenarios to achieve optimal gains.

5.1.4.2 Data Configuration

5.1.4.2.1 Data Preparation

[Table 5-3](#) and [Table 5-4](#) describe the parameters used for function activation and optimization, respectively.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_EUTRAN_ANR_SW	To enable automatic detection and addition of missing neighboring cells, automatic NCL update, and automatic external-cell PLMN list configuration, select this option.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_EUTRAN_ANR_DEL_SW	To enable automatic removal of neighboring cells, select this option. Before enabling this function, ensure that the NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter is selected.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_EUTRAN_ANR_AUTO_NO_HO_SW	To enable automatic optimization of neighbor relationship attributes based on the handover success rate, select this option.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_EUTRAN_OVR_DIST_DETECT_SW	To enable automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells, select this option.
ANR Switch	NRCCellAlgoSwitch.AnrSwitch	NR_EUTRAN_FAST_ANR_SW	To enable fast ANR, select this option. Before enabling this function, ensure that the NR_EUTRAN_ANR_SW option of the NRCCellAlgoSwitch.AnrSwitch parameter is selected.

Parameter Name	Parameter ID	Option	Setting Notes
NCell Addition CGI Report Count Thld	NRCellAnr.NCellAddCgiRptCountThld	None	To enable automatic addition of missing neighboring cells based on the number of CGI-Info reporting times, use the recommended value. This function takes effect only when this parameter is set to a value other than 1 and the NR_EUTRAN_ANR_CTRL_MODE_SW option of the NRCellAnr.AnrModeSwitch parameter is deselected.
Blacklist Control Extension Switch	gNBUECompatSw.BlacklistCtrlExtSwitch	NR_EUTRAN_ANR_SW_OFF	Select this option if inter-RAT ANR is required to be disabled for specific UEs.
Whitelist Control Extension Switch	gNBUECompatSw.WhitelistCtrlExtSwitch	NR_EUTRAN_ANR_SW_ON	Select this option if inter-RAT ANR is required to be enabled for specific UEs.
Self-Configuration Switch	gNodeBAndENB-DCgo.SelfConfigSwitch	EUTRAN_NCELL_CLASS_LABEL_SW	Select this option when neighboring cell class label management is required.
Self-Configuration Switch	gNodeBAndENB-DCgo.SelfConfigSwitch	N2L_HO_SUCCESS_RATE_INTVL_CFG_SW	Select this option in the case of measurement period modification when neighboring cell class label management, automatic optimization of neighbor relationship attributes based on the handover success rate, or automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells is enabled.
Self-Configuration Switch	gNodeBAndENB-DCgo.SelfConfigSwitch	LTE_NCELL_DELETE_TAT_OPT_SW	Select this option if automatic removal of neighboring cells is enabled and measurement based on the actual number of times a neighbor cell is used is required.

Parameter Name	Parameter ID	Option	Setting Notes
ANR Algorithm Switch	NRCellAnr. AnrAlgoSwitch	SA_LTE_NCELL_SU CC_RT_STAT_SW	Select this option when neighboring cell class label management is required. Select this option when identification of scenarios with a handover preparation failure is required in automatic optimization of neighbor relationship attributes based on the handover success rate or automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells.
ANR Switch	NRCellAlgoSwitch. AnrSwitch	AUTO_EUTRAN_FREQ_RELATION_SW	To enable automatic neighboring frequency configuration, select this option.
ANR Mode Switch	NRCellAnr. AnrModeSwitch	AUTO_EUTRAN_FREQ_RELN_MEAS_SW	To enable enhanced automatic neighboring frequency configuration, select this option. Before enabling this function, ensure that the AUTO_EUTRAN_FREQ_RELATION_SW option of the NRCellAlgoSwitch. AnrSwitch parameter is selected.
Frequency Control Switch	NRCellEutranFreq. FrequencyControlSwitch	AUTO_EUTRAN_FREQ_REL_FORBID_SW	To enable forbidding automatic neighboring frequency configuration for a specific neighboring frequency, select this option.
Mobile Country Code	gNBEutranBlacklist. Mcc	None	If a fake eNodeB is near a gNodeB, set this parameter based on the fake eNodeB information.
Mobile Network Code	gNBEutranBlacklist. Mnc	None	If a fake eNodeB is near a gNodeB, set this parameter based on the fake eNodeB information.
eNodeB ID	gNBEutranBlacklist. EnodebId	None	If a fake eNodeB is near a gNodeB, set this parameter based on the fake eNodeB information.

Table 5-4 Parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
ANR Mode Switch	NRCellAnr. AnrModeSwitch	NR_EUTRAN_ANR_CTRL_MODE_SW	Use the default value. Alternatively, set this option based on the operator's policy.
ANR Mode Switch	NRCellAnr. AnrModeSwitch	NR_EUTRAN_ANR_CTRL_MODE_HO_SW	Use the default value. Alternatively, set this option based on the operator's policy.
N2L Handover Success Rate Statistic Interval	gNodeBAnr.N2LhoSuccessRateStatInterval	None	Use the default value.
E-UTRAN Neighbor Cell Handover Succ Rate Thld	NRCellAnr. EutraNCellHoSuccRateThld	None	Use the default value.
E-UTRAN Neighbor Cell ReportCGI Number Thld	NRCellAnr. EutraNCellRptCgiNumThld	None	Use the default value.
ANR Mode Switch	NRCellAnr. AnrModeSwitch	EUTRAN_NCELL_PLMN_RR_MODE_SW	Select this option if neighboring E-UTRAN cells work in multi-operator sharing mode and broadcast PLMN IDs in rotation mode.
Over Distant Neighboring Cell Threshold	NRCellAnr. OverDistantNCellThld	None	Use the default value. This parameter applies only to the controlled mode.
No Handover Flag	NRCellEutraNRelation.NoHoFlag	None	Use the default value.
No Removal Flag	NRCellEutraNRelation.NoRmvFlag	None	Use the default value.

Parameter Name	Parameter ID	Option	Setting Notes
Auto Deleted Neighbor Cell Penalty Period	NRCellAnr. AutoDelNC ellPenalty Period	None	Use the default value.
Auto Deleted Neighbor Cell Penalty Number	NRCellAnr. AutoDelNC ellPenalty Num	None	Use the recommended value.
LTE Neighbor Cell Handover Number Thld	NRCellAnr. LteNCellH oNumThld	None	Use the default value.
NR to LTE No Handover Success Rate Thld	NRCellAnr. N2LNoHoS uccRateThl d	None	Use the default value.
NR to LTE No Handover Consecutive Period Thld	NRCellAnr. N2LNoHoC onsecPrdT hld	None	Use the default value.
Aggregation Attribute	NRCellEutr anNFreq.A ggregation Attribute	SEPARATED_FREQ_SPEC_SHR_FLAG	If a cell working a neighboring E-UTRA frequency belongs to a secondary operator in RAN sharing with special dedicated carrier mode, select this option.
Measurement Activation Flag	NRCellEutr anNFreq.M easActvFla g	None	Use the default value.
Auto E-UTRAN Freq Relation RSRP Threshold	NRCellEutr anNFreq.A utoEutranF reqRelnRsr pThld	None	Use the default value if enhanced automatic neighboring frequency configuration is enabled.

5.1.4.2.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling automatic detection and addition of missing neighboring cells, automatic NCL update, and
automatic external-cell PLMN list configuration
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_SW-1;
//Enabling fast ANR
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_SW-1&NR_EUTRAN_FAST_ANR_SW-1;
//Enabling automatic addition of missing neighboring cells based on the number of CGI-Info reporting
times when uncontrolled mode is used
MOD NRCELLANR: NrCellId=0, NCellAddCgiRptCountThld=3;
//Enabling neighboring cell class label management
MOD GNODEBALGO: SelfConfigSwitch=EUTRAN_NCELL_CLASS_LABEL_SW-1;
//Enabling UE blacklist management in inter-RAT ANR when inter-RAT ANR is required to be disabled for
specific UEs
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=NR_EUTRAN_ANR_SW_OFF-1;
//Enabling UE whitelist management in inter-RAT ANR when inter-RAT ANR is required to be enabled for
specific UEs
MOD GNBUECOMPATSW: UeInfoIndex=2, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=NR_EUTRAN_ANR_SW_ON-1;
//Enabling automatic removal of neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_SW-1&NR_EUTRAN_ANR_DEL_SW-1;
//Enabling optimized measurement for automatic removal of neighboring LTE cells when automatic
removal of neighboring cells is enabled and measurement based on the actual number of times a neighbor
cell is used is required
MOD GNODEBALGO: SelfConfigSwitch=LTE_NCELL_DEL_STAT_OPT_SW-1;
//Enabling automatic optimization of neighbor relationship attributes based on the handover success rate
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_AUTO_NO_HO_SW-1;
//Enabling optimized measurement of the success rate of handovers to neighboring LTE cells when
neighboring cell class label management is required, or identification of scenarios with a handover
preparation failure is required in automatic optimization of neighbor relationship attributes based on the
handover success rate or automatic optimization of neighbor relationship attributes based on identification
of over-distant neighboring cells
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=SA_LTE_NCELL_SUCC_RT_STAT_SW-1;
//Enabling automatic optimization of neighbor relationship attributes based on identification of over-
distant neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_OVER_DIST_DETECT_SW-1;
//Selecting the N2L_HO_SUCC_RATE_INTVL_CFG_SW option in the case of measurement period
modification when neighboring cell class label management, automatic optimization of neighbor
relationship attributes based on the handover success rate, or automatic optimization of neighbor
relationship attributes based on identification of over-distant neighboring cells is enabled
MOD GNODEBALGO: SelfConfigSwitch=N2L_HO_SUCC_RATE_INTVL_CFG_SW-1;
//Enabling automatic neighboring frequency configuration
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=AUTO_EUTRAN_FREQ_RELATION_SW-1;
//Enabling enhanced automatic neighboring frequency configuration
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=AUTO_EUTRAN_FREQ_RELN_MEAS_SW-1;
//Selecting the AUTO_NR_FREQ_REL_FORBID_SW option when forbidding automatic neighboring frequency
configuration for a specific neighboring frequency is required
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000,
FrequencyCtrlSwitch=AUTO_EUTRAN_FREQ_REL_FORBID_SW-1;
//Adding a fake eNodeB which is near the gNodeB to the blacklist
ADD GNBEUTRANBLACKLIST: EnodebId=1, Mcc="302", Mnc="220";
```

Optimization Command Examples

Command examples in TDD

```
//(Recommended) Disabling the controlled mode for inter-RAT ANR based on operator's policies
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_EUTRAN_ANR_CTRL_MODE_SW-0;
//Modifying the threshold for determining over-distant neighboring cells in controlled mode as required
MOD NRCELLANR: NrCellId=0, OverDistantNCellThld=5000;
//(Recommended) Prohibiting handovers to unconfirmed neighboring E-UTRAN cells for inter-RAT ANR
based on operator's policies when controlled mode is used
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=NR_EUTRAN_ANR_CTRL_MODE_HO_SW-0;
//Modifying the neighboring E-UTRAN cell handover success rate threshold and the threshold expressed as
the number of reportCGI measurements on neighboring E-UTRAN cells for neighboring cell class label
management as required
MOD NRCELLANR: NrCellId=0, EutraNCellHoSuccRateThld=80, EutraNCellRptCgiNumThld=100;
//Modifying the measurement period for neighboring cell class label management, automatic optimization
of neighbor relationship attributes based on the handover success rate, or automatic optimization of
neighbor relationship attributes based on identification of over-distant neighboring cells as required
MOD GNODEBALGO: N2LHoSuccRateStatInterval=24;
//Turning on EUTRAN_NCELL_PLMN_RR_MODE_SW if the neighboring E-UTRAN cells work in multi-
operator sharing mode and broadcast PLMN IDs in rotation mode
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=EUTRAN_NCELL_PLMN_RR_MODE_SW-1;
//Allowing handovers or redirections to optimized neighboring cells
MOD NRCELLEUTRANRELATION: NrCellId=0, Mcc="302", Mnc="220", EnodebId=1, CellId=0,
NoHoFlag=PERMIT_HO;
//Enabling removal of neighbor relationships through ANR
MOD NRCELLEUTRANRELATION: NrCellId=0, Mcc="302", Mnc="220", EnodebId=1, CellId=0,
NoRmvFlag=PERMIT_ANR_RMV;
//Modifying the penalty on automatically removed neighbor relationships as required
MOD NRCELLANR: NrCellId=0, AutoDelNCellPenaltyPeriod=10080, AutoDelNCellPenaltyNum=100;
//Modifying the threshold expressed as the number of handovers to a neighboring LTE cell and the
thresholds expressed as the success rate and number of consecutive periods for deciding whether to enable
NR-to-LTE handover prohibition as required
MOD NRCELLANR: NrCellId=0, LteNCellHoNumThld=100, N2LNoHoSuccRateThld=0,
N2LNoHoConsecPrdThld=1;
//Selecting the SEPARATED_FREQ_SPEC_SHR_FLAG option if a cell working on a neighboring E-UTRA
frequency belongs to a secondary operator in RAN sharing with special dedicated carrier mode
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000, MeasurementBandwidth=MBW6,
AggregationAttribute=SEPARATED_FREQ_SPEC_SHR_FLAG-1;
//Configuring the measurement activation flag of a neighboring E-UTRAN frequency when automatic
neighboring frequency configuration is enabled
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=37900, MeasActvFlag=ACTIVE;
//Modifying the RSRP threshold of enhanced automatic neighboring frequency configuration as required
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=37900, AutoEutranFreqRelnRsrpThld=-120;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling UE blacklist management in inter-RAT ANR
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
BlacklistCtrlExtSwitch=NR_EUTRAN_ANR_SW_OFF-0;
//Disabling UE whitelist management in inter-RAT ANR
MOD GNBUECOMPATSW: UeInfoIndex=2, ParamCtrlType=ALL,
WhitelistCtrlExtSwitch=NR_EUTRAN_ANR_SW_ON-0;
//Disabling fast ANR
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_FAST_ANR_SW-0;
//Disabling optimized measurement for automatic removal of neighboring LTE cells
MOD GNODEBALGO: SelfConfigSwitch=LTE_NCELL_DEL_STAT_OPT_SW-0;
//Disabling automatic removal of neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_DEL_SW-0;
//Deselecting the N2L_HO_SUCC_RATE_INTVL_CFG_SW option
MOD GNODEBALGO: SelfConfigSwitch=N2L_HO_SUCC_RATE_INTVL_CFG_SW-0;
//Disabling neighboring cell class label management
MOD GNODEBALGO: SelfConfigSwitch=EUTRAN_NCELL_CLASS_LABEL_SW-0;
//Disabling automatic detection and addition of missing neighboring cells, automatic NCL update, and
```

```
automatic external-cell PLMN list configuration
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_SW-0;
//Disabling automatic addition of missing neighboring cells based on the number of CGI-Info reporting
times when uncontrolled mode is used
MOD NRCELLANR: NrCellId=0, NCellAddCgiRptCountThld=1;
//Disabling optimized measurement of the success rate of handovers to neighboring LTE cells in SA
networking
MOD NRCELLANR: NrCellId=0, AnrAlgoSwitch=SA_LTE_NCELL_SUCC_RT_STAT_SW-0;
//Disabling automatic optimization of neighbor relationship attributes based on the handover success rate
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_ANR_AUTO_NO_HO_SW-0;
//Disabling automatic optimization of neighbor relationship attributes based on identification of over-
distant neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=NR_EUTRAN_OVER_DIST_DETECT_SW-0;
//Disabling forbidding automatic neighboring frequency configuration
MOD NRCELLEUTRANNFREQ: NrCellId=0, DLEarfcn=1000,
FrequencyCtrlSwitch=AUTO_EUTRAN_FREQ_REL_FORBID_SW-0;
//Disabling enhanced automatic neighboring frequency configuration
MOD NRCELLANR: NrCellId=0, AnrModeSwitch=AUTO_EUTRAN_FREQ_RELN_MEAS_SW-0;
//Disabling automatic neighboring frequency configuration
MOD NRCELLALGOSWITCH: NrCellId=0, AnrSwitch=AUTO_EUTRAN_FREQ_RELATION_SW-0;
//Removing an eNodeB from the blacklist
RMV GNBEUTRANBLACKLIST: EnodebId=1, Mcc="302", Mnc="220";
```

5.1.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.3 Activation Verification

Automatic Detection and Addition of Missing Neighboring Cells/Automatic NCL Update/Automatic External-Cell PLMN List Configuration/Automatic Neighboring Frequency Configuration

To check whether the functions have taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Add Neighboring Cell**, **Add External Cell**, **Update External Cell**, **Add External Cell PLMN List**, **Delete External Cell PLMN List**, or **Update Neighboring Frequency Measurement Configuration**. If any of the events is logged, these functions have taken effect.

----End

Automatic Removal of Neighboring Cells

To check whether the function has taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Delete Neighboring Cell** or **Delete External Cell**. If either event is logged, automatic removal of neighboring cells has taken effect.
- End

Automatic Optimization of Neighbor Relationship Attributes Based on the Handover Success Rate

To check whether the function has taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Set NoHo Flag**. If the event is logged and the value of **SaHoSuccTimes/SaHoTimes** is less than or equal to that of **NRCellAnr.N2LNoHoSuccRateThld** in **Event Description**, automatic optimization of neighbor relationship attributes based on the handover success rate has taken effect.
- End

Automatic Optimization of Neighbor Relationship Attributes Based on Identification of Over-Distant Neighboring Cells

To check whether the function has taken effect, query SON logs on the MAE-Access as follows:

- Step 1** On the MAE-Access, choose **SON > SON Log > Synchronize NE Log**.
- Step 2** On the **Synchronize NE Log** tab page, click **Manual Synchronization**. In the displayed dialog box, select NEs and click **OK**.
- Step 3** After the synchronization is completed, click **Query SON Log**. On the **Query SON Log** tab page, click **Query**. Select **NR ANR Log** from the **Log Category** drop-down list.
- Step 4** In the **Event Name** area, select **Set NoHo Flag**. If the event is logged and **CellRachRadiusSum** and **InterCellDistance** are not **N/A** in **Event Description**, automatic optimization of neighbor relationship attributes based on identification of over-distant neighboring cells has taken effect.
- End

Optimization Mode

The following uses neighboring cell addition as an example to describe activation verification.

- Prerequisites:
 - a. Cell A does not exist in the NRT of cell B.
 - b. Automatic detection and addition of missing neighboring cells is enabled.
- Expected result:
 - The uncontrolled mode is used if the **LST NRCELLEUTRANRELATION** command output shows that cell A exists in the NRT of cell B.
 - The controlled mode is used if the **LST NRCELLEUTRANRELATION** command output shows that cell A still does not exist in the NRT of cell B and the **Neighboring Cell Relationship to Be Handled** area on the **Optimization Advice** tab page of the MAE-Access shows that cell A needs to be added as a neighboring cell of cell B. (To open the **Optimization Advice** tab page, choose **SON > NR ANR**.)

5.1.4.4 Network Monitoring

To monitor the running status about inter-RAT ANR, query SON logs on the MAE-Access. For details about how to query SON logs on the MAE-Access, see [5.1.4.3 Activation Verification](#).

Use the KPIs in [5.1.2.2 Impacts](#) to monitor changes in the service drop rate and average UE throughput after enhanced automatic neighboring frequency configuration is enabled.

5.2 Inter-RAT PLMN ID Management

5.2.1 Principles

Concepts, application scenarios, involved MOs of inter-RAT PLMN ID management are the same as those of intra-RAT PLMN ID management. For details, see [4.2.1 Principles](#).

Up to two PLMN types can be configured for neighboring cells of a given RAT. This principle is the same as that in intra-RAT PLMN ID management. However, inter-RAT PLMN management has the following processing differences from intra-RAT PLMN ID management:

- If two PLMN types are configured, the PLMN for which no type is configured is considered to have the third type by default.
- If only one PLMN type is configured:
 - If only a PLMN blacklist or greylist is configured, the PLMN for which no type is configured is considered whitelisted by default.
 - If only a PLMN whitelist is configured, the PLMN for which no type is configured is considered greylisted by default.
- If no PLMN types are configured, the PLMN for which no type is configured is considered whitelisted by default.

5.2.2 Network Analysis

5.2.2.1 Benefits

Benefit analysis for inter-RAT PLMN ID management is the same as that for intra-RAT PLMN ID management. For details, see [4.2.2.1 Benefits](#).

5.2.2.2 Impacts

The impacts between this function and other functions are described in [5.2.3.2.3 Function Impacts](#).

This function has no impact on the network.

5.2.3 Requirements

5.2.3.1 Licenses

None

5.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.2.3.2.1 Prerequisite Functions

None

5.2.3.2.2 Mutually Exclusive Functions

None

5.2.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
Inter-RAT ANR	NR_EUTRAN_A NR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	The gNodeB checks whether the gNBNCeIlPlmnList.PlmnListType parameter corresponding to the PLMN to which a cell belongs is set to BLACKLIST before adding the cell as a neighboring cell through inter-RAT ANR. The gNodeB adds the neighboring cell only if the parameter is not set to BLACKLIST .

5.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.2.3.4 Others

None

5.2.4 Operation and Maintenance

5.2.4.1 Data Configuration

5.2.4.1.1 Data Preparation

Table 5-5 describes the parameters used for function activation. No parameters are involved in function optimization.

Table 5-5 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBNCeIlPlmnList.Mcc	Set this parameter to the correct MCC.
Mobile Network Code	gNBNCeIlPlmnList.Mnc	Set this parameter to the correct MNC.
RAT Type	gNBNCeIlPlmnList.RatType	Set this parameter to EUTRAN .
PLMN List Type	gNBNCeIlPlmnList.PlmnListType	If the gNodeB is on the border of different operators, either of the following configuration is applied: • Set this parameter to GREYLIST for the PLMNs of the operators that own the neighboring cells to which UEs cannot be handed over or redirected. • Set this parameter to WHITELIST for the PLMNs of the operators that own the neighboring cells to which UEs can be handed over or redirected. With this setting, the PLMNs of the operators that own the neighboring cells to which UEs cannot be handed over or redirected are considered greylisted. If abnormal UEs report invalid PLMN IDs of neighboring cells, set this parameter to BLACKLIST for the invalid PLMNs.

5.2.4.1.2 Using MML Commands

Before using MML commands, refer to **5.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling inter-RAT PLMN ID management  
ADD GNBCELLPLMNLIST: RatType=EUTRAN, Mcc="302", Mnc="220", PlmnListType=WHITELIST;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling inter-RAT PLMN ID management  
RMV GNBCELLPLMNLIST: RatType=EUTRAN, Mcc="302", Mnc="220";
```

5.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.2.4.2 Activation Verification

Activation verification is the same for inter-RAT PLMN ID management and intra-RAT PLMN ID management. For details, see [4.2.4.2 Activation Verification](#).

5.2.4.3 Network Monitoring

None

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- *NSA Mobility Management*
- *SA Mobility Management in Connected Mode*
- *Multi-Frequency Smart Selection*
- *High Speed Mobility*
- *DRX*
- *Flexible User Steering*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *EPC-based NSA Basics*
- *Multi-Operator Sharing*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Carrier Aggregation*
- *PCI Conflict Detection and Self-Optimization*
- *Energy Conservation and Emission Reduction*
- *Multi-RAT Coordinated Energy Saving*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *Mobility Load Balancing in Connected Mode*
- *MRO*
- *License Control Item Lists*
- *SON Management in MAE Product Documentation (EulerOS, TaiShan)*
- *RAN Sharing in eRAN Feature Documentation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"